THE HORIZONTAL GEOMETRY OF PRODUCTION NETWORKS

by VALERIO DIONISI*

November 2026

[click here for the latest version]

Abstract

Modern economies are organised as dense production networks, yet the transmission of shocks is conventionally understood through observed Input-Output linkages and vertical cascade effects. This paper argues that propagation additionally operates through horizontal relations: the extent to which sectors share upstream suppliers or downstream buyers. These interdependencies shape a latent network of connections linking also otherwise unconnected sectors. Because it moves Domar weights, horizontal propagation is hidden in Hulten's (1978) theorem while governing the effect of sectoral volatility on aggregate output, since horizontal reallocation among sectors shifts the average level of GDP. This channel complements vertical propagation, vanishes under Cobb-Douglas technologies, gains relevance as networks expand, and survives the diversification argument. Using Input-Output data, I show that sectoral responses vary systematically with horizontal linkages, forging novel implications for the transmission of sectoral and aggregate shocks.

JEL: C67, D57, E32, J21

KEYWORDS: production networks, Input-Output economy, horizontal transmission, network distance, Hulten theorem, sectoral comovement, aggregate fluctuations

* Department of Economics, Management and Statistics, University of Milano-Bicocca, *email*: valerio.dionisi@unimib.it. I am deeply grateful to Emiliano Santoro and Francesco Zanetti for their steady feedback and deep suggestions, and to Andrea Colciago for his guidance throughout. My sincere gratitude goes also to Andrey Alexandrov, Barbara Annicchiarico, Gaetano Bloise, Ugo Bolletta, Edoardo Briganti, Enzo Dia, Catalin Dragomirescu-Gaina, Andrea Ferrero, Gino Gancia, Basile Grassi, Jakob Grazzini, Isabelle Mejean, Alessio Moro, Daniele Siena, Gabriel Smagghue and Riccardo Zago for valuable and detailed reflections. I also thank participants to workshops and seminars at 25th Research in International Economics and Finance (RIEF) meeting (Université Paris-Saclay), ASSET Annual Conference 2026 (University of Cagliari), 39th Annual Irish Economic Association (Central Bank of Ireland), Catholic University of Milan, "La Sapienza" University of Rome, "Tor Vergata" University of Rome, University of Milano-Bicocca and University of Oxford. Last but not least, my thanks to Davide Debortoli for encouraging me to explore the world of production networks, and to Mishel Ghassibe for his enlightening lectures. The idea for this paper occurred after a visiting research period at the economics department of Universitat Pompeu Fabra (2023), while the theory has been pioneered when I was visiting the economics department of University of Oxford (2025). I thank both faculties for their dynamic environment as exchange of ideas, and for their hospitality. *first version*: July 3, 2025.

1. INTRODUCTION

A hallmark of contemporary production systems is the comovement of economic activity across sectors. Through Input-Output linkages, production networks connect these sectors, allowing idiosyncratic shocks to propagate across the economy and to generate synchronized adjustments in production that contribute to aggregate dynamics. In this framework, a shock travels *vertically*, cascading from suppliers to customers along parallel supply chains.[1] Yet, a shock often travels through hidden network corridors. Comovement nonetheless emerges between productions that never trade with each other, because they share a downstream buyer (*e.g.*, February 2, 2026: the HBM4 memory breakthrough reshuffles demand across competing chip providers) or an upstream supplier (*e.g.*, the 2025 AI-driven surge in cloud demand for DRAM and HBM chips strained capacity for other users).[2]

This paper argues that production networks connect sectors not only through existing trade linkages, but also by demand and supply similarities embedded in their common structure of inter-sectoral trade. I conceptualize these interdependencies as *network economic distances* (*e.g.*, Conley and Dupor 2003). Specifically, sectors are close in input demand when they source intermediate inputs from the same upstream suppliers, and close in input supply when they sell to the same downstream buyers. Viewed through this lens, an Input-Output structure carries two geometries. It reflects *vertical* interdependencies, describing who supplies whom and how intensively (*e.g.*, Long and Plosser 1983; Shea 2002), as much as *horizontal* interdependencies, where network participants are additionally linked through structural demand and supply relations from a resembling Input-Output system.

The two geometries propagate idiosyncratic shocks in different ways. Vertically, a shock cascades from suppliers to customers through input requirements, with recursive feedback and amplification along supply chains (*e.g.*, Acemoglu, Carvalho, et al. 2012). Horizontally, it does not cascade: it reallocates. A buyer facing a higher input price can reallocate downstream demand across its suppliers, while upstream congestion at a supplier, driven by increased demand from one of its buyer, can induce its other buyers to turn to alternative sources. This is why horizontal geometry is easy to overlook: since buyers reallocate optimally, these shifts leave aggregate output unchanged at first order, and the economy seems to respond to shocks through the weight of sectors alone. Yet reallocation is not neutral: it moves workers across sectors and it changes how aggregate output responds to shocks.

Network-based comovement is therefore endogenous. Since sectors mutually adjust in response due to shared economic relationships, horizontal geometry becomes a source of systematic comovement across economically-proximate sectors. The transmission of shocks depends not only on the sequentiality proper of observed connections, but also on the exposure to the same Input-Output neighbourhoods: sectors that share a buyer compete for the same demand; sectors that share a supplier draw on the same production capacity. The question is not only whether shocks propagate through production networks, but also whether the horizontal configuration of these networks matters to determine who is exposed to whom, and by how much.

Indeed, building a theory of horizontal transmission that separates it from vertical propagation and traces its implications for sectoral employment and aggregate output, and validating its predictions with U.S. sectoral data, constitute the central contributions of this paper.

---

[1] A positive shock to a sector improves its productive performance, triggering: *(i)* a *downstream effect*, as its buyers expand their demand for the cheaper commodity; *(ii)* an *upstream effect*, as its higher production raises its demand for intermediate inputs, benefiting its suppliers; and *(iii) network spillovers*, as these effects ripple beyond immediate trading partners.

[2] Refer to Supplemental Appendix E for a discussion about some documented real-world examples.

Section 2 lays down the horizontal geometry. The same Input-Output matrix can be read in two ways: along supply chains, by composing relationships from supplier to customer, or across nodes, by comparing their profiles of suppliers and of buyers. The first reading builds the familiar Leontief inverse; the second builds two distance matrices, one for common suppliers and one for common buyers. Distance has an intuitive reading as common exposure: sectors that buy from the same suppliers inherit the same cost shocks, and those selling to the same buyers face similar shifts in demand. As Lemma 1 establishes, a common neighbour makes two sectors economically connected even if they never trade, and the same structure generally places a pair of sectors at different distances on the side of suppliers and on that of customers.

Section 3 turns horizontal geometry into a transmission object. It first isolates a single shared partner, and finds that the two horizontal channels differ in nature. A common supplier transmits only through its own price: when one of its customers expands, the supplier becomes more expensive if its capacity is congested and cheaper under increasing returns, so that comovement between co-customers is a vertical round trip whose sign depends on the supplier's costs (Proposition 1). A common buyer, instead, generates genuine horizontal competition: when one upstream supplier becomes cheaper, the downstream buyer reallocates a given budget, and the shocked sector and its rivals move in opposite directions (Proposition 2). The elasticity of substitution among intermediate inputs does not alter this opposition; it only decides who wins: the shocked sector if inputs are substitutes, its rivals if they are complements. Demand distances therefore make comovement ambiguous, whereas supply distances make it transparent.

Idiosyncratic shocks are then allowed to propagate across the whole production network. By Theorem 1, general equilibrium reallocation unfolds in three steps: supply chains carry a shock to the cost of every commodity, each buyer reallocates its budget among intermediate inputs whose costs have moved unevenly, and parallel supply chains carry the resulting change in demand back to suppliers. Vertical structure is the carrier, horizontal geometry is where reallocation happens. Under Cobb-Douglas technologies buyers never reallocate (*e.g.*, Acemoglu, Carvalho, et al. 2012) so that horizontal transmission of supply shocks is absent (Corollary 1). Horizontal geometry is also an independent dimension of the network: economies with identical sectoral sizes, and hence identical aggregate responses at first order, can make the same pair of sectors rivals in one case and apart in the other (Corollary 2). Finally, horizontal linkages proliferate as the production network expands: every new demand-supply connection makes the new supplier horizontally related to all the suppliers its buyer already has (Corollary 3).

Because horizontal transmission moves sales shares, it also moves workers. Sectors spend a constant share of revenue on labour, so that employment moves relative to the aggregate only if Domar weights move, that is, only if buyers change the shares of their budgets they spend on the intermediate input of each sector (Proposition 3). Labour reallocation is thus zero-sum: a shock moves the employment of a sector and of its close common-buyer neighbours in opposite directions, whereas horizontally-distant sectors comove together with the shocked sector. For aggregate output, instead, Hulten's (1978) theorem seems to leave no room for horizontal geometry: at first order, aggregate output responds to sectoral shocks through the Domar size of sectors alone. Theorem 2 reveals, however, that horizontal geometry is not absent, but hidden and plays a fundamental allocative role. Shared Input-Output structures matter through the endogenous response of Domar weights to shocks: the horizontal reallocation of intermediate input shapes how sectoral output shares adjust, thereby moving aggregate GDP beyond its benchmark where idiosyncratic shocks average out (Lucas 1977) whatever the distribution of Domar weights. This horizontal dimension gives rise to a non-vanishing effect of micro volatility on mean aggregate output. Positive and negative shocks may cancel out in the aggregate, but their dispersion within downstream buyers' input bundles does not.

Section 4 brings the two geometries to U.S. Input-Output data. Vertical propagation is concentrated in a few hubs supplying a wide range of activities, whereas horizontal relations are diffuse, as most sectors share partners with many others. Removing transactions below a progressively higher threshold steadily weakens vertical transmission, while horizontal propagation remains stable and becomes the dominant channel: horizontal relations rest on many shared partners, and survive sparsification as Corollary 3 predicts. A Bartik-style design then studies whether horizontal geometry shapes the transmission of aggregate disturbances beyond vertical exposure. The relevant channel depends on the nature of the disturbance and its sectoral exposures: aggregate productivity shocks, which move relative prices, travel mainly through common downstream buyers, whereas government spending shocks, which are demand impulses, travel mainly through common upstream suppliers. Horizontal exposure thus contains information on shock transmission that conventional Leontief-based linkages do not capture.

Section 5 studies sectoral comovement directly, relying on the two-stage approach of Barattieri and Cacciatore (2023) through which I estimate the way identified employment shocks propagate across sectors by type and degree of network "economic" distance. The empirical results are threefold. First, an increase in the intermediate inputs circulating through a sector raises its employment, with minimal and alternating effects from similar changes in distant sectors. Second, sectors that are close in supply distance exhibit opposite employment comovement, as the common-buyer channel predicts: a rise in employment in a nearby sector tends to reduce employment in its rivals. For sectors close in demand distance, responses are ambiguous, as the common-supplier channel predicts, although they mostly point to a dampening of vertical transmission. Third, positive comovement emerges among sectors that are horizontally distant, where vertical and aggregate forces prevail.

Overall, theory and evidence point in the same direction: horizontal linkages complement the canonical Leontief inverse, vertical transmission of shocks. Production networks shape the propagation of economic activity not only through who trades with whom, but also through who competes for the same buyers and draws on the same suppliers.

Literature review.– The objective of this paper is to advance both theoretically and empirically the understanding of how Input-Output economies work. On the theoretical side, it builds on the modern and rapidly expanding literature on production networks. In the wake of Gabaix (2011)'s "granular hypothesis", the seminal observation of Long and Plosser (1983) – that comovements across sectors are not dictated by a shared disturbance but rather by sectoral interdependencies –, has been modernized through a series of papers of the early 2010s (*e.g.*, Acemoglu, Carvalho, et al. 2012; Carvalho and Gabaix 2013; Carvalho 2014; Barrot and Sauvagnat 2016). This renewed perspective gave rise to studies on production networks related to efficient economies (*e.g.*, Baqaee and Farhi 2019; vom Lehn and Winberry 2022; Liu and Tsyvinski 2024), monetary policy and nominal rigidities (*e.g.*, La'O and Tahbaz-Salehi 2022; Rubbo 2023; Ghassibe and Nakov 2025), inefficiencies (*e.g.*, Jones 2011; Grassi 2017; Baqaee 2018; Baqaee and Farhi 2020), policy-oriented issues (*e.g.*, Liu 2019; Grassi and Sauvagnat 2019; Lane 2025), endogenous network formation (*e.g.*, Oberfield 2018; Acemoglu and Azar 2020; Ghassibe 2024; Taschereau-Dumouchel 2026), economic growth (*e.g.*, Hausmann and Hidalgo 2011; Gualdi and Mandel 2019; McNerney et al. 2022), and international contexts (*e.g.*, Caliendo et al. 2022; Huo et al. 2025; Qiu et al. 2026). A common mechanism explored in these works emphasizes vertical propagation.[3] My contribution broadens this perspective by

[3] As noted by Shea (2002), a positive sectoral supply shock has a price (decrease in its price level, and reduction in its nominal expenditure for intermediate inputs) and a quantity (its supply to other sectors increase, sector's production rises, thereby increasing its input demand) effect. These opposing forces affect upstream suppliers but, under Cobb-Douglas technology, they

introducing horizontal linkages from common upstream or downstream Input-Output relationships. These novel interdependencies forge horizontal relations, allowing asymmetric shocks to propagate across the network in ways beyond vertical chains.

On the empirical side, an increasing number of studies examine the role of sectoral production networks on macroeconomic outcomes (*e.g.*, Acemoglu, Akcigit, et al. 2015; Ghassibe 2021; Barattieri and Cacciatore 2023; Barattieri, Cacciatore, and Traum 2023), with some illustrative spillovers underlying horizontal transmission (*e.g.*, Boehm and Oberfield 2018; Dragomirescu-Gaina and Elia 2021; Huremović, Nutarelli, et al. 2025). From this standpoint, the paper contributes novel empirical evidence on the role of Input-Output linkages in shaping interlinked dynamics, particularly in explaining the comovement of employment across sectors. Synchronized employment patterns are an important characteristic of business cycles, as most sectors tend to move together over time (*e.g.*, Christiano and Fitzgerald 1998). Several papers explore the nature and the sources of this comovement. Cooper and Haltiwanger (1990) find that sectoral employment levels are positively correlated, in ways not fully explained by aggregate shocks. Cassou and Vázquez (2014) attribute high employment comovement to similar shock transmission channels. In Yedid-Levi (2016) comovement is stronger along the extensive (number of workers) rather than the intensive (hours per worker) margin. Room for structural changes is in Kim (2020), where positive technology shocks in manufacturing raise employment in both manufacturing and services. While these studies provide valuable insights, they do not explicitly address how production networks shape employment comovement. Empirically, the paper contributes by: *(i)* advancing the comprehension of shocks' transmission by exploiting network-based economic distances; and *(ii)* providing unprecedented evidence on short-run employment comovement from a production network perspective.

2. Foundational Theory

The economy is populated by a finite set of sectors, $i, j, s \in \Phi = \{1, \ldots, m\}$, with Φ_i denoting the set of all others not including sector-i. Each produces a single commodity using labour and a set of circulating intermediate inputs from other sectors. The whole economic system is thus represented by an $m \times 1$ vector of sectoral production outputs, $\boldsymbol{y} = \left[y_i > 0\right]$, an $m \times 1$ vector of sectoral employment levels, $\boldsymbol{n} = \left[n_i > 0\right]$, each associated to its output elasticity $\alpha_i \in (0,1)$ with labour paid a common wage w, and by an $m \times m$ square matrix, $\boldsymbol{X} = \left[x_{ij} \geq 0\right]$, indicative of the inter-sectoral trade of intermediate input quantities, with $x_{ij} = 0$ stating that sector-i does not make use of the commodity produced by sector-j at price p_j, stacked into $\boldsymbol{p} = [p_1, \ldots, p_m]$. Finally, an $m \times m$ Input-Output matrix, $\boldsymbol{\mathcal{A}} = \left[\alpha_{ij} \geq 0\right]$, collects the intensity of the commodities produced by sectors as intermediate inputs in others. Its i-th row, $\boldsymbol{a}_i^{row} = \left[\alpha_{i1}, \ldots, \alpha_{ij}, \ldots, \alpha_{im}\right]$, is sector-$i$'s input profile: whom it buys from, and how intensively. Its i-th column, $\boldsymbol{a}_i^{col} = \left[\alpha_{1i}, \ldots, \alpha_{ji}, \ldots, \alpha_{mi}\right]'$, is sector-$i$'s output profile: who buys from it, and how intensively. Provided that $\text{Radius}\left[\boldsymbol{\mathcal{A}}\right] < 1$, denote the Leontief inverse as $\boldsymbol{\mathcal{L}} = \left(\boldsymbol{I} - \boldsymbol{\mathcal{A}}\right)^{-1} = \left[\ell_{ij} \geq 0\right]$, with $\boldsymbol{I}$ being an $m \times m$ identity matrix.

A competitive final producer combines the commodities sold to final users, $Y = \prod_{i \in \Phi} c_i^{\omega_i}$, where $\boldsymbol{c} = \left[c_i > 0\right]$ is the $m \times 1$ vector of final consumption levels and $\boldsymbol{\omega}$ is an $m \times 1$ vector of shares $\{\omega_i\}_{\forall i \in \Phi} \in (0,1)$, with $\boldsymbol{1}'\boldsymbol{\omega} = 1$, weighting the relative importance of each sectoral commodity in aggregate output. A representative household gets utility from consumption and

offset each other leaving intermediate inputs demand unchanged, thereby affecting only downstream customers. In general, only demand shocks can propagate upstream (*e.g.*, Acemoglu, Akcigit, et al. 2015; Barrot and Sauvagnat 2016; Ferrari 2024).

leisure dis-utility from working: $\mathcal{U} = f^u\left[c_i\,,\, n_i\right]_{\forall i\in\Phi}$, with consumption of all sectoral commodities and labour supplied characterized by prices $\boldsymbol{p}$ and w. Moreover, since the household spends a share ω_i of final expenditure on commodity-i, Domar weights are $\boldsymbol{\lambda}' = \boldsymbol{\omega}'\boldsymbol{\mathcal{L}}$.[4]

A representative firm characterizes each sector-i. Given an exogenous Hicks-neutral productivity z_i, the production function is

$$y_i = z_i\, f^y\left[n_i\,,\; x_i := \left\{x_{ij}\right\}_{\forall j\in\Phi}\right]$$

ASSUMPTION 1 (Production technology requirements) *The production function: (i) has constant returns to scale in n_i and x_i so that $\alpha_i + \sum_{j\in\Phi}\alpha_{ij} = 1$; (ii) is differentiable, continuous, strictly quasi-concave, homogeneous of degree one, and increasing in z_i, n_i, and x_i; (iii) rules out $f^y\left[0\,,\,\cdot\right]$ as labour input is essential to production; (iv) has at least two positive elements x_i, so that $f^y\left[\cdot\,,\,0\right]$ cannot exist; consequentially, (v) uses a part of the sector's own production, $x_{ii} > 0$, $\forall i\in\Phi$, so that a production plan with a bundle of intermediate inputs as $f^y\left[\cdot\,,\,x_{ij}\right]$, given any $j\in\Phi_i$, is precluded.*

The first two sub-assumptions are common in production technology (*e.g.*, Acemoglu, Carvalho, et al. 2012). By way of *A1.iii*, workers cannot be fully substituted by any combination of intermediate inputs, and changes in the purchased quantity of intermediate inputs would determine subsequent variations in the labour force size. The assumptions characterizing the bundle of intermediates, *A1.iv* and *A1.v*, ensure the sector not only to participate in the network, but also that part of the production process is roundabout: for sector-i delivering its commodity y_i requires to be at least vertically connected (buying from one upstream sector), while imposing that a fraction of intermediate inputs must be directly produced within the sector.

Each perfectly competitive representative firm maximizes profits given input prices, w and p_j, $\forall j\in\Phi$. For any functional form of y_i satisfying Assumption 1, by Shephard's lemma the share of commodity-j in sector-i's unit cost $\kappa_i = f^\kappa\left(\boldsymbol{p}, w\right)$ is $\alpha_{ij} = \frac{\partial\log\kappa_i}{\partial\log p_j} \geq 0$, and the labour share is $\alpha_i = \frac{\partial\log\kappa_i}{\partial\log w}$, so that the constraint $\alpha_i + \sum_{j\in\Phi}\alpha_{ij} = 1$ holds exactly for any $\left\{\boldsymbol{p}, w\right\}$.

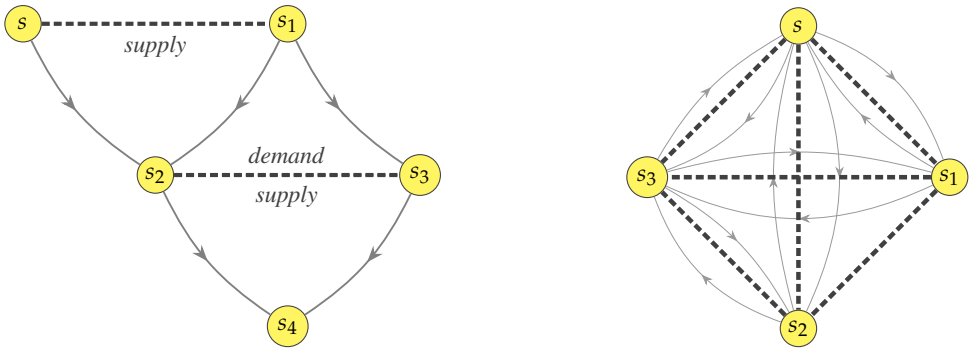
Vertical and horizontal contractions.– In a networked economy, comovement of production inputs is endogenously induced. Each sector both relies on and supplies to others, so that changes in one sector affect not only its own optimal input composition but also the demand and supply conditions faced by others. Sectors that trade with similar partners therefore face shared network exposure, providing a natural basis for a rigorous comparison.

Consider a pair of sectors, say $\{i, j\}$, related to a common one, say s. For the pair, the same Input-Output matrix can be contracted over a common partner in three ways:

- $\left[\boldsymbol{\mathcal{A}}^2\right]_{ij} = \sum_{s\in\Phi}\alpha_{is}\,\alpha_{sj}$ denotes a recursive production chain;
- $\left[\boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{A}}'\right]_{ij} = \sum_{s\in\Phi}\alpha_{is}\,\alpha_{js}$ denotes the common upstream neighbourhood;
- $\left[\boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{A}}\right]_{ij} = \sum_{s\in\Phi}\alpha_{si}\,\alpha_{sj}$ denotes the common downstream neighbourhood.

The first is a recursive chain (i buys from s, which buys from j), generalisable to k-stages beyond the common sector-s, $\left[\boldsymbol{I}\right]_{ij} + \left[\boldsymbol{\mathcal{A}}\right]_{ij} + \left[\boldsymbol{\mathcal{A}}^2\right]_{ij} + \cdots + \left[\boldsymbol{\mathcal{A}}^k\right]_{ij}$; the second is a connection through a common supplier (both i and j buy from s); the third is a connection through a

[4] Domar weights are defined as $\lambda_i = \frac{p_i y_i}{\sum_{i\in\Phi} wn_i}$, that is sectoral sales relative to nominal GDP (labour income only, since profits are zero and labour is the only factor). Multiplying the market-clearing condition for sector-i, $y_i = c_i + \sum_{j\in\Phi} x_{ji}$, by $\frac{p_i}{\sum_{i\in\Phi} wn_i}$, using $p_i c_i = \omega_i \sum_{i\in\Phi} wn_i$ and, by definition of cost shares, $p_i x_{ji} = \alpha_{ji}\, p_j y_j$, gives $\lambda_i = \omega_i + \sum_{j\in\Phi}\lambda_j\alpha_{ji}$. In matrix form, $\boldsymbol{\lambda}' = \boldsymbol{\omega}' + \boldsymbol{\lambda}'\boldsymbol{\mathcal{A}}$. Since every row of $\boldsymbol{\mathcal{A}}$ sums to $1-\alpha_i < 1$, $\boldsymbol{I} - \boldsymbol{\mathcal{A}}$ is invertible, and $\boldsymbol{\lambda}' = \boldsymbol{\omega}'\left(\boldsymbol{I} - \boldsymbol{\mathcal{A}}\right)^{-1} = \boldsymbol{\omega}'\boldsymbol{\mathcal{L}}$.



(A) DISTANCE DEFINITION (B) DISTANCE NETWORK

FIGURE 1: HORIZONTAL INPUT-OUTPUT GEOMETRIES

Note: the figure presents a stylized production network and its corresponding network of sectoral distances. Production network linkages are depicted with oriented thin lines, while dashed thick lines indicate Input-Output economic distance relationships.

common buyer (s buys from both). Under this perspective, network connections extend beyond their flows of intermediate inputs. These additional interdependencies highlight the complexity of inter-sectoral dynamics and complement the *Leontief inverse*, a reminiscent result from Leontief (1936): it captures how sectors not directly trading are indirectly related across stages of production – *e.g.*, a sector buying from an upstream one introjects such sector's purchases as well the other of more upstream sectors. Here, powers of $\boldsymbol{\mathcal{A}}$ compose relationships along the supply chains and build $\boldsymbol{\mathcal{L}} = \sum_{k\geq 0} \boldsymbol{\mathcal{A}}^k$: they operationalise the depth of *vertical* linkages through layers of existing Input-Output relationships. Common connections, by contrast, consider how network participants are mapped horizontally: sectors are additionally related because they are relying on the same suppliers or serving the same customers. Here, Gram products compare the profiles of two sectors at the same production stage: they are *horizontal*, and they link also sectors that need not trade with each other at all. These broader notions of linkages provide a richer understanding about systemic interdependencies and explain horizontal similarities in demand and supply that classical Input-Output analysis overlook. They reveal how seemingly unrelated sectors can experience synchronized fluctuations, and how the structure of common upstream or downstream connections shapes the propagation of independent and idiosyncratic shocks. In other words, the notion of network interdependencies is not confined to "who trades with whom", but also emerges from "who depends on whom".

DEFINITION 1 (Horizontal proximity and network distance) *In a production network, the common-supplier and common-buyer proximity matrices are*

$$\boldsymbol{\mathcal{P}}^{cs} = \boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{A}}' \qquad \text{and} \qquad \boldsymbol{\mathcal{P}}^{cb} = \boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{A}}$$

so that $\mathcal{P}^{cs}[i,j] = \langle \boldsymbol{a}_i^{row}, \boldsymbol{a}_j^{row} \rangle$ *and* $\mathcal{P}^{cb}[i,j] = \langle \boldsymbol{a}_i^{col}, \boldsymbol{a}_j^{col} \rangle$. *As resulting from Subsection 2.1, the associated distances are* $\mathcal{D}^q[i,j] = \left(\mathcal{P}^q[i,i] + \mathcal{P}^q[j,j] - 2\,\mathcal{P}^q[i,j]\right)^{1/2}$, *for* $q = \{cs, cb\}$, *that is* $\mathcal{D}^{cs}[i,j] = \left\|\boldsymbol{a}_i^{row} - \boldsymbol{a}_j^{row}\right\|$ *and* $\mathcal{D}^{cb}[i,j] = \left\|\boldsymbol{a}_i^{col} - \boldsymbol{a}_j^{col}\right\|$, *collected in* $\boldsymbol{\mathcal{D}}^q$.

Paraphrasing Conley and Dupor (2003), two types of network "economic" distances between sectors, sketched in Panel 1a of Figure 1, can be defined: *(a) common-supplier*, when sectors are buying their intermediate inputs from similar upstream sectors; and *(b) common-buyer*, when sectors are selling their commodities as intermediate inputs to similar downstream sectors. Panel 1b sketches a stylized economy associated to its distance-based network: distances effectively double Input-Output linkages and bound disconnected sectors.

EXAMPLE 1 (Distance in the network) *Consider an economy populated by four sectors,* $\{s, s_1, s_2, s_3\} \in \Phi$*, where some trade with all others, while some do not. In Panel 1b, focus on the pair* $\{s_1, s_2\}$*: they do not mutually trade, so* $\alpha_{s_1 s_2} = 0 = \alpha_{s_2 s_1}$*. Nevertheless, their linkages with all the remaining sectors generate a common structure of inter-sectoral trade. Yet, while the underlying trade and distance structures are unique, the horizontal metrics vary under alternative specifications of network distance, as Examples 2-3 clarify.*

Indeed, the same primitive Input-Output structure governing the trade of commodities not only determines how connections spread through production chains, but also how sectors are positioned relative to each other in the space of shared interdependencies. A single production network generates two distinct but tightly connected objects, different features of the same primitive structure: one capturing recursive chains, the other describing the overlap of network linkages. On one hand, its Leontief inverse captures a vertical system: it is the equilibrium mapping encoding how commodities are produced by the means of other commodities (Sraffa 1960) through successive layers of production along length-k production chains. On the other hand, the same matrix defines a horizontal embedding: sectors are related insofar as they share similar demand and supply relationships, interacting with common segments of the network; that is, a collection of profiles $\left\{\boldsymbol{a}_i^{row}, \boldsymbol{a}_i^{col}\right\}_{i\in\Phi}$ encoding a geometric representation of their production commonalities in the Input-Output space.

The discussion is in terms of the primitive coefficient matrix $\boldsymbol{\mathcal{A}}$, but it applies also to its Leontief inverse $\boldsymbol{\mathcal{L}}$, since it carries over the primitive structure into its cumulative form rather than creating a new one. The entries of $\boldsymbol{\mathcal{A}}$ are the direct Input-Output coefficients, while its powers aggregate their propagation across production chains of length-k.

REMARK 1 *The Leontief inverse inherits the dual representation, since* $\boldsymbol{\mathcal{L}} = \sum_{k\geq 0} \boldsymbol{\mathcal{A}}^k$.

So, this vertical-horizontal duality is intrinsic to any Input-Output system. The primitive matrix $\boldsymbol{\mathcal{A}}$ and its Leontief inverse $\boldsymbol{\mathcal{L}}$ are two representations of the same network: constructing horizontal similarities from either objects is logically equivalent, although the two carry different economic interpretations. Section 3 shows which one governs the transmission of shocks.

2.1. RECOVERING DEMAND AND SUPPLY MATRICES

Consider the case in which two sectors, say $\{i, j\}$, are buying intermediate inputs from the same sector, say s. Let the price of the common input to vary by $d\log p_s$, holding all other prices fixed. The two sectors' costs respond as $d\log \kappa_{i,s} = \alpha_{is} \cdot d\log p_s$ and $d\log \kappa_{j,s} = \alpha_{js} \cdot d\log p_s$. The comovement induced by the common sector-s, per unit of variance of $d\log p_s$, is therefore the covariance of the two responses, identified by the sectoral proximity:

$$\mathcal{P}^{cs}_{s\rightarrow}[i,j] \;=\; \text{Cov}\left[d\log \kappa_{i,s}\,,\, d\log \kappa_{j,s}\right] \;=\; \alpha_{is}\alpha_{js} \;\geq\; 0 \tag{1}$$

Three properties follows. First, eq. (1) selects genuinely shared linkages, since $\mathcal{P}^{cs}_{s\rightarrow}[i,j] > 0$ if and only if both sectors purchase from sector-s, for strictly positive connections $\alpha_{is}\alpha_{js}$. Second, it weights by the network intensity of both: sharing a supplier from which each buys a large share of its inputs counts more than sharing one that is marginal for either. Third, since these α's are cost shares, $\mathcal{P}^{cs}_{s\rightarrow}[i,j]$ is the probability that one unit of expenditure drawn at random from sector-i and one drawn independently from sector-j end up at sector-s.

Iteratively, the above condition can be rewritten accounting for all sectors from which the pair is buying: $\boldsymbol{\mathcal{P}}^{cs}[i,j] = \left[\alpha_{ii}\alpha_{ji}\,,\, \alpha_{ij}\alpha_{jj}\,,\, \ldots\,,\, \alpha_{im}\alpha_{jm}\right]$, which is the $1\times m$ vector of the pair's shared intermediate usage when sectors $\{i, j\}$ are buying from others. The vector carries two

distinct pieces of information. Its support identifies the set of positive entries through which suppliers the pair is connected, and is the extensive margin; and its magnitudes identifying how much each channel weighs, and are the intensive margin. Summing across channels and stacking such condition across all sectors delivers the $m \times m$ common-supplier proximity matrix $\boldsymbol{\mathcal{P}}^{cs}$, whose generic element is the sum of eq. (1) over common nodes, $\mathcal{P}^{cs}[i,j] = \sum_{s\in\Phi} \alpha_{is}\alpha_{js}$. The greater the value of each cell, the closer the sectors are within the production network. Co-exposure is increasing in both trade intensities: when either α_{is} or α_{js} is small, a shock transmitted within the pair $\{i,j\}$ through s moves at least one of the two costs little; the induced comovement is weak, signalling weak co-exposure; conversely, when both intensities are large the same shock moves both strongly, and $\mathcal{P}^{cs}_{s\rightarrow}[i,j]$ is large (the same logic applies to the common-supply case, eq. 3). Using the demand composition of sector-i, $\boldsymbol{a}_i^{row} = \left[\alpha_{i1},\ldots,\alpha_{im}\right]$, each entry is an inner product between profiles, $\mathcal{P}^{cs}[i,j] = \left\langle \boldsymbol{a}_i^{row}, \boldsymbol{a}_j^{row} \right\rangle$, so that $\boldsymbol{\mathcal{P}}^{cs} = \boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{A}}'$.

Matrix $\boldsymbol{\mathcal{P}}^{cs}$ measures economic *proximity*: large entries denote strongly co-exposed pairs. The corresponding economic *distance* follows from the standard Gram-to-metric conversion

$$\mathcal{D}^{cs}[i,j]^2 = \mathcal{P}^{cs}[i,i] + \mathcal{P}^{cs}[j,j] - 2\,\mathcal{P}^{cs}[i,j] = \left(\left\|\boldsymbol{a}_i^{row} - \boldsymbol{a}_j^{row}\right\|_2\right)^2 \tag{2}$$

where the second equality follows by expanding the squared norm through bi-linearity of the inner product. Explicitly,

$$\mathcal{D}^{cs}[i,j] = \sqrt{\left[\alpha_{ii} - \alpha_{ji}\right]^2 + \left[\alpha_{ij} - \alpha_{jj}\right]^2 + \cdots + \left[\alpha_{im} - \alpha_{jm}\right]^2}$$

thereby representing a certain network "economic" distance between two sectors, with larger values denoting how dissimilar they are in terms of shared Input-Output structure when buying from upstream sellers. Henceforth, the corresponding $m \times m$ matrix $\boldsymbol{\mathcal{D}}^{cs}$ collects these distances, whose characterizing elements are symmetric, $\mathcal{D}^{cs}[i,j] = \mathcal{D}^{cs}[j,i] \geq 0$, outside the main diagonal (the distance from i to j is the same as from j to i), with equality if and only if the two sectors have the same Input-Output profile, and $\mathcal{D}^{cs}[i,i] = 0$. Proximity and distance are indeed two readings of one dimension: $\boldsymbol{\mathcal{P}}^{cs}$ is foundational on the theory side, where larger entries mean closer sectors, while its distance-based conversion delivers the metric for $\boldsymbol{\mathcal{D}}^{cs}$ used in the empirics of Sections 4-5, where larger entries mean the opposite.

EXAMPLE 2 (Common-supplier metrics) *Consider the pair $\{s_1, s_2\}$ in an Input-Output matrix mirroring Example 1, both purchasing from $\{s, s_3\}$, and assume fictional trade intensities for sector-s_1 to be $\alpha_{s_1 s} = 0.2$ and $\alpha_{s_1 s_3} = 0.1$, while that of sector-s_2 are $\alpha_{s_2 s} = 0.1$ and $\alpha_{s_2 s_3} = 0.2$, with own-use intensities $\alpha_{s_1 s_1} = \alpha_{s_2 s_2} = 0.1$. Accordingly, the co-exposure vector expressing their common-supplier relation is $\boldsymbol{\mathcal{P}}^{cs}[s_1, s_2] = [.02, .02]$ when buying from sector-s and sector-s_3, respectively. Then $\mathcal{D}^{cs}[s_1, s_2] = \sqrt{.04} = .2$. Common-supplier proximity is positive even in the absence of a direct linkage between the two sectors in the pair.*

Consider now the case in which two sectors, say $\{i,j\}$, are selling their own commodity as an intermediate input to the same sector, say s. By definition of cost share, and since total cost equals revenue, the value of sector-i's sales to s is $p_i x_{si} = \alpha_{si} \cdot p_s y_s$. A variation in the buyer's scale $p_s y_s$ at constant shares therefore moves each supplier's sales to s, and the covariance of the two responses per unit of shock variance (given the common buyer's size) delivers

$$\mathcal{P}^{cb}_{\rightarrow s}[i,j] \;=\; \mathrm{Cov}\left[d\log\left[p_i x_{si}\right]/p_s y_s,\; d\log\left[p_j x_{sj}\right]/p_s y_s\right] \;=\; \alpha_{si}\alpha_{sj} \;\geq\; 0 \tag{3}$$

Note that this proximity records exposure to the same buyer, not the sign of transmission: a change in the buyer's scale moves the two suppliers together, whereas a change in the relative

price of one of them makes the buyer reallocate a given budget and moves them apart (Proposition 2). The same distinction applies to common suppliers, whose customers' costs always comove positively, while their sales need not (Subsection 3.4).

Collecting across all common buyers, $\boldsymbol{\mathcal{P}}^{cb}[i,j] = \left[\alpha_{ii}\alpha_{ij}, \ldots, \alpha_{mi}\alpha_{mj}\right]$, and stacking delivers the common-buyer proximity matrix $\boldsymbol{\mathcal{P}}^{cb}$, with generic element $\mathcal{P}^{cb}[i,j] = \sum_{s\in\Phi}\alpha_{si}\alpha_{sj} = \left\langle \boldsymbol{a}_i^{col}, \boldsymbol{a}_j^{col}\right\rangle$, where $\boldsymbol{a}_i^{col} = \left[\alpha_{1i}, \ldots, \alpha_{mi}\right]'$ is the supply profile of sector-i. In matrix form $\boldsymbol{\mathcal{P}}^{cb} = \boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{A}}$; its $m \times m$ distance matrix $\boldsymbol{\mathcal{D}}^{cb}$ follows from eq. (2) applied to supply profiles.

EXAMPLE 3 (Common-buyer metrics) *Consider the pair $\{s_1, s_2\}$ in an Input-Output matrix mirroring Example 1, both selling to $\{s, s_3\}$, and assume fictional trade intensities for sector-s_1 to be $\alpha_{ss_1} = 0.2$ and $\alpha_{s_3s_1} = 0.1$, while that of sector-s_2 are $\alpha_{ss_2} = 0.4$ and $\alpha_{s_3s_2} = 0.3$, with own-use intensities as in Example 2. Accordingly, the co-exposure vector expressing their common-buyer relation is $\boldsymbol{\mathcal{P}}^{cb}[s_1, s_2] = [.08, .03]$ when selling to sector-s and sector-s_3, respectively. Then $\mathcal{D}^{cb}[s_1, s_2] = \sqrt{.10}$. Common-buyer proximity is positive even in the absence of a direct trade linkage between the two sectors in the pair.*

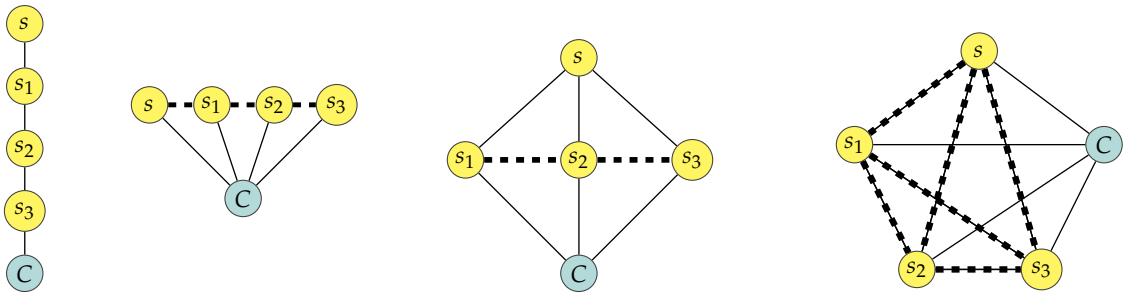
An important aspect to consider when introducing economic distances in the production network is that, under the same configuration of the inter-sectoral trade structure, sectors can be in a different "distance relation" depending on whether they are buying from or selling to other sectors. Asking when the two sides coincide amounts to asking when co-exposure fails to distinguish $\boldsymbol{\mathcal{A}}$ from $\boldsymbol{\mathcal{A}}'$, which is the content of the following result.

LEMMA 1 (Properties) *Common-supplier and common-buyer matrices satisfy:*

- *$\boldsymbol{\mathcal{P}}^{cs}$ and $\boldsymbol{\mathcal{P}}^{cb}$ are non-negative, symmetric and positive semi-definite;*
- *proximity $\mathcal{P}^{q}[\cdot]$ is coupled with $\mathcal{D}^{q}[\cdot]$, a Euclidean metric on Input-Output profiles: $\mathcal{D}^{q}[i,j] = 0$ if and only if the two profiles coincide; $\boldsymbol{\mathcal{D}}^{cs}$ and $\boldsymbol{\mathcal{D}}^{cb}$ have a zero diagonal and, unless null, are not positive semi-definite; for non-negative profiles normalized to sum to one at the common node, $\mathcal{D}^{q}[i,j] \in \left[0, \sqrt{2}\right]$ for $q = \{cs, cb\}$;*
- *proximity and distance matrices make all nodes simultaneously connected, with each cell strictly positive if the pair shares at least one neighbour, so that both matrices are full;*
- *they differ entry-wise unless the corresponding bilateral trade with the common neighbour coincides: $\boldsymbol{\mathcal{P}}^{cs} = \boldsymbol{\mathcal{P}}^{cb}$ if and only if $\boldsymbol{\mathcal{A}}$ is a normal matrix, as when $\boldsymbol{\mathcal{A}}$ is symmetric.*

Proof: for the first claim, note that for any $\boldsymbol{u} \in \mathbb{R}^m$, then $\boldsymbol{u}'\boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{A}}'\boldsymbol{u} = \|\boldsymbol{\mathcal{A}}'\boldsymbol{u}\|^2 \geq 0$ and $\boldsymbol{u}'\boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{A}}\boldsymbol{u} = \|\boldsymbol{\mathcal{A}}\boldsymbol{u}\|^2 \geq 0$; symmetry is immediate. Since $\mathcal{P}^{cs}[i,j] = \sum_{s\in\Phi}\alpha_{is}\alpha_{js}$ is a sum of non-negative terms, it is positive if and only if $\alpha_{is}\alpha_{js} > 0$ for some s, *i.e.*, if i and j share a supplier; likewise for $\mathcal{P}^{cb}$ and buyers. For the second claim, by bilinearity, $\mathcal{P}^{q}[i,i] + \mathcal{P}^{q}[j,j] - 2\,\mathcal{P}^{q}[i,j]$ equals the squared Euclidean norm of the difference between the two profiles, which defines a metric on profiles. $\boldsymbol{\mathcal{D}}^{q}$ is symmetric with zero trace; if it has a non-zero entry, its eigenvalues sum to zero and are not all zero, so at least one is negative. For non-negative profiles summing to one, $\|\boldsymbol{a}_i^{row} - \boldsymbol{a}_j^{row}\|^2 = \|\boldsymbol{a}_i^{row}\|^2 + \|\boldsymbol{a}_j^{row}\|^2 - 2\langle \boldsymbol{a}_i^{row}, \boldsymbol{a}_j^{row}\rangle \leq 2$, since $\|\boldsymbol{a}_i^{row}\|^2 = \sum_{s\in\Phi}\alpha_{is}^2 \leq \left(\sum_{s\in\Phi}\alpha_{is}\right)^2 = 1$ and $\langle \boldsymbol{a}_i^{row}, \boldsymbol{a}_j^{row}\rangle \geq 0$, and analogously for columns. The third claim follows from eqs. (1)-(3). The fourth claim holds by definition of a normal matrix, if $\boldsymbol{\mathcal{A}}$ is normal, then $\boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{A}}' = \boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{A}}$; for a given pair, $\mathcal{P}^{cs}[i,j] = \mathcal{P}^{cb}[i,j]$ whenever $\alpha_{is}\alpha_{js} = \alpha_{si}\alpha_{sj}$ for all $s \in \Phi$, as with a symmetric $\boldsymbol{\mathcal{A}}$. ■

The final property formalizes that the same Input-Output structure places a pair of sectors at different horizontal positions depending on whether one looks at their suppliers or buyers. Their relation is uniquely determined if the two buy and sell the same relative quantity of intermediate inputs with a common third sector, and their input demand and input supply co-exposures coincide. Absolute quantities may differ; what matters is that, for each sector individually, its purchases or sales with the common sector match in own-share relative terms. The first two



(A) SEQUENTIAL (B) RHOMBOID (C) STAR

FIGURE 2: BENCHMARK ECONOMIES

Note: the figure illustrates some benchmark economic systems and their associated horizontal geometry (dashed) highlighting common Input-Output structures (solid). Sectors are in yellow, while the teal circles with C represent the final consumer.

properties, instead, emerge in Examples 2-3 and Figure 1. Proximity is the object of the theory (larger means closer); distance is a metric used in Sections 4-5 to sort sector pairs (larger means farther). The two are inversely related for given concentration of the two profiles, as in eq. (2).

2.2. STYLIZED ECONOMIES AND HORIZONTAL GEOMETRY

Figure 2 illustrates how the vertical and horizontal dimensions combine. I consider three benchmark configurations: *(i) sequential economies*, where connections occurs purely along a vertical chain (sector-by-sector towards the final consumer) or, alternatively, purely horizontally (distinct sectors supplying only the final consumer); *(ii) rhomboid economies*, where sectors depend on a common upstream supplier while simultaneously selling to the final consumer; and *(iii) star economies*, with all sectors mutually connected while serving the final consumer.

Sequential economies.– Consider the "snake" in the left graph of Figure 2a. Connections are purely vertical, the upstream commodities flow sequentially along the supply chain, with no scope for horizontal interdependencies. By contrast, in the "half-spider" on the right graph, sectors sell directly to the final consumer without trading among themselves. This structure is a characteristic of standard multi-sector models with no inter-sectoral trade, and common-buyer horizontal interdependencies are central: even without explicit trade linkages, paired sectors are tied together by their shared exposure to final demand. Adjustments in one sector reshapes consumers' expenditure across all others, moving horizontally through final demand reallocation and generating simultaneous responses absent any network structure.

Rhomboid economies.– A simple network in Figure 2b combines both sequential economies. This structure blends vertical and horizontal propagation, as commodities travel vertically (via sector-specific supply chains) and horizontally (across sectors simultaneously exposed to common supplier and final demand). Rhomboid economies thus epitomize both types of horizontal interdependencies, as even sectors not directly trading with each other become tightly synchronized through their shared dependence on upstream and downstream markets.

Star economies.– Figure 2c classifies three archetypes. In fully symmetric and balanced star networks, where all sectors buy and sell with identical intensity, all Input-Output profiles coincide and network distances collapse to zero: every pair of sectors is maximally close, so that horizontal relations are uniform rather than absent. Domar weights are balanced and first-order aggregate volatility vanishes as sectoral shocks average out (*e.g.*, Lucas 1977; Acemoglu, Carvalho, et al. 2012), but reallocation of downstream budget across suppliers is active, shifting the average level of GDP (Theorem 2). In partially symmetric star networks, trade intensities in buying and selling are equal but sector-specific; vertical and horizontal dimensions coexist,

though the two distance measures coincide. Finally, asymmetric star networks (*i.e.*, different trade intensities across sectors) generate parallel vertical and horizontal channels, with comovement emerging endogenously from network distances: horizontal interdependencies in production and shared exposure to final demand weave a dense web of horizontal interdependencies, and sectors with differing upstream and downstream connections experience unrelated effects.

3. An Inspection into the Horizontal Transmission

Outlined horizontal geometry introduces novel implications for the propagation of asymmetric, independent and idiosyncratic shocks across the network. Crucially, the transmission depends not only on the size of existing connections among sectors, but also on their network "economic" distance: in Figure 3, a shock originating in a given sector propagates vertically through downstream buyers along the supply chain, while concurrently transmitting horizontally.

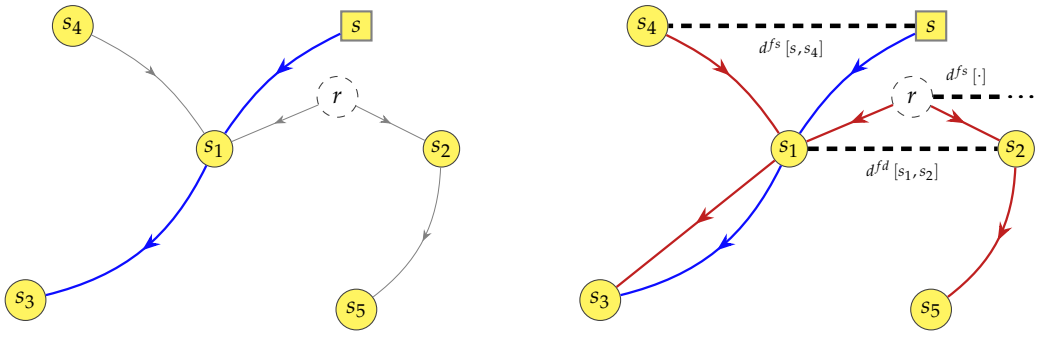
Example 4 (Propagating behaviour under horizontal distances) *Consider an economy with* $\{s, s_1, s_2, s_3, s_4, s_5, r\} \in \Phi$ *sectors, where* r *represents the rest of the Input-Output network. Some trade with others, while some others do not:* s_1 *is buying from* $\{s, s_4, r\}$ *while selling to* s_3*;* s_2 *trades from* r *and to* s_5*. Suppose a supply shock to* s*. In Panel 3a, the shock propagates downstream: it directly affects* s_1 *and indirectly reaches* s_3*, yielding a purely vertical transmission with others unaffected.*[5] *Unknown remain the potential effects of induced adjustments in* s_1*, even though it shares linkages with others. Panel 3b illustrates both vertical and horizontal propagation: a shock to* s *not only affects its immediate trading connections but also alters its interaction with* s_4*: changes in* s_1 *triggered by the initial shock generate an horizontal effect that links* $\{s, s_4\}$ *through their common common-buyer connection with* s_1*. By analogy, this reasoning falls under supply distances of* $\{s, s_4\}$ *with the rest of other sectors-*r *selling to* s_1*,* $\mathcal{D}^{cb}[\cdot]$*. With due common-supplier scheme, analogous is the logic underlying* $\mathcal{D}^{cs}[s_1, s_2]$*, simultaneously buying from* r*. Subsequent adjustments in* $\{s_4, r\}$ *feed back into* s_1*, compounding the indirect propagation towards* s_3*, and into* s_2*: reverberating the initial shock on* s*, via* $\{s_1, r\}$ *to* s_2*, an additional vertical effect further downstream, from* s_2 *to* s_5*, is generated.*

Figure 3 summarizes the guiding question of the paper. A shock to sector-s propagates downstream along supply chains (Panel 3a). Along the way it changes the prices and quantities faced by the buyers and suppliers of every sector it reaches, and thereby the conditions of sectors that share those buyers and suppliers (Panel 3b). This section characterizes the sign and the magnitude of these horizontal effects.

3.1. Distance-lead comovement

The framework developed so far places no restriction on how easily intermediate inputs can be substituted. The horizontal geometry of a network, however, can embed two aspects: *(i)* vertical propagation through the supply chain may remain the dominant channel, and what may appear as a horizontal transmission can in fact be the by-product of vertical propagation; and *(ii)* intermediate inputs complementarity in production may constrain the transmission of shocks. This

[5] Upstream responses vanish because price and quantity effects offset each other (*e.g.*, Acemoglu, Carvalho, et al. 2012); with horizontal mechanisms they do not (Theorem 1 and corollaries). Yet, the horizontal reallocation uncovers these latent transmission channels via common network structures instead of tracking non-linearities in production (*e.g.*, Baqaee and Farhi 2019). Note that the example presents the more intuitive case of a supply shock propagating downstream but the specular, bottom-to-top logic applies in the case of a demand shock propagating upstream.



(A) VERTICALITY (B) VERTICALITY AND DISTANCES

FIGURE 3: HORIZONTALITY OF A TRANSMISSION

Note: stylized example of how a sectoral shock propagates across the production network. Coloured and oriented bold-curves trace the propagation mechanism: vertical transmission is in blue, while additional transmission paths from the horizontal geometry are in red. Oriented thin-grey curves indicate further existing Input-Output connections, and network economic distances through common nodes are represented by dashed lines. The underlying network structure is identical in both panels.

tension between horizontal interdependence at the network level and substitutability at the input level sets the stage for a more precise characterization of how sectoral comovement occurs. In the context of common-buyer, alleviating the tension is doable since competition for common downstream buyers yields a direct and unambiguous form of horizontal interdependence. Conversely, the structure of upstream connections renders common-supplier linkages ambiguous to interpret, since the coexistence of a dual transmission makes the direction and the strength of induced comovement subtle. Below, comovement is characterised at a single shared node – *i.e.*, at "first round": the shock is traced to the shared partner and back, holding fixed the prices of all others – and shows that demand and supply distances differ along these two dimensions.

Two local mechanisms.– From now on, the production function of Assumption 1 is a Cobb-Douglas aggregate of labour and of a Constant Elasticity of Substitution (CES) intermediate bundle with substitution $\sigma_i > 0$. Since the household has Cobb-Douglas preferences over commodities, with constant expenditure shares $\{\omega_i\}_{i\in\Phi}$, all horizontal transmission runs through the production network. Throughout, refer to representative household with index-0.

Let $\widehat{\alpha}_{sj} = \frac{\alpha_{sj}}{1-\alpha_s}$ denote the share of sector-j in buyer-s's intermediate spending, $\widehat{\boldsymbol{a}}_s^{row} = \left[\widehat{\alpha}_{s1}, \ldots, \widehat{\alpha}_{sm}\right] = \frac{\boldsymbol{a}_s^{row}}{1-\alpha_s}$ the corresponding row vector – buyer-s's intermediate portfolio, whose entries sum to one –, and $\boldsymbol{\mathcal{C}}_s = \text{Diag}\left[\widehat{\boldsymbol{a}}_s^{row}\right] - \left(\widehat{\boldsymbol{a}}_s^{row}\right)'\widehat{\boldsymbol{a}}_s^{row}$ its covariance matrix, where Diag $[\cdot]$ is the $m \times m$ diagonal matrix with the entries of its argument on its diagonal: for any vector $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m$, then $\text{Var}_{\widehat{\boldsymbol{a}}_i^{row}}[\boldsymbol{u}] = \boldsymbol{u}'\boldsymbol{\mathcal{C}}_i\boldsymbol{u}$ and $\text{Cov}_{\widehat{\boldsymbol{a}}_i^{row}}[\boldsymbol{u}, \boldsymbol{v}] = \boldsymbol{u}'\boldsymbol{\mathcal{C}}_i\boldsymbol{v}$ are the variance and covariance of their entries across buyer-s's inputs, weighted by the shares. Let also η_i be the elasticity of demand faced by buyer-i for its own output, so that $d\log y_i = -\eta_i\, d\log p_i$, and let the marginal cost of a shared supplier-s have elasticity $\varphi_s = \frac{\partial \log p_s}{\partial \log y_s}$. Moreover, $e_{si} = \frac{\lambda_i\,\alpha_{is}}{\lambda_s}$ is the share of s's sales to buyer-i, $e_{s0} = \frac{\omega_s}{\lambda_s}$ the share sold to final demand, and consider an exogenous expansion in the scale of buyer-j from supplier-$s$at given prices.

PROPOSITION 1 (Common-supplier transmission) *At first round, the purchases of downstream sector-i from the shared upstream supplier-s respond as*

$$\delta_{s\to}[i,j] = \frac{\partial \log x_{is}}{\partial \log x_{js}} = -\chi_{is}\,\widetilde{\varphi}_s\,e_{sj} \qquad \textit{for} \quad i \neq j$$

with $\chi_{is} = \sigma_i\left(1-\widehat{\alpha}_{is}\right) + \alpha_i\,\widehat{\alpha}_{is} + \eta_i\,\alpha_{is} > 0$ *and* $\widetilde{\varphi}_s = \varphi_s\big/\left[1+\varphi_s\left(e_{s0}+\sum_{m\in\Phi}e_{sm}\chi_{ms}\right)\right]$.

Sector-i's output responds as $\frac{\partial \log y_i}{\partial \log x_{js}} = -\eta_i \lambda_j \sum_{s\in\Phi} \alpha_{is} \frac{\widetilde{\varphi}_s}{\lambda_s} \alpha_{js}$.

Proof: let me derive each step separately. Start by fixing the prices of all commodities other than s and its buyers, and neglect the feedback of buyers' own prices on their input mix, that is let $\mathbf{1}_{ij}$ equal one if $i = j$ and zero otherwise. *Step 1.* Each buyer-i of s prices at unit cost, so that $d\log p_i = \alpha_{is}\, d\log p_s$. *Step 2.* For $i \neq j$, CES cost minimization implies $d\log \widehat{\alpha}_{is} = (1-\sigma_i)(1-\widehat{\alpha}_{is})\, d\log p_s$, and $d\log[p_i y_i] = (1-\eta_i)\,\alpha_{is}\, d\log p_s + \mathbf{1}_{ij}\, d\log x_{js}$. *Step 3.* Subtracting $d\log p_s$ from the sum of the two terms, and using $\alpha_{is} = (1-\alpha_i)\,\widehat{\alpha}_{is}$, gives $d\log x_{is} = \mathbf{1}_{ij}\, d\log x_{js} - \chi_{is}\, d\log p_s$. *Step 4.* Market clearing, $y_s = c_s + \sum_{i\in\Phi} x_{is}$, with final demand $c_s = \frac{\omega_s}{p_s} wN$, gives $d\log y_s = -e_{s0}\, d\log p_s + \sum_{m\in\Phi} e_{sm}\, d\log x_{ms}$, and $d\log p_s = \varphi_s\, d\log y_s$. Combining, $d\log p_s = \varphi_s\big[e_{sj}\, d\log x_{js} - d\log p_s\big(e_{s0} + \sum_{m\in\Phi} e_{sm}\chi_{ms}\big)\big]$, so that, provided a stability condition $1 + \varphi_s\big(e_{s0} + \sum_{m\in\Phi} e_{sm}\chi_{ms}\big) > 0$, then $d\log p_s = \widetilde{\varphi}_s\, e_{sj}\, d\log x_{js}$. *Step 5.* By Step 3, $d\log x_{is} = -\chi_{is}\,\widetilde{\varphi}_s\, e_{sj}\, d\log x_{js}$ for $i \neq j$. For the second part, the expansion of j raises its demand for every supplier-s by $d\log x_{js}$, so that, treating each supplier's market as above, $d\log p_s = \widetilde{\varphi}_s\, e_{sj}\, d\log x_{js}$ for every s; summing over all shared suppliers, the unit cost of i changes by $\sum_{s\in\Phi} \alpha_{is}\, d\log p_s$, its output by $-\eta_i$ times this amount, and $e_{sj} = \frac{\lambda_j \alpha_{js}}{\lambda_s}$. ■

Comovement between downstream sectors arises through upstream adjustments in supply. Intuitively, when the demand for j's output expands, it purchases more from s. If s faces rising marginal costs, its price increases (with pass-through $\widetilde{\varphi}_s e_{sj}$), raising the i's unit cost by $\alpha_{is}\, d\log p_s$ and lowering its output by η_i.[6] Every other buyer of s contracts, both by substituting away from s (by σ_i) and by shrinking its own scale (by η_i). Hence, the common-supplier channel is horizontal in origin but vertical in mechanism: from i's viewpoint, the shock is a vertical cost shock originated at s; it is horizontal only in that it was caused by j. Common-supplier comovement reflects not automatic crowding-out, but the interplay of upstream price pass-through (from vertical propagation), and downstream demand adjustments (indirectly generated by horizontal interdependencies). Both features make comovement between sectors that share upstream suppliers hard to read.

The common-buyer channel behaves differently. Let s be a common buyer of i and j, and consider a shock that changes the price of j, holding other prices fixed. Let $p_i y_i$ denote the total sales of i, the following holds.

PROPOSITION 2 (Common-buyer transmission) *At first round, the sales of upstream sector-i to the shared downstream buyer-s respond as*

$$\delta_{\to s}[i,j] = \frac{\partial \log\left[p_i\, x_{si}\right]}{\partial \log p_j} = \left[\,(\sigma_s - 1) + (1-\alpha_s)\,(1-\eta_s)\,\right]\widehat{\alpha}_{sj}$$

and, summing over the common buyers $s \in \Phi_{i,j} = \Phi \setminus \{i,j\}$, the corresponding component of i's total sales responds as $\frac{\partial \log p_i y_i}{\partial \log p_j} = \frac{1}{\lambda_i}\sum_{s\in\Phi_{i,j}} \lambda_s\left[\,(\sigma_s-1) + (1-\alpha_s)\,(1-\eta_s)\,\right]\alpha_{si}\,\widehat{\alpha}_{sj}$.

Proof: fix all prices other than p_j. The price of s changes by $d\log p_s = \alpha_{sj}\, d\log p_j = (1-\alpha_s)\,\widehat{\alpha}_{sj}\, d\log p_j$. Hence, the scale term is $d\log[p_s y_s] = (1-\eta_s)\,(1-\alpha_s)\,\widehat{\alpha}_{sj}\, d\log p_j$. Under CES and a common buyer, with $d\log p_i = 0$, the substitution term is $d\log\widehat{\alpha}_{si} = (\sigma_s - 1)\,\widehat{\alpha}_{sj}\, d\log p_j$. Adding the two terms gives the first part. Total sales are $p_i y_i = p_i c_i + \sum_{s\in\Phi} p_i x_{si}$, where final demand spending on i is a constant share of income and does not respond to p_j, and the share of i's sales going to s is $\frac{p_i x_{si}}{p_i y_i} = \lambda_s \frac{\alpha_{si}}{\lambda_i}$; summing with these weights over the common buyers $\forall s \neq i,j$ gives the second part. ■

Comovement between upstream suppliers arises from competition for downstream buyers. The common-buyer channel separates cleanly into a horizontal and a vertical component. The $\sigma_s - 1$ term is competition between upstream suppliers for the intermediate budget of their common buyer: when the price of j falls, s reallocates a given budget across its suppliers. This

[6] Under Assumption 1, with labour as the only primary factor, marginal costs are constant, so that $\varphi_s = 0$. A non-zero φ_s requires a departure from this benchmark: rising marginal costs, $\varphi_s > 0$, as with capacity constraints, which Subsection 3.4 and Supplemental Appendix B micro-found with a factor in fixed supply, or falling ones, $\varphi_s < 0$, as with increasing returns.

downstream reallocation necessarily comes at the expense of the competing supplier-i and has no vertical counterpart. The second term is vertical: the lower price of j lowers the price of s and changes its scale, and hence its demand for all of its inputs in the same proportion, whether or not they compete with j. With the buyer's budget given ($\eta_s = 1$), the vertical term vanishes and transmission is purely horizontal: the rivals' sales to s move by $(\sigma_s - 1)\,\widehat{\alpha}_{sj}\,d\log p_j$ and j's own by $(1-\sigma_s)\left(1-\widehat{\alpha}_{sj}\right)d\log p_j$. Upstream reallocation of downstream expenditure is then zero-sum: the shocked sector and its rivals move in opposite directions whatever $\sigma_s \neq 1$. Hence the response of rivals to a shock in an horizontally-related sector reveals whether shared buyers treat their intermediate inputs as substitutes or complements.

A networked economy inherently embeds complementarities in intermediate inputs, as sectors depend on overlapping sets of buyers and suppliers. Importantly, the discrepancy between the ambiguous effects in demand and the clear responses in supply is in line with non-linear network theories (*e.g.*, Atalay 2017; Baqaee and Farhi 2019), where downstream complementarities – and hence complementarities between *purchased* intermediate inputs at a common node – matter the most. This dimension is embodied into the common-buyer distances through the downstream pass-through effect, which create horizontal interdependencies among sectors selling to the same buyers: when one supplier experiences a positive shock, downstream complementarities of common buyers cause other suppliers to adjust their production in response, and the propagation operates directly through the interdependence of upstream inputs used together in downstream production. Such amplification is largely absent under common-supplier distances, since the upstream pass-through reflects only substitution patterns across common upstream suppliers. The propositions deliberately hold fixed everything beyond the common node. Three questions remain: whether the common-buyer mechanism survive when shocks travel along the entire length of supply chains and downstream buyers' budgets adjust; what it implies for sectoral employment (the other input in production) and aggregate output; and when common suppliers, silent under constant marginal costs, transmit shocks.

3.2. General equilibrium: widespread horizontal geometry

In general equilibrium, across the complete Input-Output system, three movements happen at once: two refer to the "(anti-)cascade effects" operating vertically; the other is fully horizontal. A shock initially travels along supply chains, lowering the cost of every commodity that embodies the commodity produced by the shocked sector. Each downstream buyer then reallocates its intermediate expenditure across upstream inputs whose relative costs have moved unevenly (Proposition 2). Finally, due to these adjustments, the size of each buyer's budget is itself endogenous: travelling back upstream, a supplier's sales are what its own buyers decide to spend on it. The next result shows that, in the benchmark economy (*i.e.*, exploiting the full production network structure), these three forces organize into a product of three matrices.

Theorem 1 (On the vertical-horizontal factorization) *To first order, Domar weights respond to sectoral supply shocks as*

$$d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'\,\boldsymbol{\mathcal{H}}\,\boldsymbol{\mathcal{L}}\,d\log \boldsymbol{z} \qquad \textit{with} \quad \boldsymbol{\mathcal{H}} = \sum_{s\in\Phi}(\sigma_s - 1)\,\lambda_s\,(1-\alpha_s)\,\boldsymbol{\mathcal{C}}_s$$

Equivalently, $\frac{\partial \lambda_i}{\partial \log z_j} = \sum_{s\in\Phi}(\sigma_s-1)\,\lambda_s\,(1-\alpha_s)\,\mathrm{Cov}_{\widehat{\boldsymbol{a}}_s^{row}}\left[\boldsymbol{\ell}_i^{col}, \boldsymbol{\ell}_j^{col}\right]$, *where* $\boldsymbol{\ell}_j^{col}$ *is the* j*-th column of* $\boldsymbol{\mathcal{L}}$ *and the covariance is across buyer-*s*'s intermediate inputs with probabilities* $\widehat{\boldsymbol{a}}_s^{row}$.

Proof: Domar weights $\boldsymbol{\lambda}' = \boldsymbol{\omega}'\boldsymbol{\mathcal{L}}$ solve $\boldsymbol{\lambda}' = \boldsymbol{\omega}' + \boldsymbol{\lambda}'\boldsymbol{\mathcal{A}}$ at every equilibrium. Differentiating, with $\boldsymbol{\omega}$ constant, $d\boldsymbol{\lambda}' = \boldsymbol{\lambda}' d\boldsymbol{\mathcal{A}}\,\boldsymbol{\mathcal{L}}$: the vector $\boldsymbol{\lambda}' d\boldsymbol{\mathcal{A}}$ is the change in spending on each commodity due to buyers changing their shares at given size, and $\boldsymbol{\mathcal{L}}$ passes it upstream. Let $\boldsymbol{\pi} = -d\log \boldsymbol{p} = \boldsymbol{\mathcal{L}}\,d\log \boldsymbol{z}$ be the cost reduction of each commodity. Since $\alpha_{sj} = (1-\alpha_s)\widehat{\alpha}_{sj}$ with α_s

constant, CES minimization gives $\lambda_s\, d\alpha_{sj} = (\sigma_s - 1)\lambda_s(1-\alpha_s)\left(\widehat{\alpha}_{sj}\,\pi_j - \widehat{\alpha}_{sj}\,\widehat{\boldsymbol{a}}_s^{row}\boldsymbol{\pi}\right)$. Summing over buyers, $\left[\boldsymbol{\lambda}' d\boldsymbol{\mathcal{A}}\right]_j = \left[\boldsymbol{\pi}'\boldsymbol{\mathcal{H}}\right]_j$, because $\left[\boldsymbol{\pi}'\,\mathrm{Diag}\left[\widehat{\boldsymbol{a}}_s^{row}\right]\right]_j = \pi_j\widehat{\alpha}_{sj}$ and $\left[\boldsymbol{\pi}'\left(\widehat{\boldsymbol{a}}_s^{row}\right)'\widehat{\boldsymbol{a}}_s^{row}\right]_j = \left(\widehat{\boldsymbol{a}}_s^{row}\boldsymbol{\pi}\right)\widehat{\alpha}_{sj}$. Hence $d\boldsymbol{\lambda}' = d\log\mathbf{z}'\boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$ and, $\boldsymbol{\mathcal{H}}$ being symmetric, $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\,d\log\mathbf{z}$. Covariance follows from $\left[\boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\right]_{ij} = \sum_{s\in\Phi}(\sigma_s-1)\lambda_s(1-\alpha_s)\left(\boldsymbol{\ell}_i^{col}\right)'\boldsymbol{\mathcal{C}}_s\boldsymbol{\ell}_j^{col}$. Rows of $\boldsymbol{\mathcal{C}}_s$ sum to zero, since $\boldsymbol{\mathcal{C}}_s\mathbf{1} = \left(\widehat{\boldsymbol{a}}_s^{row}\right)' - \left(\widehat{\boldsymbol{a}}_s^{row}\right)'\left(\widehat{\boldsymbol{a}}_s^{row}\mathbf{1}\right) = \mathbf{0}$; hence so do rows of $\boldsymbol{\mathcal{H}}$. Finally, since $(1-\alpha_s)\,\mathrm{Diag}\left[\widehat{\boldsymbol{a}}_s^{row}\right] = \mathrm{Diag}\left[\boldsymbol{a}_s^{row}\right]$ and $(1-\alpha_s)\left(\widehat{\boldsymbol{a}}_s^{row}\right)'\widehat{\boldsymbol{a}}_s^{row} = \frac{(\boldsymbol{a}_s^{row})'\boldsymbol{a}_s^{row}}{1-\alpha_s}$, then $\boldsymbol{\mathcal{H}} = \mathrm{Diag}\left[\boldsymbol{\mathcal{A}}'\boldsymbol{\omega}\right] - \boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{W}}\boldsymbol{\mathcal{A}}$ with $\omega_s = (\sigma_s-1)\lambda_s$ and $\boldsymbol{\mathcal{W}} = \mathrm{Diag}\left[(\sigma_s-1)\frac{\lambda_s}{1-\alpha_s}\right]_{s\in\Phi}$, so that reallocation is $\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{L}}'\,\mathrm{Diag}\left[\boldsymbol{\mathcal{A}}'\boldsymbol{\omega}\right]\boldsymbol{\mathcal{L}} - \boldsymbol{\mathcal{T}}'\boldsymbol{\mathcal{W}}\boldsymbol{\mathcal{T}}$ with $\boldsymbol{\mathcal{T}} = \boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{L}} = \boldsymbol{\mathcal{L}} - \boldsymbol{I}$. ■

Reading from right to left, the vertical-horizontal factorization retraces the three movements. The Leontief inverse representation $\boldsymbol{\mathcal{L}}$ transports a shock vertically, cascading from sector-i to the cost of every downstream commodity embedding it, $\boldsymbol{\ell}_i^{col}$. The horizontal operator $\boldsymbol{\mathcal{H}}$ then reallocates each buyer's intermediate budget, $\lambda_s(1-\alpha_s)$, according to how unevenly the costs of its intermediate inputs have moved: it is the substitution term $(\sigma_s - 1)$ of Proposition 2, applied at every downstream and further downstream buyer at once. Finally, $\boldsymbol{\mathcal{L}}'$ transports the change in the demand for each commodity back up to the suppliers of that commodity: this anti-cascade adjustment is how the scale effects of Proposition 2 are resolved in general equilibrium, where a buyer's budget is not given but equals what its own buyers spend on it. The vertical structure enters only through $\boldsymbol{\mathcal{L}}$; everything that happens at shared buyers is in $\boldsymbol{\mathcal{H}}$. Its off-diagonal entries display the horizontal reallocation in sector-i after a shock in j,

$$\mathcal{H}_{i,j} = -\sum_{s\in\Phi}(\sigma_s - 1)\,\lambda_s\,(1-\alpha_s)\,\widehat{\alpha}_{si}\,\widehat{\alpha}_{sj} \qquad \text{for} \quad i \neq j$$

as common-buyer economic proximities weighted by the size of each buyer's intermediate budget and by its substitutability: $\boldsymbol{\mathcal{P}}^{cb} = \boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{A}}$ is their unweighted counterpart, and the first-tier term coincides with Proposition 2 with the buyer's budget given. Moreover, each row of $\boldsymbol{\mathcal{H}}$ sums to zero: within every buyer's bundle, what one supplier gains its rivals lose. Writing $\boldsymbol{\mathcal{L}} = \boldsymbol{I} + (\boldsymbol{\mathcal{L}} - \boldsymbol{I})$, where $\boldsymbol{\mathcal{L}} - \boldsymbol{I} = \boldsymbol{\mathcal{A}} + \boldsymbol{\mathcal{A}}^2 + \cdots + \boldsymbol{\mathcal{A}}^k = \sum_{k\geq 1}\boldsymbol{\mathcal{A}}^k$ collects the requirements embodied in intermediate inputs at each stage-k, and the reallocation operator $\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$ decomposes as $\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{H}} + \left[(\boldsymbol{\mathcal{L}} - \boldsymbol{I})'\boldsymbol{\mathcal{H}} + \boldsymbol{\mathcal{H}}(\boldsymbol{\mathcal{L}} - \boldsymbol{I})\right] + (\boldsymbol{\mathcal{L}} - \boldsymbol{I})'\boldsymbol{\mathcal{H}}(\boldsymbol{\mathcal{L}} - \boldsymbol{I})$: the response of sector-i to a shock in sector-j is "direct reallocation" between the two sectors at their common buyers (between sectors i and j both supplying at s), plus "indirect reallocation" between each of these sectors and the other sectors' commodities embodied in their respective intermediate inputs bundle (between i's output and the intermediate inputs required by j, and vice versa), plus "embodied reallocation" between the intermediate-input bundles embodying the two sectors (between the intermediate inputs being the production requirements of i and j). Equivalently, since from the proof $\boldsymbol{\mathcal{H}} = \mathrm{Diag}\left[\boldsymbol{\mathcal{A}}'\boldsymbol{\omega}\right] - \boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{W}}\boldsymbol{\mathcal{A}}$, the reallocation operator is $\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{L}}'\,\mathrm{Diag}\left[\boldsymbol{\mathcal{A}}'\boldsymbol{\omega}\right]\boldsymbol{\mathcal{L}} - \boldsymbol{\mathcal{T}}'\boldsymbol{\mathcal{W}}\boldsymbol{\mathcal{T}}$, whose second term (the common-buyer component) involves $\boldsymbol{\mathcal{T}} = \boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{L}} = \boldsymbol{\mathcal{L}} - \boldsymbol{I}$, the $m \times m$ matrix of buyers' total requirements, and the $m \times m$ matrix $\boldsymbol{\mathcal{W}} = \mathrm{Diag}\left[(\sigma_s - 1)\frac{\lambda_s}{1-\alpha_s}\right]_{s\in\Phi}$: two sectors are related when the same buyers require both, directly or indirectly. This is the logical step linking theory and measurement of Sections 4-5: the operator that governs the horizontal transmission is a weighted common-buyer Gram matrix computed on total requirements. The covariance form is similar to the second-order network centrality covariance operator of Baqaee and Farhi's (2019) Proposition 7. Both describe the first-order response of Domar weights, that is, the second-order response of aggregate output (Theorem 2). The factorization in Theorem 1 makes this object directly observable along the network's vertical and horizontal dimensions, $\boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$, turning a covariance across each buyer's inputs into a common-buyer Gram matrix that Input-Output data measure up to elasticities.

The theorem therefore separates the network response to a shock into two conceptually distinct dimensions. The vertical structure determines which commodities are affected through the

propagation of costs along supply chains, while the horizontal structure determines how these changes in relative costs reallocate downstream expenditure across common upstream suppliers. This decomposition yields two implications that are useful for interpreting the role of horizontal geometry. The first concerns the direction of these reallocations: whether common-buyer relationships generate rivalry or complementarity. The second concerns their informational content: horizontal geometry is not encoded in the Domar weights and therefore cannot be recovered from the vertical structure alone.

COROLLARY 1 (Signs) *Let sector-s be the shared node of sectors* $\{i, j\}$*. Then:*

- *under Cobb-Douglas intermediates* ($\sigma_s = 1$) *it holds* $\boldsymbol{\mathcal{H}} = \mathbf{0}$*;*
- *if* $\sigma_s \geq 1$ ($\sigma_s \leq 1$)*,* $\forall s \in \Phi$*, then* $\boldsymbol{\mathcal{H}}$ *and* $\boldsymbol{\mathcal{R}}$ *are positive (negative) semi-definite, and all off-diagonal entries of* $\boldsymbol{\mathcal{H}}$ *are non-positive (non-negative) and, for any pair* $\{i, j\}$ *with a common buyer, first-tier co-suppliers are rivals (complements) in sales.*

Proof: if $\sigma_s = 1$ for all $s \in \Phi$, then downstream expenditure shares do not adjust, and every weight in $\boldsymbol{\mathcal{H}}$ is zero. Moreover, each $\boldsymbol{\mathcal{C}}_s$ is positive semi-definite since, for any $\boldsymbol{u} \in \mathbb{R}^m$, it holds $\boldsymbol{u}'\boldsymbol{\mathcal{C}}_s\boldsymbol{u} = \mathrm{Var}_{\widehat{\boldsymbol{a}}_s^{row}}[\boldsymbol{u}] \geq 0$; its diagonal entries, $\widehat{\alpha}_{si}(1 - \widehat{\alpha}_{si})$, are non-negative and its off-diagonal entries, $-\widehat{\alpha}_{si}\widehat{\alpha}_{sj}$, non-positive. A combination with non-negative (non-positive) weights inherits these properties (with reversed signs), and $\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$ inherits semi-definiteness by congruence. ■

Corollary 1 turns Proposition 2 into a general-equilibrium statement and makes the first implication precise. The horizontal operator inherits a transparent sign structure from the CES substitution mechanism.[7] With Cobb-Douglas intermediate bundle, expenditure shares on inputs are fixed and invariant to the shocks (*e.g.*, Carvalho and Tahbaz-Salehi 2019) and horizontal transmission disappears altogether. With elasticities above (below) one, co-suppliers are rivals (complements): a shock raises the sales weight of some suppliers at the expense of their horizontal common-buyer neighbours, or vice versa. In both cases sectoral reallocations have opposite signs, and the shock moves the first-tier Domar weights of a sector and of its common-buyer neighbours in opposite directions, whatever the size of the elasticities: a shock reallocates sales shares between a sector and its common-buyer neighbours, and the size of the elasticity only decides who gains across horizontally-related sectors.

COROLLARY 2 (Domar weights do not span horizontal geometry) *There exist networked economies satisfying Assumption 1, with identical elasticities and identical Domar weights – hence identical first-order responses of aggregate output to any vector of shocks –, in which the same pair of sectors shares a common buyer in one economy and no common buyer in the other.*

Proof: by market clearing, Domar weights $\boldsymbol{\lambda}' = \boldsymbol{\omega}' + \boldsymbol{\lambda}'\boldsymbol{\mathcal{A}}$ depend on the flows $\lambda_s\alpha_{si}$ only through buyers' budgets, $\lambda_s(1 - \alpha_s)$, and suppliers' intermediate sales, $\lambda_i - \omega_i$. Let s, with $\lambda_s > 0$ and $\sigma_s \neq 1$, be the only common buyer of $\{i, j\}$, and m_1 a buyer that spends at least $\lambda_s\alpha_{sj}$ on some $m_2 \notin \{i, j\}$ but nothing on i. Shifting s's purchases from j to m_2, and the same value of m_1's purchases from m_2 to j, leaves these margins, hence Domar weights and $d\log Y = \boldsymbol{\lambda}' d\log \mathbf{z}$, unchanged, but i and j no longer share a buyer: $\mathcal{H}_{i,j}$ goes from $-(\sigma_s - 1)\lambda_s(1 - \alpha_s)\widehat{\alpha}_{si}\widehat{\alpha}_{sj} \neq 0$ to zero. The construction respects Assumption 1, since s keeps buying from i and from itself. ■

[7] When also households have CES preferences over final goods of elasticity $\sigma_0 \neq 1$, final demand would become a further common buyer, of size one and with no labour share, and each result below would gain a household term with weight $\sigma_0 - 1$: $\boldsymbol{\mathcal{H}}$ would add $(\sigma_0 - 1)\left\{\mathrm{Diag}[\boldsymbol{\omega}] - \boldsymbol{\omega}\boldsymbol{\omega}'\right\}$, the rivals' sales to final demand in Proposition 2 would move by $(\sigma_0 - 1)\omega_j d\log p_j$, the unweighted counterpart of $\boldsymbol{\mathcal{H}}$ would become $\boldsymbol{\mathcal{P}}^{cb} + \boldsymbol{\omega}\boldsymbol{\omega}'$, and the sign conditions of Corollary 1 would extend to σ_0. The household channel is active even without intermediate inputs – it is the horizontal channel of standard multi-sector models – and adds to the second-order term of Theorem 2 a component with the sign of $\sigma_0 - 1$ and absolute value at least $\frac{1}{2}\nu^2|\sigma_0 - 1|(1 - \|\boldsymbol{\lambda}\|^2)$. It reinforces, and does not alter, the conclusions on the supply side. See more in Supplemental Appendix A.

The corollary establishes a dimensional separation: within an Input-Output economy, the horizontal geometry of the network – captured by common-supplier and common-buyer shared relationships – is not spanned by the Domar weights, the sufficient statistics of its vertical importance. While these summarize how large sectors are, the horizontal structure identifies which sectors share upstream suppliers or downstream buyers. In other words, distance matrices are transformations of the Input-Output matrix that Domar weights, and hence first-order aggregate statistics, cannot recover, and they reveal otherwise hidden propagation channels.

Theorem 1 also explains why horizontal relations become increasingly important as the network expands. At the first tier, a shock to sector-i travels vertically to each buyer of i, through the positive entries of $\boldsymbol{\mathcal{A}}$, and horizontally to each sector with which i shares a buyer, through the non-zero off-diagonal entries of $\boldsymbol{\mathcal{H}}$. The two sets of relations grow at different rates, because vertical relations are shared: when a buyer adds a supplier, the new supplier becomes a rival of every supplier the buyer already has.

COROLLARY 3 (Horizontal proliferation) *If $\sigma_s \neq 1$ for all $s \in \Phi$, and $\sigma_s - 1$ has the same sign across buyers, then it holds $\mathcal{H}_{i,j} \neq 0$ if and only if $\mathcal{P}^{cb}[i,j] > 0$, and:*

- *if buyer-s starts purchasing from supplier-i, $\boldsymbol{\mathcal{A}}$ gains one positive entry, α_{si}, while $\mathcal{H}_{i,j}$ gains a non-zero term for every $j \in \Phi_i$ with $\alpha_{sj} > 0$;*
- *if buyer-s stops purchasing from supplier-i, $\boldsymbol{\mathcal{A}}$ loses one positive entry, while $\mathcal{H}_{i,j}$ vanishes only for the suppliers $j \in \Phi_i$ whose only common buyer with i was s;*
- *relative to buyer-s's intermediate purchases, $\lambda_s(1-\alpha_s)$, the horizontal reallocation $\boldsymbol{\mathcal{H}}$ from s is $|\sigma_s - 1|\left(1 - \|\widehat{\boldsymbol{a}}_s^{row}\|^2\right)$, with $\|\widehat{\boldsymbol{a}}_s^{row}\|^2$ the Herfindahl index of s's purchases.*

Proof: since $\sigma_s - 1$ has the same sign across sectors, all non-zero terms share it and cannot offset each other: by Theorem 1, for $i \neq j$, $\mathcal{H}_{i,j} = -\sum_{s\in\Phi}(\sigma_s - 1)\lambda_s(1-\alpha_s)\widehat{\alpha}_{si}\widehat{\alpha}_{sj}$. A buyer with $\widehat{\alpha}_{si}\widehat{\alpha}_{sj} > 0$ purchases intermediate inputs, so that all non-zero terms share the same sign and cannot offset each other: $\mathcal{H}_{i,j} \neq 0$ if and only if $\alpha_{si}\alpha_{sj} > 0$ for some s, that is, for $\mathcal{P}^{cb}[i,j] = \sum_{s\in\Phi}\alpha_{si}\alpha_{sj} > 0$. For the first claim, if α_{si} turns positive, the shares of s's other suppliers remain positive, and $\mathcal{H}_{i,j}$ gains the term $-(\sigma_s - 1)\lambda_s(1-\alpha_s)\widehat{\alpha}_{si}\widehat{\alpha}_{sj} \neq 0$ for every $j \in \Phi_i$ with $\alpha_{sj} > 0$. For the second claim, if α_{si} drops to zero, only the term of buyer-s disappears from $\mathcal{H}_{i,j}$, which therefore remains non-zero if and only if i and j share another buyer. For the third claim, the off-diagonal entries of $\mathrm{Diag}\left[\widehat{\boldsymbol{a}}_s^{row}\right] - \left(\widehat{\boldsymbol{a}}_s^{row}\right)'\widehat{\boldsymbol{a}}_s^{row}$ are $-\widehat{\alpha}_{si}\widehat{\alpha}_{sj}$, and, since $\sum_{m\in\Phi}\widehat{\alpha}_{sm} = 1$, their absolute values sum to $\sum_{m\in\Phi}\widehat{\alpha}_{sm}\left(1-\widehat{\alpha}_{sm}\right) = 1 - \|\widehat{\boldsymbol{a}}_s^{row}\|^2$; multiplying by $|\sigma_s - 1|\lambda_s(1-\alpha_s)$ and dividing by $\lambda_s\sum_{m\in\Phi}\alpha_{sm} = \lambda_s(1-\alpha_s)$ gives the result. With equal shares, $\|\widehat{\boldsymbol{a}}_s^{row}\|^2$ is the inverse of the number of s's suppliers. ■

Horizontal linkages are built out of vertical ones, but not one for one. Each vertical relation of a buyer supports a horizontal relation with each of the buyer's other suppliers, and each horizontal relation is supported by every buyer that the two sectors share. Counted once for every buyer, horizontal relations grow with the square of buyers' portfolios, vertical relations only linearly. The first claim is the margin of this asymmetry. As the network expands, every new link adds one vertical relation and many horizontal ones, so that horizontal relations proliferate even as supply chains themselves expand. The second is its mirror image. As transactions are removed, vertical relations are lost one at a time, whereas a horizontal relation is lost only when the last buyer that supports it is gone, so that horizontal relations survive sparsification. The same counting applies to common suppliers: when s adds supplier-i, it becomes a neighbour, in $\boldsymbol{\mathcal{P}}^{cs}$, of every other buyer of i. Finally, the last claim shows that the asymmetry is not only a matter of counting: the horizontal reallocation that s generates, $|\sigma_s - 1|\left(1 - \|\widehat{\boldsymbol{a}}_s^{row}\|^2\right)$, is zero for a buyer with a single supplier (whose Herfindahl index is one) and rises towards $|\sigma_s - 1|$ as s spreads its purchases evenly over more suppliers. In other words, as a buyer spreads a given intermediate budget over an increasing number of upstream suppliers, the vertical intensity of its purchases does not change, whereas the reallocation it generates among them rises. A sectoral shock reaches the buyers of the shocked sector vertically, but all the other suppliers of these

buyers horizontally: in larger and denser networks, the relations through which a shock spreads by reallocation within clusters of horizontally-related sectors grow faster than those through which it cascades along supply chains.

3.3. Sectoral employment and aggregate output

Theorem 1 describes how a shock reallocates sales shares across sectors. I now draw its implications for the two objects: the cross-section of sectoral employment, and aggregate output. A single operator links the two: the matrix $\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$ that moves Domar weights also moves workers across sectors, so that horizontal transmission becomes directly observable in employment. And since Domar weights are the slopes of aggregate output, the same matrix governs how aggregate output bends when many sectors are hit at once.

Sectoral employment.– Horizontal transmission reallocates a fixed production factor across sectors. In the model, a sector never substitutes between workers and its intermediate bundle as a whole. It spends a constant fraction α_i of its revenue on labour, with property *A1.iii* of Assumption 1 ensuring this fraction to be positive. Sectoral employment can therefore move relative to the aggregate only if its Domar weight moves (provided in Theorem 1).

Proposition 3 (Employment reallocation) *To first order, sectoral employment responds as*

$$d\log \boldsymbol{n} = d\log N\,\mathbf{1} + \boldsymbol{\Lambda}^{-1}\boldsymbol{\mathcal{R}}^{v}\,d\log \mathbf{z} + \boldsymbol{\Lambda}^{-1}\boldsymbol{\mathcal{R}}^{h}\,d\log \mathbf{z}$$

where $\boldsymbol{\Lambda} = \mathrm{Diag}\,[\boldsymbol{\lambda}]$, $\boldsymbol{\mathcal{R}}^{v} = \boldsymbol{\mathcal{L}}'\big(\boldsymbol{I}\circ\boldsymbol{\mathcal{H}}\big)\boldsymbol{\mathcal{L}}$ *and* $\boldsymbol{\mathcal{R}}^{h} = \boldsymbol{\mathcal{L}}'\big(\boldsymbol{\mathcal{H}} - \boldsymbol{I}\circ\boldsymbol{\mathcal{H}}\big)\boldsymbol{\mathcal{L}}$*, with* $\circ$ *the element-wise product, so that* $\boldsymbol{I}\circ\boldsymbol{\mathcal{H}}$ *retains the diagonal of* $\boldsymbol{\mathcal{H}}$ *and* $\boldsymbol{\mathcal{R}}^{v} + \boldsymbol{\mathcal{R}}^{h} = \boldsymbol{\mathcal{R}}$*. Moreover, if* $\sigma_s \geq 1$ *(*$\sigma_s \leq 1$*),* $\forall s \in \Phi$*, all entries of* $\boldsymbol{\mathcal{R}}^{v}$ *and the diagonal entries of* $\boldsymbol{\mathcal{R}}$ *are non-negative (non-positive), and all entries of* $\boldsymbol{\mathcal{R}}^{h}$ *are non-positive (non-negative).*

Proof: employment is $n_i = \alpha_i \frac{p_i y_i}{w} = \alpha_i \lambda_i N$, with $\sum_{i\in\Phi} n_i = N$, so that $d\log n_i = d\log N + \frac{d\lambda_i}{\lambda_i}$, and $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{R}}\,d\log\mathbf{z}$ by Theorem 1; splitting $\boldsymbol{\mathcal{H}}$ into its diagonal ($\boldsymbol{I}\circ\boldsymbol{\mathcal{H}}$) and its off-diagonal ($\boldsymbol{\mathcal{H}} - \boldsymbol{I}\circ\boldsymbol{\mathcal{H}}$) gives the decomposition. Since $\sum_{i\in\Phi}\alpha_i\lambda_i = \frac{1}{wN}\sum_{i\in\Phi} w n_i = 1$ at every equilibrium and $\boldsymbol{\alpha} = [\alpha_1,\ldots,\alpha_m]$ is constant, then $\boldsymbol{\alpha}'d\boldsymbol{\lambda} = 0$ for every shock, that is $\boldsymbol{\alpha}'\boldsymbol{\mathcal{R}} = \mathbf{0}$. If $\sigma_s \geq 1$, $\forall s \in \Phi$, the diagonal entries of $\boldsymbol{\mathcal{H}}$, $\mathcal{H}_{i,i} = \sum_{s\in\Phi}(\sigma_s - 1)\,\lambda_s\,(1-\alpha_s)\,\widehat{\alpha}_{si}\big(1-\widehat{\alpha}_{si}\big)$, are non-negative, its off-diagonal entries are non-positive (Corollary 1), and $\boldsymbol{\mathcal{L}} \geq \mathbf{0}$. Hence $\mathcal{R}^{v}_{i,j} = \sum_{s\in\Phi}\ell_{si}\,\mathcal{H}_{s,s}\,\ell_{sj} \geq 0$ and $\mathcal{R}^{h}_{i,j} = \sum_{s\neq m}\ell_{si}\,\mathcal{H}_{s,m}\,\ell_{mj} \leq 0$, while $\mathcal{R}_{i,i} = \big(\boldsymbol{\ell}_i^{col}\big)'\boldsymbol{\mathcal{H}}\,\boldsymbol{\ell}_i^{col} \geq 0$, since $\boldsymbol{\mathcal{H}}$ is positive semi-definite. The case $\sigma_s \leq 1$ reverses all signs. ■

The decomposition separates three forces. The common component (*i.e.*, the change in aggregate employment $d\log N$) shifts all sectors in the same proportion and carries no information on the network. The two network components are the employment counterpart of Theorem 1: pre-multiplying by $\boldsymbol{\Lambda}^{-1}$ turns a change in sales shares into a percentage change in employment, so that the same shift in spending matters more for a small sector than for a large one. Both components are switched on by substitution at buyers, and both vanish under Cobb-Douglas intermediates, but they create different relations between sectors. The vertical component transports the own response of each commodity along supply chains: when a commodity becomes cheaper, it gains share at its buyers if they treat their inputs as substitutes, and loses share if they treat them as complements, and every sector embodied in it moves with it. Hence $\mathcal{R}^{v}_{i,j} \neq 0$ only if some commodity embodies both i and j, that is, only if the two sectors lie along a common supply chain. The horizontal component is the rivalry between distinct commodities at their common buyers, transported to the sectors they embody: $\mathcal{R}^{h}_{i,j} \neq 0$ only if some buyer requires i and j through different inputs. By $\boldsymbol{\alpha}'\boldsymbol{\mathcal{R}} = \mathbf{0}$, the two components offset each other in the aggregate: a shock reallocates workers between supply chains that compete for the same budgets, and the jobs gained along the chains that embody the shocked sector equal

those lost along the chains of its rivals.[8] Three predictions follow, for substitutes and complements alike. First, by Corollary 1, a shock that moves the employment of a sector moves that of its first-tier common-buyer neighbours in the opposite direction, unless they also lie along a common supply chain: shared budgets make reallocation zero-sum, and the elasticities only determine who gains. Second, sectors that are horizontally distant but vertically connected, for which $\mathcal{R}^h$ is small relative to $\mathcal{R}^v$, comove positively through the production network: they gain or lose together with the shocked sector, and the elasticities only determine whether they expand or contract. Third, sectors that are distant along both dimensions comove only through the common component. In general, comovement between two sectors is positive when their vertical connection outweighs their horizontal proximity, and opposite otherwise.

Aggregate output: what Hulten's theorem hides.– Since horizontal reallocation from sectoral shocks move workers across sectors (Proposition 3), one might expect this reallocation to move aggregate output as well. To first order, it does not. In efficient economies Hulten's (1978) theorem, under fixed aggregate labour ($d \log N = 0$),

$$d \log Y = \boldsymbol{\lambda}' d \log \mathbf{z}$$

holds at every equilibrium. Aggregate output responds through Domar weights alone, whatever the arrangement of the Input-Output relationships behind them: economies with equal Domar weights but different horizontal geometries (as in Corollary 2) would respond identically to any vector of shocks. The reason is an envelope argument: the reallocation described by Theorem 1 is an optimal response of buyers to new relative prices in which moving labour and spending between intermediate uses that at the optimum have the same marginal value leaves output unchanged to first order. At the aggregate level, horizontal geometry seems to vanish. This is not the case: it is hidden inside Hulten. While the Hulten's argument holds at every equilibrium, the weights it uses are themselves equilibrium objects, and they move with the shocks. Indeed, the theorem postulated by Hulten gives the slope of aggregate output with respect to sectoral productivities, $\frac{\partial \log Y}{\partial \log \mathbf{z}'} = \boldsymbol{\lambda}'$; Theorem 1 gives how this slope changes, $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{R}}\, d \log \mathbf{z}$.

Horizontal geometry therefore leaves the first-order response unaffected but determines how that response changes with the shock itself, that is, the curvature of aggregate output – how the economy's production structure transforms dispersion in intermediate input prices into aggregate gains or losses. The logic is that of a single buyer. To first order, the cost of a buyer's intermediate bundle responds to input prices as their share-weighted average, so that how the bundle is composed does not matter. To second order, when intermediate input prices move unevenly, a buyer whose intermediate inputs are complements is held back by the inputs whose price fell least, and its cost falls by less than the average; a buyer whose inputs are substitutes shifts towards the input that became cheapest, and its cost falls by more. Aggregate output inherits these buyer-level curvatures, weighted by the size of each buyer's intermediate budget.

Consider a vector of independent and identically distributed sectoral supply shocks, $\boldsymbol{\varepsilon} = d \log \mathbf{z} \overset{\text{iid}}{\sim} (0, \nu^2)$, across $m = |\Phi|$ sectors; hold aggregate labour fixed, and let $\log Y^{det}$ be the deterministic level of real GDP (the Lucas's 1977 diversification: no role for idiosyncratic shocks). Positive and negative shocks offset each other in the first-order term, $\boldsymbol{\lambda}'\boldsymbol{\varepsilon}$; they do not offset each other in the dispersion of input prices within buyers' bundles. Through the curvature, dispersion shifts the mean of aggregate output.

THEOREM 2 (On horizontal geometry and aggregate output) *The impact of the network's horizontal structure on aggregate GDP is as follows:*

[8] Since $\boldsymbol{\alpha}'\boldsymbol{\mathcal{R}} = \mathbf{0}$, the employment-weighted sum of relative responses is zero, so that $\boldsymbol{\alpha}'\boldsymbol{\mathcal{R}}^v = -\boldsymbol{\alpha}'\boldsymbol{\mathcal{R}}^h$.

- *to first order,* $d \log Y = \boldsymbol{\lambda}' \boldsymbol{\varepsilon}$*, so that* $\operatorname{Var}[d \log Y] = \nu^2 \|\boldsymbol{\lambda}\|^2$ *does not depend on* $\boldsymbol{\mathcal{H}}$ *and vanishes as* $m \to \infty$ *when* $\lambda_i \asymp \frac{1}{m}$*;*
- *to second order,* $d^2 \log Y = \boldsymbol{\varepsilon}' \boldsymbol{\mathcal{L}}' \boldsymbol{\mathcal{H}} \boldsymbol{\mathcal{L}} \boldsymbol{\varepsilon} = \sum_{s \in \Phi} (\sigma_s - 1) \lambda_s (1 - \alpha_s) \operatorname{Var}_{\widehat{\boldsymbol{a}}_s^{row}} [\boldsymbol{\mathcal{L}} \boldsymbol{\varepsilon}]$*;*
- *consequently, it holds* $\mathbb{E}[\log Y] - \log Y^{det} = \frac{1}{2} \nu^2 \operatorname{Trace} [\boldsymbol{\mathcal{L}}' \boldsymbol{\mathcal{H}} \boldsymbol{\mathcal{L}}] + o[\nu^2] \neq 0.$

Proof: provided in Appendix A. ■

The three claims follow the argument above. The first establishes the implication of Hulten's (1978) theorem for aggregate volatility. Since the first-order response depends on the network only through Domar weights, their distribution determines whether idiosyncratic shocks average out. When balanced, $\lambda_i \asymp \frac{1}{m}$, the variance of aggregate output decays at rate $\frac{1}{m}$, as in Lucas (1977). When the network makes a few sectors large suppliers, directly or along supply chains, Domar weights are concentrated, $\|\boldsymbol{\lambda}\|$ decays more slowly, and diversification fails, as in Acemoglu, Carvalho, et al. (2012). In either case, holding Domar weights fixed, the downstream composition of intermediate input bundles is irrelevant: horizontal geometry plays no role and cannot be inferred from first-order aggregate volatility, nor does it alter its rate of decay.

The second part reveals what the Hulten's logic hides. The curvature of aggregate output is a size-weighted sum, across buyers, of the dispersion of their intermediate input prices, after shocks have travelled down the supply chains. Vertical structure, $\boldsymbol{\mathcal{L}}$, determines how shocks translate into intermediate input-price changes; horizontal geometry $\boldsymbol{\mathcal{H}}$ determines how these changes are distributed across the shared suppliers of each buyer. The resulting dispersion is then translated into cost changes according to common-buyers' elasticities: complementarity makes buyers' costs fall less than their average input-price change, while substitutability allows them to shift towards cheaper inputs and makes them fall by more. Thus, even with identical Domar weights and vertical structures, economies with different horizontal geometries can exhibit different aggregate output curvatures. In particular, if all intermediate inputs of a buyer become cheaper in the same proportion, that buyer contributes nothing to curvature, regardless of the magnitude of the shock. The overall reallocation term $\boldsymbol{\mathcal{L}}' \boldsymbol{\mathcal{H}} \boldsymbol{\mathcal{L}}$ therefore captures how the horizontal structure of the network transforms sectoral shocks in aggregate output curvature.

Finally, the third claim draws the implication for the level of GDP. Lucas's diversification eliminates aggregate volatility, $\operatorname{Var}[d \log Y] \to 0$, without eliminating micro volatility on *mean* output. Averaging shocks across many sectors does not necessarily reduce the resulting reallocation at each buyer, and horizontal transmission moves average output away from its deterministic benchmark: $\mathbb{E}[\log Y] - \log Y^{det} \neq 0$. As the number of sectors increases, positive and negative shocks can offset each other in the aggregate (Lucas 1977), but the resulting dispersion within each buyer's bundle need not vanish.

COROLLARY 4 (Lower bound of aggregate output) *If* $\sigma_s \geq 1$ *(*$\sigma_s \leq 1$*),* $\forall s \in \Phi$*, the second-order term is at least (at most)* $\frac{1}{2} \nu^2 \underline{\alpha}^2 \sum_{s \in \Phi} (\sigma_s - 1) \lambda_s (1 - \alpha_s) \left(1 - \|\widehat{\boldsymbol{a}}_s^{row}\|^2\right)$*, where* $\underline{\alpha} = \min_{i \in \Phi} \alpha_i$ *and* $\|\widehat{\boldsymbol{a}}_s^{row}\|^2$ *is the Herfindahl index of buyer-*s*'s intermediate purchases. Hence, if* $|\sigma_s - 1| \geq \underline{c} > 0$*,* $\underline{\alpha} \leq \alpha_i \leq \overline{\alpha}$ *with* $0 < \underline{\alpha} \leq \overline{\alpha} < 1$*, and* $\|\widehat{\boldsymbol{a}}_s^{row}\|^2 \leq \overline{h} < 1$*, for all sectors and uniformly in* M*, the absolute value of the second-order term is at least* $\frac{1}{2} \nu^2 \underline{c}\, \underline{\alpha}^2 (1 - \overline{\alpha}) \left(1 - \overline{h}\right)$*, whether Domar weights are balanced or concentrated.*

Proof: provided in the proof of Theorem 2. ■

The effect on output remains bounded away from zero, whether Domar weights are balanced or concentrated, provided that intermediate input elasticities are not unitary, sectoral labour

shares are bounded away from zero and one, and buyers' upstream purchases from the production network are diversified. As such, the magnitude and sign of this effect depend jointly on horizontal geometry and elasticities: with complementarities, which empirical estimates support (*e.g.*, Atalay 2017; Boehm, Flaaen, et al. 2019), micro volatility lowers average output because of the network's horizontal structures.[9]

3.4. Demand shocks and common suppliers

The analysis so far concerns supply shocks. They move prices along supply chains, and common downstream buyers turn these price changes into a reallocation of their budgets: horizontal transmission is upstream rivalry at each node. Demand shocks raise a different question. A shift in final demand towards sector-j travels upstream, raising the sales of the direct and indirect suppliers of j; under constant marginal costs it stops there because the prices of these suppliers do not move, with the other sectors that buy from them are left untouched (Proposition 1 with $\varphi_s = 0$). The ingredient that turns a demand shock into horizontal transmission is *upstream congestion*: suppliers whose capacity cannot expand at constant cost. Supplemental Appendix B introduces it by letting each sector use a factor in fixed supply – with revenue share θ_s, marginal cost rises at rate $\varphi_s = \frac{\theta_s}{1-\theta_s}$ with output –, and studies demand shocks as shifts in final expenditure, $d\boldsymbol{\omega}$ with $\mathbf{1}'d\boldsymbol{\omega} = 0$. Theorem B.1 and Corollary B.1 deliver a duality:

$$\begin{array}{lccc} \textit{(supply shock)} & d\log \boldsymbol{p} = -\boldsymbol{\mathcal{L}}\, d\log \boldsymbol{z} & \text{and} & d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\, d\log \boldsymbol{z} \\ \textit{(demand shock)} & d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}' d\boldsymbol{\omega} & \text{and} & d\log \boldsymbol{p} = \boldsymbol{\mathcal{S}}\, d\boldsymbol{\omega} \end{array}$$

where $\boldsymbol{\mathcal{S}} = \boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{K}}\boldsymbol{\mathcal{L}}'$, with $\mathcal{S}_{i,j} = \sum_{s\in\Phi} \ell_{is}\frac{\theta_s}{\lambda_s}\ell_{js} + O\left[\theta^2\right]$, is a common-supplier Gram matrix on total requirements, weighted by the congestion of each supplier relative to its Domar size. Supply shocks move prices vertically and quantities horizontally; demand shocks move quantities vertically and prices horizontally. In other words, common-buyer relationships make upstream suppliers rivals over a shared downstream budget, whereas common-supplier ties make downstream customers rivals over a shared upstream productive capacity: a demand shock to j raises the costs of every sector-i that draws on the same congested supplier-s, in proportion to $\mathcal{S}_{i,j}$. The two geometries are not alternatives. Constant marginal costs and Cobb-Douglas technologies are the knife-edge cases that silence one of them, and, when both are active, they feed into each other, congested suppliers carrying prices and common buyers carrying quantities.

Rivalry over a budget and rivalry over a capacity have different consequences for employment, which tracks sales. A budget is zero-sum in sales by construction: what one supplier gains, the others lose, and the elasticity only decides who wins (Corollary 1). A capacity is zero-sum only in costs. The co-customers of j pay more for the shared intermediate input, but whether they sell more or less depends on how their own buyers react to a higher price, which the shared supplier does not determine. Corollary B.2 makes this precise: the Domar weight of sector-i, and hence its employment relative to the common component, responds to a demand shock to sector-j, $d\xi_j$, through its component specific to j and to a deterministic first-order in

[9] Theorem 2 shares its starting point with Baqaee and Farhi (2019): Hulten's (1978) theorem is a first-order result, and departures from it are governed by how Domar weights respond to shocks. The results differ in how this object is read. The authors summarize it through unobservable sufficient statistics and emphasize how easily production factors can be reallocated. In my framework labour is fully mobile and the object factorizes into vertical transport (the Leontief inverse) and horizontal reallocation of buyers' budgets (whose geometry is observed in Input-Output data and whose scale is set by elasticities). Elasticities switch the allocative channel on and scale it, while horizontal geometry determines where reallocation takes place and how much of it occurs. Importantly, the last claim of the theorem adds that these effects on mean aggregate output survives the Lucas's (1977) diversification argument.

congestion, as

$$\frac{\partial \lambda_i}{\partial \xi_j} = \ell_{ji} - \sum_{s \in \Phi} \mathcal{R}_{i,s} \frac{\theta_s}{\lambda_s} \ell_{js}$$

The cost increase at each congested supplier-s of j acts on i as a negative supply shock at s, and reaches its sales only through $\mathcal{R}_{i,s}$, that is, through substitutability at the buyers of i. If all elasticities lie on the same side of one, then, for a co-customer that is neither upstream of j nor a common-buyer neighbour of its congested suppliers, $\mathcal{R}_{i,s}$ has the sign of $\sigma - 1$ (Proposition 3), and the sign is that of $\theta_s (1 - \sigma_s)$: employment falls if its buyers treat inputs as substitutes, rises if they treat them as complements, and does not move if it sells only to final demand, whereas its price always rises. At first round, Proposition 1 signs comovement in quantities by congestion alone; in employment, the sign is that of $\theta_s (1 - \sigma)$. Ambiguity is therefore structural rather than statistical. It combines a supply-side parameter of the shared supplier with a demand-side parameter of the co-customer's buyers, both of which vary across clusters of the network, and it is compounded by the vertical term ℓ_{ji}, through which a demand shock raises the employment of the shocked sector and of its suppliers together (Proposition B.1).

Finally, congestion also qualifies the aggregate message of Theorem 2. To first order, demand shocks do not move aggregate output, but only its slope – *i.e.*, the Domar weights (Theorem B.2). For supply shocks, the reallocation that bends aggregate output now meets rising marginal costs at the suppliers of the winners, and its curvature becomes

$$\widetilde{\boldsymbol{\mathcal{R}}} = \boldsymbol{\mathcal{R}} - \boldsymbol{\mathcal{L}}' \boldsymbol{\mathcal{H}} \boldsymbol{\mathcal{S}} \boldsymbol{\mathcal{H}} \boldsymbol{\mathcal{L}} \qquad \text{with} \qquad \text{Trace}\left[\widetilde{\boldsymbol{\mathcal{R}}}\right] \leq \text{Trace}\left[\boldsymbol{\mathcal{R}}\right]$$

The correction is the second round of the duality: winners at common buyers congest their suppliers, whose price increases fall on all the sectors that share them, and are reallocated again at common buyers. Congestion therefore lowers the effect of micro volatility on mean output whatever the elasticities, dampening the gains under substitutes and amplifying the losses when intermediates are complements. Part of the reallocation of downstream buyers' budgets is absorbed by congestion, and its cost is borne by the sectors that share the congested suppliers.

4. Data and Horizontal Anatomy

From the United States' Bureau of Economic Analysis (BEA), available data display information, for 3-digit U.S. 2017 North American Industrial Classification System (NAICS) sectors, on value-added, employment, net exports, and inter-sectoral Input-Output tables. The (balanced) panel data built covers 65 private sectors over the period 1998-2022. For additional details refer to Supplemental Appendix C.

Distance measures are derived from the *Total Requirements Table*, which records both direct and indirect Input-Output trade along each supply chain. The theory motivates this choice. By Theorem 1, the common-buyer component of the reallocation operator, $-\boldsymbol{\mathcal{T}}'\boldsymbol{\mathcal{W}}\boldsymbol{\mathcal{T}}$, is a weighted common-buyer Gram matrix on total requirements, $\boldsymbol{\mathcal{T}} = \boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{L}} = \boldsymbol{\mathcal{L}} - \boldsymbol{I}$: two commodities are related when the same buyers require both, directly or indirectly. On the supplier side, the horizontal operator of demand shocks is $\boldsymbol{\mathcal{S}} = \boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{K}}\boldsymbol{\mathcal{L}}'$ (Subsection 3.4), a congestion-weighted common-supplier Gram matrix on the rows of the Leontief inverse, whose unweighted counterpart is $\boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{L}}'$. In both cases two sectors are close when they share partners both locally and throughout the depth of their supply chains. Accordingly, the distance between two sectors reflects the extent to which they share similar buyers/suppliers across the whole production network. Each cell $\mathcal{D}^q[i,j] \in [0, \sqrt{2}]$ of matrix $\boldsymbol{\mathcal{D}}^q$, for $q = \{cs, cb\}$ and $\forall i, j \in \Phi$, identifies the Euclidean distance (square root of the sum for squared differences of network intensities; see

Section 2) between dyads of sectors in demand or supply relative to common sectors.

The empirical estimation of Section 5 classifies sectors by network economic distances at the *extensive margin*, $\boldsymbol{\mathcal{D}}^{q}_{ext}$, whose generic entry is $\mathcal{D}^{q}_{ext}[i,j] = \{1,2,\ldots,\mathcal{D}_{max}\}$, with lower values denoting closer sectors. Such network distance-d among any two sectors $\{i,j\}$ is computed as a *shortest path problem* on the adjacency matrix of *intensive margin* distances, $\boldsymbol{\mathcal{D}}^{q}$, implementing an algebraic version (*i.e.*, using matrix multiplication) of the traditional Dijkstra or Roy-Floyd-Warshall algorithms, since it is: *(i)* suitable for discrete-value adjacency matrices; and *(ii)* fairly efficient for medium-sized network structures. Supplemental Appendix C reports the technical version of the implemented algorithm which, in plain words, reads as follows.

(Shortest path algorithm) *Central idea is to find, for each single node, all the other nodes that can be reached in one step, then in two steps, and so on. Each time a node is reached for the first time, it records the number of steps it took to get there – this is the shortest path approach. In other words, find the shortest number of steps it takes to get from every node to every other nodes in a network. In particular:*

(a) identify the direct (one-step) connections between any pair of nodes;

(b) for not-connected pairs, increase the path length by 1 in each round (two steps, then three steps, etc.). Multiply the graph by itself to discover which new pairs are now connected;

(c) if two nodes becomes connected, record the length as the shortest distance;

(d) repeat this process until no new reachable pairs of nodes is found. For any pair of nodes that are still uncategorised, set their distance to infinity.

4.1. Propagating patterns

I now analyse the mechanics of the horizontal transmission. Exploiting the observed propagation matrices, Figure 4 compares the importance of each sector in the vertical network, built on the Leontief inverse, with its importance in the horizontal network, built on network distances, for the 2007 table. In particular, it visualizes the sectoral propagation patterns using (PageRank) centrality measures from spectral norms of the vertical and horizontal propagation matrices, to be read in terms of how important each sector is for the transmission of shocks considering the dual network structures. The vertical importance from the Leontief transmission highlights few key sectors that act as propagation hubs: these retain a systemic role in providing inputs – *e.g.*, energy (211), risk intermediation (524), knowledge-intensive services (5412OP), and organizational support (561) –, underpinning a wide range of production activities. Asymmetric shocks originating in these sectors are recursively amplified along parallel supply chains, generating strong cascading effects while other sectors transmit shocks only weakly. In contrast, the horizontal propagation system spreads shocks almost uniformly. Most sectors share buyers or suppliers with many others, so that horizontal transmission is diffuse rather than hub-driven. This is the pattern from Corollary 3 : every buyer that diversifies its purchases makes all of its suppliers horizontally related, so that horizontal relations are spread over many pairs rather than concentrated on a few hubs. Overall, the image demonstrates the distinction between cascade-driven amplification along key nodes and diffuse propagation across economically similar sectors: vertical chains concentrate and amplify shocks from central nodes whereas horizontal interdependencies induce diffuse comovement without amplification by key sectors.

Table 1 in Appendix B, instead, documents the same contrast along a different dimension, since it investigates how vertical and horizontal geometries respond to varying levels of network sparsity. Progressively removing weaker transactions (increasing the threshold-τ for the volume of total sectoral shares), makes the network sparser and steadily lowers the spectral norm of the

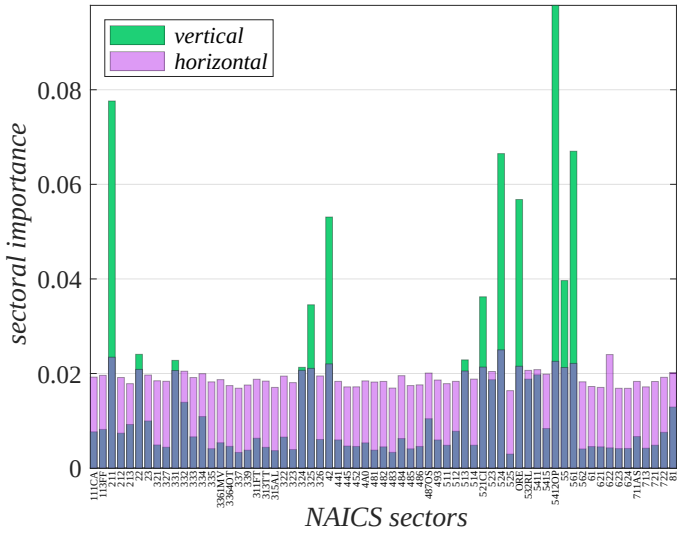


FIGURE 4: SECTORAL INFLUENCE IN SHOCK TRANSMISSION

Note: importance of each sector in reverberating idiosyncratic shocks through their vertical and horizontal network structures as dictated by the U.S. Input-Output propagation matrices in year 2007. *Source*: BEA and own calculations.

vertical operator, as supply chains are cut, thereby reflecting the diminishing role of cascade effects. Differently, the spectral norm of the horizontal propagation remains remarkably stable across thresholds, and it gradually becomes the predominant channel of shock transmission as τ increases, so that horizontal interdependencies dominate in sparser networks. These results illustrate that, when weak linkages (*i.e.*, observed trade intensities) are trimmed, horizontal interdependencies exceed the vertical ones. They sustain substantial propagation as they rest on the overlap of many moderate links rather than on a few strong ones, and therefore survive sparsification. Table 1 thus lends empirical support to Corollary 3: horizontal relations, which rest on many shared partners, survive the removal of weak transactions, whereas supply chains are cut. As Input-Output economies become sparser, shocks increasingly travel along clusters of horizontally related sectors rather than along elongated vertical chains.

4.2. NETWORK EXPOSURE AND AGGREGATE SHOCKS

The discussion so far has shown that the two geometries of the U.S. production network are distinct propagation objects, a precondition for the exercises that follow. The headline treatment of the network, however, is centred on the idea that horizontal geometries capture dimensions of economic exposures that are unconditionally distinct from vertical Input-Output linkages. If horizontal linkages capture an economically meaningful dimension of interdependence, they should also explain differences in sectoral responses to shocks "above and beyond" the information already embedded in Leontief inverse exposures, retaining a predictive power that conditionally complements them. Moreover, by Propositions 1-2 and the duality of Subsection 3.4, the relevant horizontal channel should depend on the nature of the shock: common-buyer distance transmits price movements, while common-supplier distance transmits demand impulses.

A Bartik (1991) shift-share design comes to the rescue. Consider an aggregate shock that affects all sectors simultaneously. Since the shock is widespread, identification cannot come from time-series variation alone. Instead, the relevant source of variation is cross-sectional: sectors may be differently exposed to the same aggregate disturbance because of their difference in Input-Output exposures. Formally, I first construct a measure of exposure by combining the

weight of each network structure with a measure of predetermined sectoral sensitivity to the aggregate disturbance,

$$exposure_i^{(\mathcal{V},\mathcal{D}^q)} = \sum_{j \in \Phi_i} weight_{ij}^{(\mathcal{V},\mathcal{D}^q)} \; sensitivity_j \qquad for \quad q = \{cs, cb\}$$

This formulation encompasses all network structures, in that $weight_{ij}^{\mathcal{V}}$ are Leontief weights (the total requirement of sector-i per unit of sector-j's output), and $weight_{ij}^{\mathcal{D}^q}$ are common-supplier and common-buyer distances. A sector is highly exposed when it is strongly connected to sectors that are themselves highly exposed to the aggregate. Exposure thus measures how embedded a sector is to the other sectors' sensitivities relative to the aggregate disturbance.

A natural concern is that horizontal and vertical exposures may be highly collinear and partly reflect similar objects since both belong to the same primitive Input-Output matrix (Section 2). Hence, the component of horizontal exposure that is orthogonal to vertical one is residualised from the auxiliary projection

$$exposure_i^{\mathcal{D}^q} = \beta_{exposure}^{cons} + \beta_{\mathcal{V}} \; exposure_i^{\mathcal{V}} + \gamma_i' \mathcal{Z}_i + r_i^{\mathcal{D}^q}$$

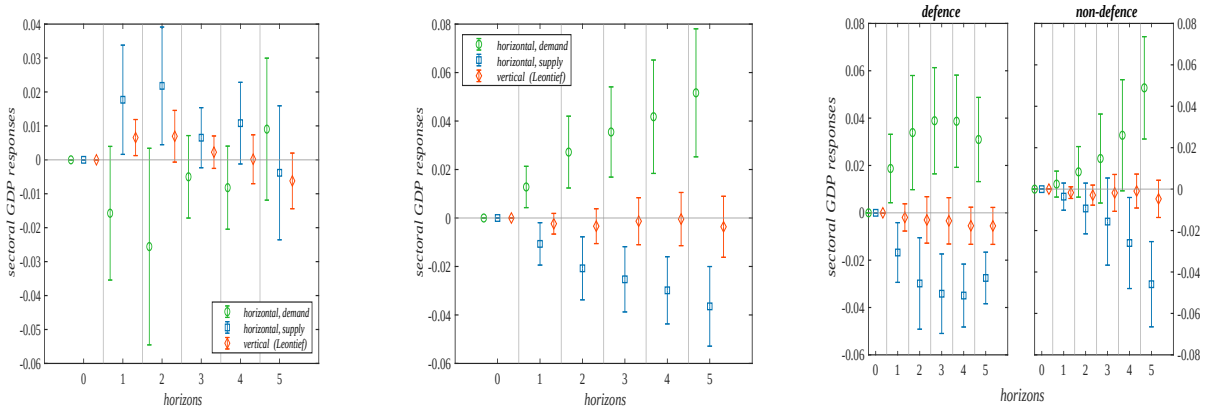
where $\mathcal{Z}_i$ are sectoral controls. Then, let G_t denote an aggregate shock. The Bartik exercise is the Local Projection (LP) of the estimated $r_i^{\mathcal{D}^q}$ on the cumulative sectoral output

$$\begin{aligned} \Delta y_{i,t+h} \; &= \beta_h^{\mathcal{D}^{cs}} \left(G_t \times \widehat{r}_i^{\mathcal{D}^{cs}} \right) + \beta_h^{\mathcal{D}^{cb}} \left(G_t \times \widehat{r}_i^{\mathcal{D}^{cb}} \right) \\ &+ \beta_h^{\mathcal{V}} \left(G_t \times exposure_i^{\mathcal{V}} \right) + \phi_{t+h} + \phi_i + v_{t+h}^{y_i} \end{aligned}$$

over $h = \{1, \ldots, H\}$ time-horizons, where ϕ_i and ϕ_{t+h} denote sector and time fixed effects, respectively, and $v_{t+h}^{y_i}$ the horizon-specific error term. The interaction term captures the identified intensity with which the aggregate disturbance affects each sector through the network. Importantly, the aggregate shock itself is absorbed by time fixed effects, so that each coefficient β does not measure the dynamic response of the economy to the aggregate shock. Instead, each measures whether sectors with greater vertical/horizontal network exposure systematically react more ($\beta > 0$) or less ($\beta < 0$) strongly to the same shock than sectors with lower exposure. The sequence of coefficients traces the evolution of these differences over horizons. Consequently, the identifying variation is cross-sectional: a positive coefficient implies that highly exposed sectors react more strongly to the aggregate disturbance, whereas a negative coefficient indicates that network exposure dampens the transmission of the shock; the sequence of such coefficients across horizons traces the evolution of these cross-sectional differences over time. Therefore, the key empirical content of Figure 5 is not the dynamic unconditional effect of horizontal exposures, but rather the per-period conditional effect of horizontal distances after controlling for vertical Input-Output exposure.

Aggregate TFP shock.– I first consider an aggregate TFP disturbance, obtained as the unexpected component of aggregate total factor productivity after removing persistence and low-frequency trends, with sectoral sensitivity measuring how close each sector is to sectors that are historically sensitive to aggregate TFP innovations.[10] The estimated cross-sectional LP coef-

[10] Using the annual series from Fernald (2014), I extract the TFP innovation as the fitted residual from the AR(1) regression with time trend: $\log TFP_t = \beta_{tfp}^{cons} + \beta_{tfp} \log TFP_{t-1} + \beta_{time} \cdot t + \varepsilon_t^{tfp}$. This series is also exploited to construct sectoral exposures. First, I estimate sector-specific historical sensitivities to aggregate TFP innovations, capturing how strongly the sector tends to react when aggregate productivity unexpectedly changes. For government spending, instead, pre-determined exposures are constructed from the 1997 U.S. Input-Output table as the government purchases in the sectoral supply of intermediate inputs, capturing the extent to which a sector is connected to government demand and therefore likely to benefit from



(A) TFP (B) GOVERNMENT SPENDING

FIGURE 5: "NETWORKING UP" AGGREGATE SHOCKS

Note: cross-sectional coefficients measure how differently sectors respond positively (amplification) or negatively (attenuation), due to vertical and horizontal network exposures, to the same aggregate shock, as in one standard deviation. Bars correspond to intervals at 90% of bootstrapped confidence level, clustered by sectors. *Source*: various and own calculations.

ficients in Panel 5a reveal a sharp contrast across transmission channels. While the Leontief-based exposure has some predictive power at short horizons, its magnitude is smaller and its effects dissipate rapidly. By contrast, the common-buyer distance measures generate larger and more persistent differences in sectoral responses: sectors that are more closely connected, through common downstream buyers, to TFP-sensitive sectors react more strongly to aggregate productivity shocks over several horizons. Common-supplier horizontal exposure does not generate statistically significant effects. Both patterns are those predicted by the theory. A positive aggregate TFP shock hits asymmetrically due to pre-determined exposures, increasing productivity more in some sectors than others, thereby lowering the relative price of the most sensitive sectors. Through the horizontal supply channel (Proposition 2 and Theorem 1), more productive sectors expand production and capture a larger share of downstream demand. Their common buyers reallocate a given budgets, and the sign of the response of the other suppliers reveals the buyers' substitutability: a positive coefficient means that a sector gains, relative to others, when its common-buyer neighbours become more productive – *i.e.*, that shared buyers treat their intermediate inputs as complements at annual frequency, in line with the estimates of Atalay (2017) and Boehm, Flaaen, et al. (2019). This competitive displacement implies that aggregate demand is satisfied by increasingly efficient producers, so that total economic activity rises despite unchanged downstream demand, as production is concentrated in higher-productivity sectors. By contrast, the horizontal common-supplier channel does not play a significant role, since TFP shocks operate through productivity differences rather than own demand expansion. In fact, a productivity shock lowers the relative price of the most sensitive sectors without directly raising anyone's intermediate input demand, making upstream propagation through shared suppliers weak and non-systematic: by Proposition 1, common-supplier transmission requires a demand impulse at shared suppliers and is therefore second order.

Government spending shock.– Panel 5b, instead, plots the coefficients for fiscal disturbances, constructed similarly to the TFP case. The results reveal a markedly different pattern across propagation channels. Government spending shocks diffuse not only through directly affected industries but also through common Input-Output relationships, and appear to diffuse across sectors sharing common suppliers, generating positive horizontal spillovers in consis-

expansions in aggregate public expenditure.

tency with the intrinsic common-supplier nature of these shocks, while sectors sharing common buyers exhibit weaker responses, consistent with competitive reallocations of downstream demand. Reassuringly, the results are also robust to disentangling aggregate government expenditure into defence and non-defence components, whose pre-determined spending categories are concentrated in targeted industries: despite the different cross-sectional allocation of network exposures, the pattern of sectoral responses is preserved. Spending is a demand impulse that raises the scale of directly exposed sectors. By the duality found in Supplemental Appendix B (Corollary B.2), the impulse raises the costs of the sectors that share congested suppliers with them, and moves their sales through substitution at their own buyers: if these treat inputs as complements – the same elasticity pattern inferred from TFP shocks –, co-customers are crowded in, as found, without requiring shared suppliers to expand at decreasing marginal cost. By Proposition 2, the same impulse reaches common-buyer neighbours only through the price of the exposed sectors: under rising marginal costs, their prices rise together with their output, and complementarity at shared buyers implies that their rivals contract, as found. Both responses thus rest on the same two ingredients, congestion and complementarity, which Subsection 3.4 identifies as the joint determinants of comovement under demand shocks.

Taken together, the two exercises reinforce the broader message of the paper: that the horizontal geometry is a distinct dimension of an Input-Output system whose information is not contained in Leontief-based exposures (Section 2), and whose relevant channel depends on the nature of the shock, as the duality Subsection 3.4 and Propositions 1-2 predict. Aggregate productivity (supply) shocks primarily propagate through the common-buyer horizontal structure since they move relative prices faced by shared downstream buyers, while aggregate fiscal (demand) shocks operate through a distinct network dimension, namely the common-supplier horizontal geometry, since they move quantities and travel through shared upstream suppliers. The fact that the dominant propagation channel varies across shocks suggests that horizontal geometries do not merely replicate classical Input-Output effects, but instead capture economically meaningful dimensions of sectoral interdependence that become relevant under different types of aggregate disturbances, and contain additional information on shock propagation that is not fully captured by standard vertical production linkages.

5. Empirics

Besides Input-Output tables, sector-level data for the U.S. economy are exploited to test the predictions about sectoral comovement in Section 3. Estimation relies on the two-stage procedure outlined by Barattieri and Cacciatore (2023): first, I perform a series of sectoral panel Fixed-Effect (FE) regressions to isolate the exogenous dynamics in a given input of production not due to other (sector-specific, other sectors-specific and aggregate) factors; then, each estimated residual is exploited as an identified structural shock into a Local Projection (LP) analysis, in order to assess the network-based short-run cumulative variations in sectoral employment.

5.1. Intermediate inputs and sectoral employment

Sections 2 and 3 rest on the premise that adjustments in the stock of circulating intermediate inputs lead to synchronized sectoral employment responses. In fact, in Proposition 3, relative sectoral employment responds to the reallocation of buyers' budgets, which reaches a sector directly, through its horizontal neighbours, and indirectly, through the supply chains that embody it. Considering variations within a sector and across its economic neighbours, the first step isolates changes in input sourcing plausibly orthogonal to employment dynamics. The following

panel fixed-effect regression over $i \in \Phi$ sectors is estimated:

$$\begin{aligned} x^{sh}_{i,t} = \beta^{cons}_{x} + \beta_n \, \dot{n}_{i,t} + \sum_{d} \beta_{n(d)} \, \dot{n}_t\big(\Phi_i, d\big) + \sum_{l} \beta_{n(l)} \, \dot{n}_t\big(\Phi_i, l\big) \\ + \gamma'_i \mathcal{Z}_{i,t} + \Gamma'_i \dot{\mathcal{Z}}_{i,t} + \gamma' \mathcal{Z}_t + \Gamma' \dot{\mathcal{Z}}_t + \phi_i + \; u^x_{i,t} \end{aligned} \tag{4}$$

where $x^{sh}_{i,t}$ identifies the share of sector-i's total production supplied by other sectors, while $\dot{n}_{i,t}$, $\dot{n}_t\big(\Phi_i, d\big)$, and $\dot{n}_t\big(\Phi_i, l\big)$ represent its employment growth, that of its economically closer and further sectors, and that of its sectors located upstream or downstream, respectively. The latter two measures are defined as

$$\dot{n}_t\big(\Phi_i, d\big) = \sum_{j \in \Phi_i} \mathbb{1}\big[\mathcal{D}^q_{ext}[i,j] = d\big] \, \Delta n_{j,t} \quad , \quad \forall d \in \{1, \ldots, \mathcal{D}_{max}\} \quad \text{and} \quad q = \{cs, cb\}$$

$$\dot{n}_t\big(\Phi_i, l\big) = \sum_{j \in \Phi_i} \ell_{i,j}\,(l) \; \Delta \, n_{j,t}\,(l) \qquad , \quad \forall l = \big\{up, dw\big\}$$

where $\Delta n\,(\cdot) = n_t\,(\cdot) - n_{t-1}\,(\cdot)$. As for vertical propagation, each change is weighted by the sales-normalised Leontief inverse relation, $\ell_{i,j}\,(l)$, for every sector-j located upstream or downstream of sector-i, thereby capturing direct and indirect network exposure. Changes in horizontally-related sectors enter without Leontief weights: distances are computed on total requirements (Section 4), and therefore already embed the vertical transport of Theorem 1, so that weighting them by Leontief exposures would count the vertical dimension twice.

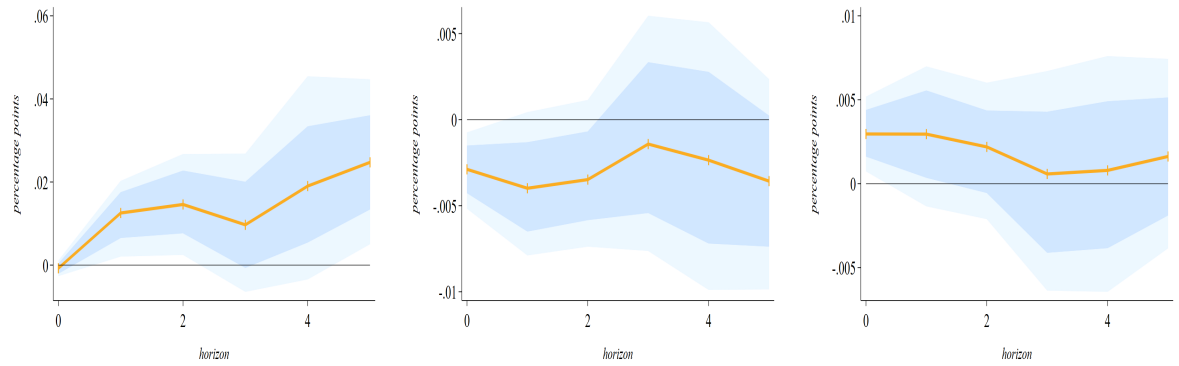
A set of control variables characterizes the second line of eq. (4). $\mathcal{Z}_{i,t}$ and $\dot{\mathcal{Z}}_{i,t}$ comprise the labour force size of the sector, both its level and growth rate of value-added, and its net exports that control for changes in demand/supply not due to internal factors. The set of aggregate controls, $\mathcal{Z}_t$ and $\dot{\mathcal{Z}}_t$, is made of aggregate value-added and aggregate employment in size as well as in growth terms. In this and in the identification in the next subsection I control for variations in aggregates since comovement between aggregate and sectoral variables is well established (*e.g.*, Stock and Watson 1999; Huffman and Wynne 1999; Rebelo 2005; DiCecio 2009). Finally, while β's are coefficients to be estimated, ϕ_i imposes fixed-effects to control for unobserved heterogeneity across sectors. Standard errors are clustered at sector-level. For each sector, the estimated residual $\widehat{u}^x_{i,t}$ identifies the exogenous variation in the share of production of sector-i that is bought from Input-Output connections.

Impulse Response Functions (IRFs) of sectoral employment to changes in the sector-specific intermediate inputs rely on Local Projections. Given $h = \{1, \ldots, H\}$ time-horizons, and for all sectors $i, j \in \Phi = \{1, \ldots, m\}$, a battery of panel FE predictive regressions of the identified sector-specific structural shock, $\widehat{u}^x_{i,t}$, on the cumulative difference in sectoral levels is performed:

$$\dot{n}^{cum}_{i,t+h} = \beta_{u^x,h} \; \widehat{u}^x_{i,t} + \phi_{t+h} + \phi_i + v^x_{i,t+h} \tag{5}$$

with $\dot{n}^{cum}_{i,t+h} = n_{i,t+h} - n_{i,t-1}$ denoting the cumulative change in sector-i employment at each horizon. Estimated coefficient $\beta_{u^x,h}$ is the response of the variation in the cumulative difference of sectoral employment at time $t+h$ given the identified shock hitting in period-t. Prediction error term $v^x_{i,t+h}$ is horizon-specific, while standard errors and bootstrapped Confidence Interval (CI) of $\beta_{u^x,h}$ are clustered by sectors. To avoid measurement errors due to unobserved heterogeneity in response, sectoral (ϕ_i) and horizon (ϕ_{t+h}) fixed-effects – that remove common trends in the comovement of sectoral employment –, are imposed. Differently from the first-stage (which controls common time trends using aggregate variables), ϕ_{t+h} removes the possibility of common trend in the *response* of sector-specific employment to idiosyncratic shocks.

Outcomes from eq. (5) estimate the cumulative responses of sectoral employment to changes



(A) SECTOR-SPECIFIC (B) $d = 1$ (C) $d = 2$

FIGURE 6: SECTORAL EMPLOYMENT RESPONSE TO INTERMEDIATE INPUTS

Note: given the *common-buyer* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposures) characterizing the North American production network in year 2007, it is shown the response of sectoral employment to a 1% increase in the sector-specific share of intermediate inputs (the share of the sector's corresponding commodity produced by alternative sectors) for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. The solid-orange line corresponds to the average response of employment across sectors, while shadow-blue and shadow-light blue areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eqs. (5)-(6). *Source*: BEA and own calculations.

in their set of intermediate inputs. Analogous estimation is performed to variations in nearby sectors, considering the dth-distance between sectors:

$$\dot{n}^{cum}_{i,t+h} = \sum_{j \in \Phi_i} \beta_{u^x_j,h}(d)\ \widehat{u}^x_{j,t}(d) + \phi_{t+h} + \phi_i + v^x_{i,t+h}(d) \tag{6}$$

Estimating eqs. (5) and (6), under *Leontief inverse* and *common-buyer* distances, delivers Figure 6. The first panel shows the cumulative (short-term) response of sector-specific employment to a 1% change in the output share of its intermediate input: averaging across sectors, the increase induces a positive statistically-significant change in sectoral employment, where each additional unit of intermediate inputs requires less than one worker so that factors of production are at least substitute, as the Cobb-Douglas specification instructs.

Changes in intermediate inputs from other sectors do not yield consistent outcomes. Results are almost always not statistically significant at 90% except when the shock hits, and the magnitude of the variation is negligible. Interesting is to note the different response between Panels 6b and 6c: for intermediate inputs changes in closer sectors there is an average negative effect, while changes in further sectors are positively related with variations in sectoral employment. Although mostly not significant, these signs are those that Proposition 3 predicts for shocks that move a sector's sales: its first-tier common-buyer neighbours lose the budgets it gains, whereas horizontally distant sectors along common supply chains move with it. In Supplemental Appendix D, under *Leontief inverse* and *common-supplier* network distances, the same average impact holds, while results for distances' effects identify stronger but opposite results: increasing for closer sectors, declining for further ones. From these common-supplier distances, the positive response of closer sectors is in line with Subsection 3.4: higher purchases at shared suppliers raise the costs of co-customers, whose sales rise when their own buyers treat inputs as complements. Such observations might suggest that certain types of intermediate inputs are more essential than others in production (*e.g.*, Carvalho and Voigtländer 2015), and that the type of economic distance matters for their role within the network.

5.2. SECTORAL COMOVEMENT OF EMPLOYMENT

I now analyse how sectoral employment adjusts in response to employment changes in other sectors, depending on their network economic distance. The previous identification and estimation scheme is developed, but now purging the dynamics of sector-specific employment with changes in its and other sectors' value-added, since employment is not exogenous to growth rates of sectoral output (*e.g.*, Hornstein and Praschnik 1997; Acemoglu, Akcigit, et al. 2015; Sandqvist 2017). Proposition 3 maps into the empirical design. The common component is absorbed by time fixed effects. Vertical exposures enter the first stage, which purges sectoral employment of the output growth of upstream and downstream sectors, weighted by the Leontief inverse. Sectoral responses to shocks at a given horizontal distance then combines the two network components: opposite comovement where common-buyer proximity dominates ($\boldsymbol{\mathcal{R}}^h$), and positive comovement where vertical connections dominate ($\boldsymbol{\mathcal{R}}^v$).

Identification.– To identify exogenous variations in sectoral employment, $n_{i,t}$, a panel fixed-effects regression over each sector $i \in \Phi$ is estimated:

$$\begin{aligned} n_{i,t} = \beta_n^{cons} + \beta_y \, \dot{y}_{i,t} + \sum_d \beta_{y(d)} \, \dot{y}_t\left(\Phi_i, d\right) + \sum_l \beta_{y(l)} \, \dot{y}_t\left(\Phi_i, l\right) \\ + \gamma_i' \mathcal{Z}_{i,t} + \Gamma_i' \dot{\mathcal{Z}}_{i,t} + \gamma' \mathcal{Z}_t + \Gamma' \dot{\mathcal{Z}}_t + \phi_i + u_{i,t}^n \end{aligned} \tag{7}$$

where $\dot{y}_{i,t}$ is the value-added growth for sector-i, $\dot{y}_t\left(\Phi_i, d\right)$ identifies the value-added growth of its related sectors according to network distances, and $\dot{y}_t\left(\Phi_i, l\right)$ represents the value-added growth of its upstream and downstream sectors. Each measure is defined as that of eq. (4) but using y_i instead of employment. Moreover, for this specification, sector-level controls, $\mathcal{Z}_{i,t}$ and $\dot{\mathcal{Z}}_{i,t}$, comprise the labour force size of the sector, its value-added level and net exports, while aggregate controls, $\mathcal{Z}_t$ and $\dot{\mathcal{Z}}_t$, are analogous to eq. (4). Estimated coefficients are the β's, given sectoral fixed-effects, ϕ_i, and sectoral standard errors. Sector-specific estimated residual $\widehat{u}_{i,t}^n$ from eq. (7) identifies the exogenous variation in sectoral employment.

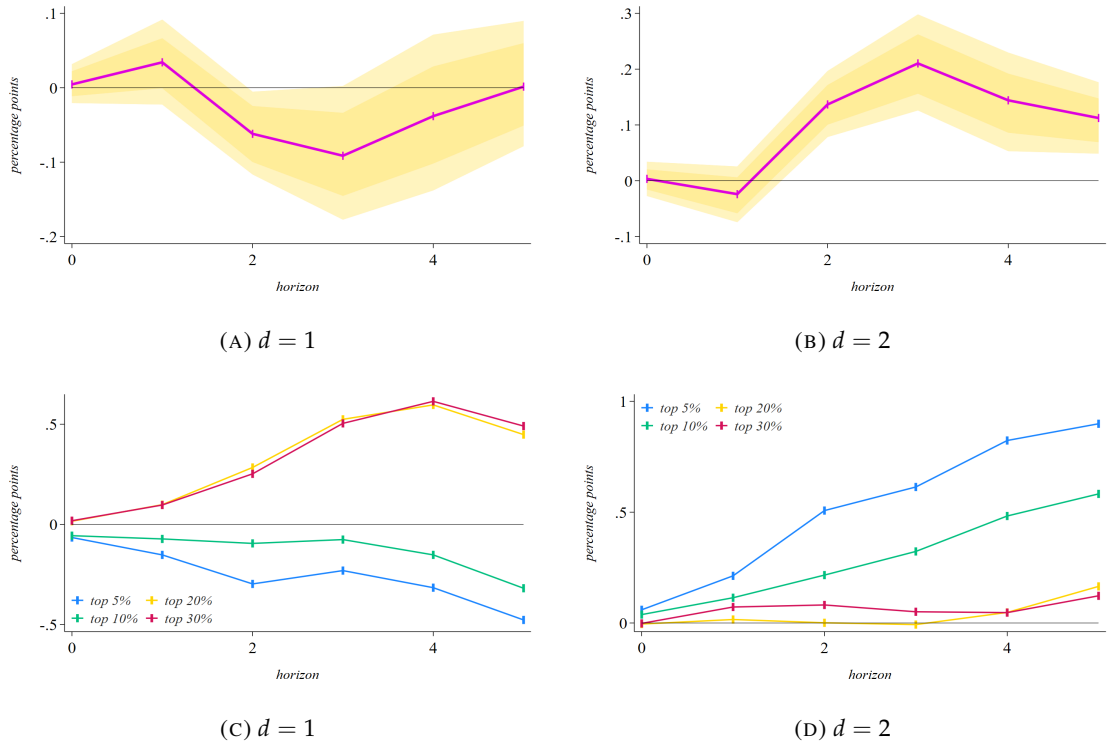
Local Projection analysis.– Sectoral comovement is defined as the response of employment in a given sector to changes in employment of other sectors:

$$\dot{n}_{i,t+h}^{cum} = \sum_{j \in \Phi_i} \beta_{u_j^n,h}\left(d\right) \; \widehat{u}_{j,t}^n\left(d\right) + \phi_{t+h} + \phi_i + v_{i,t+h}^n\left(d\right) \tag{8}$$

with $\dot{n}_{i,t+h}^{cum} = n_{i,t+h} - n_{i,t-1}$ denoting the cumulative change in sector-i employment at each horizon given the identified employment shock in closer or further sectors at the dth-distance. Coefficient $\beta_{u_j^n,h}\left(d\right)$ is the h-step-ahead response of sector-specific employment in cumulative difference due to the identified shock $\widehat{u}_{j,t}^n\left(d\right)$. Still, sector and horizon fixed effects, ϕ_i and ϕ_{t+h} respectively, are imposed to remove unobserved heterogeneity and common trends across sectors in the comovement of sectoral employment. Prediction error term $v_{i,t+h}^n\left(d\right)$ is horizon- and distance-specific in each predictive panel regression, with standard errors and bootstrapped Confidence Interval (CI) of $\beta_{u_j^n,h}\left(d\right)$ clustered by sectors.

5.2.1. Results for common-supplier distances

Considering sectors whose distance is defined in terms of buying from the same set of upstream suppliers, estimating eq. (8) yields Figure 7. By Proposition 1 and Subsection 3.4, common-supplier linkages are ambiguous to interpret, since they depend on the joint action of vertical propagation and horizontal spillovers: a positive shock in one sector raises its demand for the shared supplier and, if the latter is congested, its price, thereby raising the costs of the other



(A) $d = 1$ (B) $d = 2$

(C) $d = 1$ (D) $d = 2$

FIGURE 7: EMPLOYMENT COMOVEMENT UNDER COMMON-SUPPLIER LINKAGES

Note: given the *common-supplier* network distance (*i.e.*, demand linkages across sectors given their common upstream suppliers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposures) characterizing the North American production network in year 2007, it is shown the response of sectoral employment to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. In the top panels, the solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). In the bottom panels, each line corresponds to the average response of employment across sectors with different numbers of linkages, robust to 68% and 90% significance levels of bootstrapped Confidence Interval (CI); if a plot is not appearing, it means there is no response for the specified distance value. *Source*: BEA and own calculations.

sector; whether this crowds the other sector out or in depends on its own buyers, which shift away from it if they treat inputs as substitutes and towards it if they treat them as complements. Where these forces conflict or remain weak, observed correlations are muted or ambiguous.

These mechanisms echo in observed responses. Panels 7a and 7b show that a 1% increase in employment in closely linked sectors ($d = 1$) produces an ambiguous response, whereas more distant sectors ($d = 2$) consistently display positive comovement. However, sectoral shocks in Panel 7a do not produce any statistically significant effects, as the estimated responses fail to reach significance at the 90% confidence levels. This lack may be driven by including all sectors indiscriminately, as sectors with fewer inter-sectoral linkages might introduce noise or bias to the estimation, masking and diluting the true magnitude of measured comovement in sectoral employment: restricting the sample to sectors with most Input-Output linkages could reveal more meaningful propagation effects where network effects are more pronounced, under which horizontal transmission scales with the pair's shares in shared partners.

Nevertheless, ambiguous patterns persist when partitioning sectors into finer subsets by the number of Input-Output linkages: comovement depends on the specific cluster considered. This confirms that common-supplier horizontal interdependencies and vertical propagation operate jointly, often obscuring one another. Such mixed evidence ultimately reflects the inherent ambiguity of common-supplier distances discussed in Subsection 3.4: comovement between sectors that share suppliers combines congestion at the shared supplier with substitutability at the co-customers' buyers, both of which vary across clusters, and overlaps with vertical propagation.

This is what Supplemental Appendix B (Corollary B.2) anticipates.

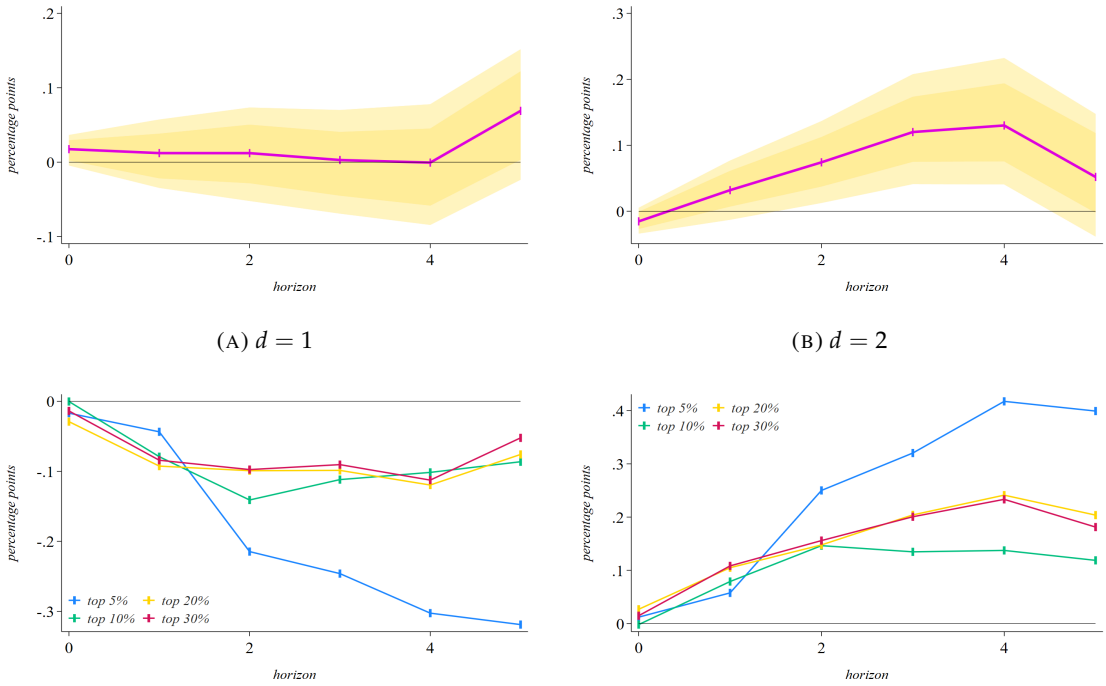
Positive comovement is most likely to emerge between sectors that are "economically" distant, for which the horizontal component of Proposition 3 is small and vertical connections dominate: since the common component is absorbed by time fixed effects, positive comovement relative to other sectors reflects common supply chains ($\boldsymbol{\mathcal{R}}^{v}$) and, under demand shocks, the upstream vertical channel of Proposition B.1 in Supplemental Appendix B. Among closely connected sectors, the sign depends on the curvature of shared suppliers' costs and the balance between vertical and horizontal channels is more delicate, producing the mixed responses observed in the data. Observed employment comovement thus mirrors an interplay of vertical and horizontal propagation, highlighting the inherent complexity in interpreting common-supplier interdependencies.

5.2.2. Results for common-buyer distances

Considering now sectors whose distance is defined in terms of selling to common downstream buyers, Figure 8 plots the responses from the estimation of eq. (8). Interpreting the effect of further ($d = 2$) sectors' variations in employment delivers the same results of the common-supplier case: from Panel 8b, a 1% increase in their employment increases sectoral employment as well, thus determining positive comovement and a positive transmission of the sector-idiosyncratic shock. This is the second prediction of Proposition 3: sectors that do not share buyers do not compete for the same budgets, so that their comovement is driven by the vertical component, which moves together the sectors that lie along common supply chains, whatever the elasticities. Different is the case of closer ($d = 1$) sectors, where it appears a positive and then a negative cumulative effect on sector-specific employment with a small magnitude.

The lack of significance in Panel 8a might have two complementary interpretations: *(i)* changes in employment in nearby sectors have no measurable impact on others when the sectors are linked through a common set of downstream buyers; and *(ii)* the common-buyer network distance seems to dampen the vertical propagation of shocks across the production network, thereby weakening the expected comovement in production inputs across sectors. In the language of Proposition 3, the horizontal component, whose sign is always opposite to that of the vertical one, offsets it for pairs that are close at common buyers. Specifically, when sectors share the same downstream buyers, a change in their relative price faced by the common buyer leads to opposite changes in their relative supply of intermediate inputs. This generates an inverse comovement in sectoral production and, consequently, an opposite comovement in sectoral employment. In the context of Panel 8a, these opposing forces roughly cancel each other out when averaged across all sectors, resulting in the non-significant response. Accordingly, clearer effects might be correctly identified when considering highly interlinked sectors. Note that the non-significance of Panel 8a aligns with Theorem 1: horizontal rivalry is concentrated where shared buyers are large relative to the pair, with weights $\lambda_s (1 - \alpha_s) \widehat{\alpha}_{si} \widehat{\alpha}_{sj}$; averaged over all pairs at $d = 1$, many of which share only minor buyers, it is diluted.

Impulse responses for the clustered analysis are presented at the bottom of Figure 8. The results for major network distances, shown in Panel 8d, largely mirror those found using the full sample, but with larger point estimates responding to the structural employment shock. More strikingly, at short supply distances (Panel 8c), an increase in employment in closely linked sectors reduces employment in the sector – as Proposition 3 and Corollary 1 predict for first-tier common-buyer neighbours –, resulting in an opposite comovement in sectoral employment. This inverse relationship grows stronger as the analysis narrows to smaller groups of sectors with increasingly dense linkages, which share larger buyers and carry larger weights in $\boldsymbol{\mathcal{H}}$:



(A) $d = 1$ (B) $d = 2$

(C) $d = 1$ (D) $d = 2$

FIGURE 8: EMPLOYMENT COMOVEMENT UNDER COMMON-BUYER LINKAGES

Note: given the *common-buyer* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposures) characterizing the North American production network in year 2007, it is shown the response of sectoral employment to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. In the top panels, he solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). In the bottom panels, each line corresponds to the average response of employment across sectors with different numbers of linkages, robust to 68% and 90% significance levels of bootstrapped Confidence Interval (CI); if a plot is not appearing, it means there is no response for the specified distance value. *Source*: BEA and own calculations.

the closer the sectors are in the production network when supplying downstream, the more pronounced these opposing employment responses become.

5.3. ROBUSTNESS

Results on comovement hold along different "economically-based" robustness checks, sequentially presented in Supplemental Appendix D. Initially, I shift the focus towards peripheral sectors within the production network. Much of the existing literature holds on the role of central sectors, typically measured using Bonacich-Katz centrality (*e.g.*, Carvalho 2014). The underlying premise is that a positive shock in a central sector transmits outward, potentially inducing broader economic expansions – *i.e.*, microeconomic shocks can scale up to macroeconomic consequences through network centrality. From a comovement perspective, periods of economic expansion (contraction) are expected only when a positive (negative) shock in a central sector effectively propagates to more peripheral sectors: fluctuating aggregate activity is not merely a direct realization of shocks to central nodes, but rather it stems from the manner in which such asymmetric shocks ripple across the network to other, more peripheral sectors. Stated differently, it is not the initial shock per se that generates business cycle fluctuations, but the subsequent consequences it produces as it transmits across the network to less central nodes. If network distances were irrelevant, comovement would be expected universally. However, should common-supplier and common-buyer linkages prove significant, the comovement

patterns ought to remain consistent with those documented thus far.

I further account for changes in sectoral inventories, since they introduce a short-run margin that may buffer the network-based transmission: when inventories are available, sectors smooth shocks through previously accumulated stocks, temporarily decoupling production adjustments from contemporaneous changes in the purchased intermediate input bundle. As a result, the transmission of shocks through shared Input-Output structures should be attenuated, breaking the simultaneity of horizontal interdependencies and weakening the role of both common-supplier and common-buyer network economic distances in shaping short-run comovement. By contrast, in the absence of inventories, changes in economically nearby sectors are expected to translate directly into the usual horizontal propagation effects.

Finally, I test the robustness by employing alternative base years for the U.S. production network. The main analysis is anchored at the 2007 Input-Output structure, which conveniently divides the sample period into two equal sub-periods. Theoretically, the results should remain valid provided that the production network structure exhibits limited evolution over time.[11] Empirically, I replicate the full set of analyses using network data from other benchmark years.

Centrality scores.– Figures D.6 and D.7 present the impulse responses of employment to common-supplier and common-buyer distances, respectively, considering only the peripheral sectors of the U.S. production network. Results are broadly consistent with those discussed in earlier sections: opposite comovement under supply distances, and positive comovement for shocks originated in further sectors. The only noteworthy exception concerns the response of employment to changes in closer sectors under common-supplier distances: rather than exhibiting the ambiguity observed for all sectors (overlapping of vertical and horizontal dimensions), inspecting horizontal interdependencies in demand for more peripheral sectors generate effects that resemble those observed under common-buyer distances, with opposite comovement occurring across sectors sharing the same network structure in terms of buying from common suppliers. This refines the observed pattern under direct common-supplier measures of Figure 7, suggesting that opposite employment comovement among sectors purchasing from similar upstream suppliers largely occurs in sectors not playing a key role in the propagation of shocks. These empirical outcomes, alongside with Sections 2-3, relate and expand Acemoglu, Carvalho, et al. (2012): shocks to key nodes reach peripheral sectors differently according to their network economic distance, a margin of heterogeneity that may shape how sectoral shocks aggregate.

Inventories.– The outlined hypothesis is strongly supported by the common-buyer patterns in Figure D.8. horizontal interdependencies vanish for sectors holding inventories, and the transmission of a positive shock reverts to its vertical logic.[12] By contrast, in contexts without inventories, the baseline results are preserved, with positive shocks generating opposite comovement across sectors with nearly similar Input-Output structures. This sharp contrast indicates that horizontal effects are operative when production adjustments are directly tied to contemporaneous changes in circulating intermediate inputs, rather than when accumulated in storage. Inventories act as a short-run buffer that temporarily deactivates the horizontal propagation channel without altering the underlying network structure. These findings underscore that the documented horizontal interdependencies reflect structural economic mechanisms arising from contemporaneous adjustments through shared linkages rather than from mechanical

[11] A condition broadly supported in the empirical literature. Carvalho (2014) and Acemoglu, Ozdaglar, et al. (2016) document relative stability of U.S. sectoral production network and its role on shock propagation. Classical Input-Output analysis (*e.g.*, Leontief 1986; Miller and Blair 2022) also emphasises the persistence of production linkages in the short to medium run.

[12] Over the sample period, inventories are observed in NAICS agriculture (11), extraction (21), and manufacturing (31-33) sectors, as well as in transportation (481-484), publishing (511), and recording (512) industries.

features of the production network.

Different base years.– All the documented results, derived using the 2007 U.S. tables, remain robust if replicated for alternative benchmark years (2002 and 2012), as depicted in Figures D.9, D.10, and D.11, supporting the inherent ambiguity of common-supplier distances and the clearer, more systematic comovement patterns under common-buyer distances. This temporal consistency is in line with the slow-moving evolution of production networks documented in the literature, on which the analysis relies.

6. Concluding Remarks

Initial inquiries into production networks chiefly centred on how disaggregation induced by inter-sectoral trade generates comovement of macroeconomic variables. Reviving this foundational perspective, this paper moves beyond traditional vertical linkages by theoretically introducing and empirically validating the horizontal geometry of Input-Output economies, measured through network "economic" distances reflecting similarities between sectors in terms of common upstream suppliers or downstream buyers. Constitutive premise is that sectors are densely interconnected not only by their observed trade flows but also through shared inter-sectoral relationships, providing a richer understanding of the complex architecture underlying any modern economy. Common-supplier distances promote an overlap of horizontal and vertical propagation, as shocks transmitted through shared upstream suppliers interact with parallel supply chain effects. By contrast, common-buyer distances counteract vertical transmission, as revenue reallocation among sectors selling to the same downstream buyers counteracts the standard propagation of shocks. These mechanisms add to the observed inter-sectoral trade intensities, offering a powerful refinement to prevailing network analysis. As for aggregate output, horizontal propagation reallocates rather than cascades. It is hidden in Hulten's (1978) theorem, which leaves first-order aggregate fluctuations to Domar weights alone, but it governs how these weights move with shocks: it shifts workers across sectors and bends aggregate output, so that micro volatility moves mean output by an amount that diversification across many sectors does not eliminate. By conceptualising these horizontal interdependencies, the paper enriches the understanding of how independent and idiosyncratic shocks permeate the production system. Ultimately, it reveals that the demand/supply architecture of an Input-Output structure and its horizontal geometry critically shape the transmission of shocks, offering novel insights into the networked nature of economic activity.

Horizontal geometry may help to assess policy effectiveness by accounting for sectoral composition and interdependencies between targeted and untargeted sectors. Fiscal policy provides a natural application, as sectors sharing common upstream suppliers or downstream buyers with those directly exposed to government spending may experience additional effects that conventional Input-Output linkages (*e.g.*, Barattieri, Cacciatore, and Traum 2023; Bouakez et al. 2025) fail to capture. The same logic applies to tariff policy, as changes in relative prices redistribute revenues across sectors (*e.g.*, Barattieri and Cacciatore 2023; Antonova et al. 2025; Clausing and Obstfeld 2025) and may generate horizontal effects among sectors serving common downstream buyers. Likewise, industrial policies targeting strategic sectors may affect economically close but untargeted sectors through shared production relationships, extending the analysis of policy-induced distortions and downstream spillovers (*e.g.*, Liu 2019; Lane 2025). Finally, horizontal geometry may enrich the analysis of monetary policy by identifying how common production relationships shape the expectations on sectoral price behaviours and the propagation of policy-induced shocks, complementing existing production network approaches to monetary transmission (*e.g.*, Pasten et al. 2020; Kalemli-Özcan et al. 2025).

Beyond these policy applications, horizontal linkages offer a theoretical framework for investigating global production networks (*e.g.*, Caliendo et al. 2022; Huo et al. 2025) through sector-country economic distances, the complexity of firm-to-firm linkages (*e.g.*, di Giovanni et al. 2018; Boehm, Flaaen, et al. 2019) by linking otherwise unconnected firms, and the endogenous formation of production networks (*e.g.*, Carvalho and Voigtländer 2015; Oberfield 2018) under node-specific comparison across a spectrum of competing suppliers. More broadly, the principle of identifying economic proximity through shared interdependencies extends beyond Input-Output economies to investment (*e.g.*, vom Lehn and Winberry 2022), credit (*e.g.*, Delli Gatti et al. 2010), and financial (*e.g.*, Huremović, Jiménez, et al. 2026) networks, suggesting that horizontal geometry provides a general analytical framework for understanding how economic shocks and policy interventions propagate through interconnected systems.

APPENDIX

A. PROOF OF THEOREM 2

Let me derive the three claims one-by-one: in what follows, part *(i)* derives the slope of aggregate output with respect to sectoral productivities; part *(ii)* how this slope changes with productivities; and part *(iii)* what this change implies for the mean of aggregate output. Throughout, $\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$, and, for any vector $\boldsymbol{u}$, $\mathrm{Var}_{\widehat{\boldsymbol{a}}_j^{row}}[\boldsymbol{u}] = \sum_{i\in\Phi}\widehat{\alpha}_{ji}u_i^2 - \left(\sum_{i\in\Phi}\widehat{\alpha}_{ji}u_i\right)^2$ is the variance of the entries of $\boldsymbol{u}$ across buyer-j's inputs, weighted by their shares in j's bundle.

(i) Step i.1: real GDP and the price index. With labour fixed at N and the wage w as numeraire, nominal GDP, wN, is constant, and Cobb-Douglas preferences give $\log P = \sum_{i\in\Phi}\omega_i \log p_i + \text{const}$, where P is the aggregate final-consumption price index and p_i is the price of commodity-i. Hence, $\log Y = \log[wN] - \log P$ implies $d\log Y = -\boldsymbol{\omega}' d\log \boldsymbol{p}$, where Y denotes real GDP and $\boldsymbol{p}$ is the vector of sectoral prices. *Step i.2: prices.* Unit costs give $d\log p_i = -d\log z_i + \sum_{j\in\Phi}\alpha_{ij}\, d\log p_j$, that is, $d\log \boldsymbol{p} = -\boldsymbol{\mathcal{L}}\, d\log \boldsymbol{z}$. *Step i.3: Hulten's theorem.* Substituting, and since Domar weights are collected into $\boldsymbol{\lambda}' = \boldsymbol{\omega}'\boldsymbol{\mathcal{L}}$, then $d\log Y = \boldsymbol{\omega}'\boldsymbol{\mathcal{L}}\, d\log \boldsymbol{z} = \boldsymbol{\lambda}' d\log \boldsymbol{z}$ holds at every equilibrium, with $\boldsymbol{\lambda}$ evaluated at current shares. *Step i.4: variance.* Since $\mathbb{E}[\boldsymbol{\varepsilon}] = \boldsymbol{0}$ and $\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'] = \nu^2\boldsymbol{I}$, where ν^2 is the common variance of sectoral shocks, then $\mathrm{Var}[\boldsymbol{\lambda}'\boldsymbol{\varepsilon}] = \boldsymbol{\lambda}'\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}']\boldsymbol{\lambda} = \nu^2\|\boldsymbol{\lambda}\|^2$, in which horizontal operator $\boldsymbol{\mathcal{H}}$ does not appear. *Step i.5: rate.* If $\lambda_i \asymp \frac{1}{m}$ uniformly across sectors, then $\|\boldsymbol{\lambda}\|^2 = \sum_{i\in\Phi}\lambda_i^2 \asymp m\cdot\frac{1}{m^2} = \frac{1}{m} \to 0$.

(ii) Step ii.1: the slope. By part *(i)*, the gradient of $\log Y$ at any vector of productivities is $\boldsymbol{\lambda}'[\boldsymbol{z}]$. *Step ii.2: the Hessian.* Differentiating once more, Theorem 1 gives

$$\frac{\partial^2 \log Y}{\partial \log \boldsymbol{z}\, \partial \log \boldsymbol{z}'} = \frac{\partial \boldsymbol{\lambda}}{\partial \log \boldsymbol{z}'} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$$

which is symmetric (as a canonical Hessian matrix). *Step ii.3: transport, then dispersion.* Along the shock direction, with $d\log \boldsymbol{z} = \boldsymbol{\varepsilon}$ evaluated at the deterministic benchmark, $d^2\log Y = \boldsymbol{\varepsilon}'\boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon} = (\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon})'\,\boldsymbol{\mathcal{H}}\,(\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon})$, where $\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon} = -d\log \boldsymbol{p}$ is the price fall of every commodity. *Step ii.4: covariance form.* For any $\boldsymbol{u}$, then $\boldsymbol{u}'\left[\mathrm{Diag}\left[\widehat{\boldsymbol{a}}_s^{row}\right] - \left(\widehat{\boldsymbol{a}}_s^{row}\right)'\widehat{\boldsymbol{a}}_s^{row}\right]\boldsymbol{u} = \sum_{i\in\Phi}\widehat{\alpha}_{si}u_i^2 - \left(\sum_{i\in\Phi}\widehat{\alpha}_{si}u_i\right)^2 = \mathrm{Var}_{\widehat{\boldsymbol{a}}_s^{row}}[\boldsymbol{u}]$. Applying it with $\boldsymbol{u} = \boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon}$ to each term of $\boldsymbol{\mathcal{H}}$:

$$d^2\log Y = \sum_{s\in\Phi}(\sigma_s - 1)\,\lambda_s\,(1-\alpha_s)\,\mathrm{Var}_{\widehat{\boldsymbol{a}}_s^{row}}\left[\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon}\right]$$

Each term is minus the curvature of buyer-s's cost index, $(1-\sigma_s)\,\mathrm{Var}_{\widehat{\boldsymbol{a}}_s^{row}}[d\log \boldsymbol{p}]$, weighted by s's intermediate spending relative to GDP.

(iii) Step iii.1: second-order expansion. Let Y^{det} denote real GDP at the deterministic benchmark, before sectoral shocks. Around this deterministic equilibrium, $\log Y = \log Y^{det} + \boldsymbol{\lambda}'\boldsymbol{\varepsilon} + \frac{1}{2}\boldsymbol{\varepsilon}'\boldsymbol{\mathcal{R}}\boldsymbol{\varepsilon} + R[\boldsymbol{\varepsilon}]$, with the remainder equal to $R[\boldsymbol{\varepsilon}] = o\left[\|\boldsymbol{\varepsilon}\|^2\right]$. As implied by the little-$o[\cdot]$ notation, the remainder is asymptotically negligible relative to $\|\boldsymbol{\varepsilon}\|^2$, meaning that $\lim_{\boldsymbol{\varepsilon}\to\mathbf{0}} \frac{R[\boldsymbol{\varepsilon}]}{\|\boldsymbol{\varepsilon}\|^2} = 0$. *Step iii.2: expectations.* The first-order term has mean zero, $\boldsymbol{\lambda}'\mathbb{E}[\boldsymbol{\varepsilon}] = 0$. The second-order term does not: by the cyclic property of the trace identity $\mathbb{E}[\boldsymbol{\varepsilon}'\boldsymbol{\mathcal{R}}\boldsymbol{\varepsilon}] = \text{Trace}\left[\boldsymbol{\mathcal{R}}\,\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}']\right]$, where Trace$[\cdot]$ denotes the sum of the diagonal entries of a square matrix, then $\mathbb{E}[\boldsymbol{\varepsilon}'\boldsymbol{\mathcal{R}}\boldsymbol{\varepsilon}] = \text{Trace}\left[\boldsymbol{\mathcal{R}}\,\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}']\right] = \nu^2\,\text{Trace}[\boldsymbol{\mathcal{R}}]$. *Step iii.3: remainder.* With $\boldsymbol{\varepsilon} = \nu\boldsymbol{\eta}$, where $\boldsymbol{\eta}$ is a normalized shock vector with bounded support and $\mathbb{E}\left[\|\boldsymbol{\eta}\|^3\right] < \infty$ – so that, for ν small enough, $\boldsymbol{\varepsilon}$ lies in a neighbourhood of the deterministic equilibrium in which the third derivatives of $\log Y$ are bounded –, Taylor expansion implies $\left|\mathbb{E}[R]\right| \leq \Xi\nu^3\,\mathbb{E}\left[\|\boldsymbol{\eta}\|^3\right] = o\left[\nu^2\right]$ for a fixed number of sectors, as $\nu^2 \to 0$, where Ξ is a constant independent of variance ν^2. Taking expectations in Step *iii.1* gives the third statement. Since $\nu^2\,\text{Trace}[\boldsymbol{\mathcal{R}}] = \sum_{s\in\Phi}(\sigma_s - 1)\lambda_s(1-\alpha_s)\,\mathbb{E}\left[\text{Var}_{\widehat{\boldsymbol{a}}_s^{row}}[\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon}]\right]$ by part *(ii)* and Step *iii.2*, the mean effect is the expected dispersion of input prices within buyers' bundles, which, unlike the first-order term, cannot average out. *Step iii.4: a lower bound.* Fix a buyer-s and a commodity-i. Column-i of $\boldsymbol{\mathcal{L}} = \boldsymbol{I} + \boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{L}}$ gives, for $j \neq i$, $\ell_{ji} = \sum_{m\in\Phi}\alpha_{jm}\ell_{mi} \leq (1-\alpha_j)\max_m \ell_{mi} < \max_m \ell_{mi}$, so that the maximum of the column is $\ell_{ii} \geq 1$ and $\ell_{ji} \leq (1-\underline{\alpha})\,\ell_{ii}$. Draw one of buyer-$s$'s inputs, j with probability $\widehat{\alpha}_{sj}$, and record ℓ_{ji}. Splitting the draws into $j = i$ and $j \neq i$, the law of total variance bounds $\text{Var}_{\widehat{\boldsymbol{a}}_s^{row}}\left[\boldsymbol{\ell}_i^{col}\right]$ from below by the dispersion between the two group means, $\widehat{\alpha}_{si}\left(1-\widehat{\alpha}_{si}\right)\left(\ell_{ii} - \overline{\ell}_{si}\right)^2$, where $\overline{\ell}_{si} = \sum_{j\neq i}\widehat{\alpha}_{sj}\ell_{ji}/\left(1-\widehat{\alpha}_{si}\right) \leq (1-\underline{\alpha})\,\ell_{ii}$, so that the gap is at least $\underline{\alpha}\,\ell_{ii} \geq \underline{\alpha}$. Summing over i, $\text{Trace}\left[\boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{C}}_s\boldsymbol{\mathcal{L}}\right] = \sum_{i\in\Phi}\text{Var}_{\widehat{\boldsymbol{a}}_s^{row}}\left[\boldsymbol{\ell}_i^{col}\right] \geq \underline{\alpha}^2\sum_{i\in\Phi}\widehat{\alpha}_{si}\left(1-\widehat{\alpha}_{si}\right) = \underline{\alpha}^2\left(1 - \|\widehat{\boldsymbol{a}}_s^{row}\|^2\right)$. If $\sigma_s - 1$ has the same sign for all s, every term of $\text{Trace}[\boldsymbol{\mathcal{R}}] = \sum_{s\in\Phi}(\sigma_s-1)\lambda_s(1-\alpha_s)\,\text{Trace}\left[\boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{C}}_s\boldsymbol{\mathcal{L}}\right]$ shares it, which gives the first part of the claim. Finally, since $\sum_{s\in\Phi}\alpha_s\lambda_s = 1$, then $\sum_{s\in\Phi}\lambda_s \geq \frac{1}{\overline{\alpha}} \geq 1$, so that, under the stated conditions, $\sum_{s\in\Phi}|\sigma_s - 1|\,\lambda_s(1-\alpha_s)\left(1 - \|\widehat{\boldsymbol{a}}_s^{row}\|^2\right) \geq \underline{c}\,(1-\overline{\alpha})\left(1-\overline{h}\right)$, a constant that does not depend on the size of the production network $m = |\Phi|$.

B. EMPIRICS

TABLE 1: DIMENSIONAL DOMINANCE

	network sparsity				
	15% $\tau = 0.5\%$	30% $\tau = 1\%$	40% $\tau = 1.5\%$	50% $\tau = 2\%$	95% $\tau = 5\%$
vertical propagation	.2133	.1995	.1803	.1497	.0762
horizontal propagation	.1371	.1559	.1718	.1731	.1746

Spectral norms for a 3-digit U.S. Input-Output network in year 2007. Due to tresholding-τ, transactions are removed and sparsity is approximated. Source: BEA and own calculations.

BIBLIOGRAPHY

Acemoglu, D., U. Akcigit, and W. Kerr (2015), "Networks and the macroeconomy: an empirical exploration", *NBER Macroeconomics Annual*, 30:273–335

Acemoglu, D. and P. D. Azar (2020), "Endogenous production networks", *Econometrica*, 88(1):33–82

Acemoglu, D., V. M. Carvalho, A. Ozdaglar, and A. Tahbaz-Salehi (2012), "The network origins of aggregate fluctuations", *Econometrica*, 80(5):1977–2016

Acemoglu, D., A. Ozdaglar, and A. Tahbaz-Salehi (2016), "Networks, shocks, and systemic risk". IN: *The Oxford Handbook of the Economics of Networks*, pp. 569–608, edited by Yann Bramoullé, A. G. and B. W. Rogers, New York: Oxford University Press.

Antonova, A., L. Huxel, M. Matvieiev, and G. Muller (2025), "The propagation of tariff shocks via production networks", *CEPR Discussion Paper n. 20305*

Atalay, E. (2017), "How important are sectoral shocks?", *American Economic Journal: Macroeconomics*, 9(4):254–280

Baqaee, D. R. (2018), "Cascading failures in production networks", *Econometrica*, 86(5):1819–1838

Baqaee, D. R. and E. Farhi (2019), "The macroeconomic impact of microeconomic shocks: beyond Hulten's Theorem", *Econometrica*, 87(4):1155–1203

— (2020), "Productivity and misallocation in general equilibrium", *Quarterly Journal of Economics*, 135(1):105–163

Barattieri, A. and M. Cacciatore (2023), "Self-harming trade policy? Protectionism and production networks", *American Economic Journal: Macroeconomics*, 15(2):97–128

Barattieri, A., M. Cacciatore, and N. Traum (2023), "Estimating the effects of government spending through the production network", *NBER Working Paper n. 31680*

Barrot, J.-N. and J. Sauvagnat (2016), "Input specificity and the propagation of idiosyncratic shocks in production networks", *Quarterly Journal of Economics*, 131(3):1543–1592

Bartik, T. J. (1991), *Who Benefits from State and Local Economic Development Policies?* Kalamazoo, MI: W.E. Upjohn Institute for Employment Research

Boehm, C. E., A. Flaaen, and N. Pandalai-Nayar (2019), "Input linkages and the transmission of shocks: firm-level evidence from the 2011 Tōhoku earthquake", *Review of Economics and Statistics*, 101(1):60–75

Boehm, J. and E. Oberfield (2018), "Growth and the fragmentation of production", *Working Paper n. 885 Meeting Papers from Society for Economic Dynamics*

Bouakez, H., O. Rachedi, and E. Santoro (2025), "The sectoral origins of heterogeneous spending multipliers", *Journal of Public Economics*, 248:105404

Caliendo, L., F. Parro, and A. Tsyvinski (2022), "Distortions and the structure of the world economy", *American Economic Journal: Macroeconomics*, 14(4):274–308

Carvalho, V. M. (2014), "From micro to macro via production networks", *Journal of Economic Perspectives*, 28(4):23–48

Carvalho, V. M. and X. Gabaix (2013), "The Great Diversification and its undoing", *American Economic Review*, 103(5):1697–1727

Carvalho, V. M. and A. Tahbaz-Salehi (2019), "Production networks: a primer", *Annual Review of Economics*, 11:635–663

Carvalho, V. M. and N. Voigtländer (2015), "Input diffusion and the evolution of production networks", *NBER Working Paper n. 20025*

Cassou, S. P. and J. Vázquez (2014), "Employment comovements at the sectoral level over the business cycle", *Empirical Economics*, 46:1301–1323

Christiano, L. J. and T. J. Fitzgerald (1998), "The business cycle: it's still a puzzle", *Economic Perspectives*, 22(4):56–83

Clausing, K. and M. Obstfeld (2025), "Tariffs as fiscal policy", *CEPR Discussion Paper n. 20677*

Conley, T. G. and B. Dupor (2003), "A spatial analysis of sectoral complementarity", *Journal of Political Economy*, 11(2):311–352

Cooper, R. and J. Haltiwanger (1990), "Inventories and the propagation of sectoral shocks", *American Economic Review*, 80(1):170–190

Delli Gatti, D., M. Gallegati, B. Greenwald, A. Russo, and J. E. Stglitz (2010), "The financial accelerator in an evolving credit network", *Journal of Economic Dynamics and Control*, 34(9):1627–1650

di Giovanni, J., A. A. Levchenko, and I. Mejean (2018), "The micro origins of international business-cycle comovement", *American Economic Review*, 108(1):82–108

DiCecio, R. (2009), "Sticky wages and sectoral labor comovement", *Journal of Economic Dynamics and Control*, 33:538–553

Dragomirescu-Gaina, C. and L. Elia (2021), "Technology shocks and sectoral labour market spill-overs", *Economics Letters*, 201:109784

Fernald, J. G. (2014), "A quarterly, utilization-adjusted series on Total Factor Productivity", *Federal Reserve Bank of San Francisco Working Paper 2012-19*

Ferrari, A. (2024), "Inventories, demand shocks propagation and amplification in supply chains", *CEPR Discussion Paper n. 19716*

Gabaix, X. (2011), "The granular origins of aggregate fluctuations", *Econometrica*, 79(3):733–772

Ghassibe, M. (2021), "Monetary policy and production networks: an empirical investigation", *Journal of Monetary Economics*, 119:21–39

— (2024), "Endogenous production networks and non-linear monetary transmission", *BSE Working Paper n. 1449*

Ghassibe, M. and A. Nakov (2025), "Business cycles with pricing cascades", *CEPR Discussion Paper n. 20605*

Grassi, B. (2017), "IO in I-O: size, industrial organization, and the Input-Output network make a firm structurally important", *IGIER Working Paper n. 619*

Grassi, B. and J. Sauvagnat (2019), "Production networks and economic policy", *Oxford Review of Economic Policy*, 35(4):638–677

Gualdi, S. and A. Mandel (2019), "Endogenous growth in production networks", *Journal of Evolutionary Economics*, 29:91–117

Hausmann, R. and C. A. Hidalgo (2011), "The network structure of economic output", *Journal of Economic Growth*, 16:309–342

Hornstein, A. and J. Praschnik (1997), "Intermediate inputs and sectoral comovement in the business cycle", *Journal of Monetary Economics*, 40(3):573–595

Huffman, G. W. and M. A. Wynne (1999), "The role of intratemporal adjustment costs in a multisector economy", *Journal of Monetary Economics*, 43(2):317–350

Hulten, C. R. (1978), "Growth accounting with intermediate inputs", *Review of Economic Studies*, 45(3):511–518

Huo, Z., A. A. Levchenko, and N. Pandalai-Nayar (2025), "International comovement in the global production network", *Review of Economic Studies*, 92(1):365–403

Huremović, K., G. Jiménez, E. Moral-Benito, J.-L. Peydró, and F. Vega-Redondo (2026), "Production and financial networks in interplay", *American Economic Review*, 116(5):1611–1647

Huremović, K., F. Nutarelli, F. Serti, and F. Vega-Redondo (2025), "Learning to import through production networks", *Working Paper*

Jones, C. I. (2011), "Intermediate goods and weak links in the theory of economic development", *American Economic Journal: Macroeconomics*, 3(2):1–28

Kalemli-Özcan, S., C. Soylu, and M. A. Yildirim (2025), "Global networks, monetary policy and trade", *NBER Working Paper n. 33686*

Kim, B.-G. (2020), "Sectoral shifts and comovements in employment", *Economics Letters*, 192:109208

La'O, J. and A. Tahbaz-Salehi (2022), "Optimal monetary policy in production networks", *Econometrica*, 90(3):1295–1336

Lane, N. (2025), "Manufacturing revolutions: industrial policy and industrialization in South Korea", *Quarterly Journal of Economics*, 140(3):1683–1741

Leontief, W. W. (1936), "Quantitative input and output relations in the economic systems of the United States", *Review of Economics and Statistics*, 18(3):105–125

— (1986), *Input-Output Economics*. New York: Oxford University Press

Liu, E. (2019), "Industrial policies in production networks", *Quarterly Journal of Economics*, 134(4):1883–1948

Liu, E. and A. Tsyvinski (2024), "A dynamic model of input-output networks", *Review of Economic Studies*, 91(6):3608–3644

Long, J. B. and C. I. Plosser (1983), "Real business cycles", *Journal of Political Economy*, 91(1):39–69

Lucas, R. E. (1977), "Understanding business cycles", *Carnegie-Rochester Conference Series on Public Policy*, 5:7–29

McNerney, J., C. Savoie, F. Caravelli, V. M. Carvalho, and D. J. Farmer (2022), "How production networks amplify economic growth", *Proceedings of the National Academy of Sciences (PNAS)*, 119(1):e2106031118

Miller, R. E. and P. D. Blair (2022), *Input-Output Analysis: Foundations and Extensions*. 3rd edition, Cambridge, UK: Cambridge University Press

Oberfield, E. (2018), "A theory of Input-Output architecture", *Econometrica*, 86(2):559–589

Pasten, E., R. Schoenle, and M. Weber (2020), "The propagation of monetary policy shocks in a heterogeneous production economy", *Journal of Monetary Economics*, 116:1–22

Qiu, Z., Y. W. Wang, L. Xu, and F. Zanetti (2026), "Monetary policy in open economies with production networks", *Journal of Monetary Economics*, 159:103918

Rebelo, S. (2005), "Real Business Cycle models: past, present and future", *The Scandinavian Journal of Economics*, 107(2):217–238

Rubbo, E. (2023), "Networks, Phillips curves, and monetary policy", *Econometrica*, 91(4):1417–1455

Sandqvist, A. P. (2017), "Dynamics of sectoral business cycle comovement", *Applied Economics*, 49(47):4742–4759

Shea, J. (2002), "Complementarities and comovements", *Journal of Money, Credit and Banking*, 34(2):412–433

Sraffa, P. (1960), *Production of Commodities by Means of Commodities: Prelude to a Critique of Economic Theory*. London: Cambridge University Press

Stock, J. H. and M. W. Watson (1999), "Business cycle fluctuations in U.S. macroeconomic time series". IN: *Handbook of Macroeconomics*, vol. 1, pt. A, pp. 3–64, edited by Taylor, J. B. and M. Woodford, North-Holland: Elsevier.

Taschereau-Dumouchel, M. (2026), "Cascades and fluctuations in an economy with an endogenous production network", *Review of Economic Studies*, 93(2):1354–1392

vom Lehn, C. and T. Winberry (2022), "The investment network, sectoral comovement, and the changing U.S. business cycle", *Quarterly Journal of Economics*, 137(1):387–433

Yedid-Levi, Y. (2016), "Why does employment in all major sectors move together over the business cycle?", *Review of Economic Dynamics*, 22:131–156

SUPPLEMENTAL APPENDIX

(Outline) *In this appendix I report all the material complementary to the main text; it is made of additional tables and figures, and of discussions on further analytical results. Section A deepens further the theoretical findings on common-buyer network distances, while Section B extensively discusses the comm-buyer/common-supplier duality of Subsection 3.4. Section C consolidates and further discusses the data presented in Section 4, with Section D complementing and enriching the exposition of the empirical findings in Section 5. Finally, Section E presents some illustrative and very recent real-world examples for horizontal transmission.*

A. THE HOUSEHOLD AS AN "ADDITIONAL SECTOR"

The theory of Sections 2-3 gives the household Cobb-Douglas preferences, so that its expenditure shares across sectors, $\{\omega_i\}_{i\in\Phi}$, are constant and all horizontal transmission runs through the production network, isolating the supply side from the final demand. Instead, this appendix asks what changes when the household has CES preferences with elasticity $\sigma_0 \neq 1$. The answer is organized around a single observation: final demand becomes one more buyer of size one, with no labour share, and with a budget that does not respond to prices. Recall that, for notational reason, the representative household is referred to with index-0. Every result of the paper then holds with one more term, weighted by $\sigma_0 - 1$, while the terms that describe the supply side keep exactly the form derived and discussed in the main text. The benchmark is the special case $\sigma_0 = 1$, in which the household term is Cobb-Douglas and vanishes. When $\sigma_0 - 1$ has the same sign as the elasticities of the production network, the household reinforces the common-buyer results; when it has the opposite sign, the two terms may offset each other, and the results of the paper describe the common-buyer component of the response.

A.1. The household as a buyer

The representative household supplies N units of labour at wage w, with $d\log N = 0$ when labour is supplied inelastically, and consumes

$$C = \left(\sum_{i\in\Phi} \gamma_{0i}^{1/\sigma_0} c_i^{(\sigma_0-1)/\sigma_0}\right)^{\sigma_0/(\sigma_0-1)} \qquad \text{at} \qquad P = \left(\sum_{i\in\Phi} \gamma_{0i}\, p_i^{1-\sigma_0}\right)^{1/(1-\sigma_0)}$$

where P is the consumer price index and real GDP is $Y = \frac{wN}{P}$. Expenditure shares, $\omega_i = \frac{p_i c_i}{wN} = \gamma_{0i}\frac{p_i^{1-\sigma_0}}{P^{1-\sigma_0}}$, are no longer constant, and respond to prices as

$$d\log\omega_i = (1-\sigma_0)\left(d\log p_i - \sum_{j\in\Phi}\omega_j\, d\log p_j\right) \tag{A.1}$$

when a good becomes cheaper relative to the others, its share of final expenditure rises if goods are substitutes in consumption ($\sigma_0 > 1$) and falls if they are complements ($\sigma_0 < 1$).

***Final demand as buyer*-0.–** Let $\Phi_0 = \{0\} \cup \Phi$, and give final demand the attributes of a buyer: size $\lambda_0 = 1$, since its budget is GDP; labour share $\alpha_0 = 0$, since it spends all of its budget on final goods; portfolio $\widehat{\boldsymbol{a}}_0^{row} = \boldsymbol{\omega}'$, that is, $\widehat{\alpha}_{0i} = \omega_i$; and demand elasticity $\eta_0 = 1$, since its nominal budget, wN, does not respond to prices. With this convention, market clearing in value includes final demand as one of the buyers of every good,

$$\lambda_i = \sum_{s\in\Phi_0} \lambda_s\,(1-\alpha_s)\,\widehat{\alpha}_{si} \qquad \text{and, summing over } i, \qquad \sum_{s\in\Phi_0} \lambda_s\,(1-\alpha_s) = \sum_{i\in\Phi} \lambda_i$$

Total downstream buyers' budget, household included, equals total sales from upstream. Let $\boldsymbol{\mathcal{C}}_s = \text{Diag}\left[\widehat{\boldsymbol{a}}_s^{row}\right] - \left(\widehat{\boldsymbol{a}}_s^{row}\right)'\widehat{\boldsymbol{a}}_s^{row}$ be the $m \times m$ covariance matrix of buyer-s's portfolio, so that $\text{Var}_{\widehat{\boldsymbol{a}}_s^{row}}[\boldsymbol{u}] = \boldsymbol{u}'\boldsymbol{\mathcal{C}}_s\,\boldsymbol{u}$ and $\text{Cov}_{\widehat{\boldsymbol{a}}_s^{row}}[\boldsymbol{u},\boldsymbol{v}] = \boldsymbol{u}'\boldsymbol{\mathcal{C}}_s\boldsymbol{v}$, and let $\boldsymbol{\ell}_i^{col}$ be the i-th column of the Leontief inverse $\boldsymbol{\mathcal{L}}$. For the household, $\boldsymbol{\mathcal{C}}_0 = \text{Diag}[\boldsymbol{\omega}] - \boldsymbol{\omega}\boldsymbol{\omega}'$.

Three properties carry over unchanged, because they depend on technology and on the household's budget, not on its preferences. First, prices: unit costs depend only on technology and on the wage, so that $d\log\boldsymbol{p} = -\boldsymbol{\mathcal{L}}\,d\log\boldsymbol{z}$. Second, the Hulten's (1978) theorem: by Shephard's lemma applied to P, $d\log P = \sum_{i\in\Phi}\omega_i\,d\log p_i$, so that $d\log Y = -\boldsymbol{\omega}'d\log\boldsymbol{p} + d\log N = \boldsymbol{\lambda}'d\log\boldsymbol{z} + d\log N$, with $\boldsymbol{\lambda}' = \boldsymbol{\omega}'\boldsymbol{\mathcal{L}}$, at every equilibrium. Third, employment: labour shares are constant, so that $n_i = \alpha_i\lambda_i N$ and $\sum_{i\in\Phi}\alpha_i\lambda_i = 1$. The household changes how Domar weights move, not how prices move: this is why every result below differs from the benchmark only through the response of Domar weights.

LEMMA A.1 (Leontief columns and buyers' dispersion) *Considering the final demand as an additional common downstream buyer:*

- *for every i and every $j \neq i$, then $\ell_{ji} \leq (1-\alpha_j)\,\ell_{ii}$;*
- *for every buyer $s \in \Phi_0$ and good $i \in \Phi$, $\text{Var}_{\widehat{\boldsymbol{a}}_s^{row}}\left[\boldsymbol{\ell}_i^{col}\right] \geq \underline{\alpha}^2\,\widehat{\alpha}_{si}(1-\widehat{\alpha}_{si})$, where $\underline{\alpha} = \min_i \alpha_i > 0$; for the household, moreover, $\sum_{i\in\Phi}\text{Var}_{\boldsymbol{\omega}}\left[\boldsymbol{\ell}_i^{col}\right] \geq 1 - \|\boldsymbol{\lambda}\|^2$.*

Proof: for the first claim, and for $j \neq i$, column-i of $\boldsymbol{\mathcal{L}} = \boldsymbol{I} + \boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{L}}$ gives $\ell_{ji} = \sum_{m\in\Phi}\alpha_{jm}\ell_{mi} \leq (1-\alpha_j)\max_m \ell_{mi} < \max_m \ell_{mi}$, since $\max_m \ell_{mi} \geq \ell_{ii} \geq 1$. The maximum of column-$i$ is therefore attained only at ℓ_{ii}, and $\ell_{ji} \leq (1-\alpha_j)\ell_{ii}$. For the second claim, draw one of buyer-s's inputs, final good-j with probability $\widehat{\alpha}_{sj}$, and record ℓ_{ji}. Splitting the draws into final good-i, with probability $\widehat{\alpha}_{si}$, and the other final goods, the law of total variance bounds the variance from below by the dispersion between the two group means, $\widehat{\alpha}_{si}(1-\widehat{\alpha}_{si})\left(\ell_{ii} - \sum_{j\neq i}\widehat{\alpha}_{sj}\ell_{ji}/\left(1-\widehat{\alpha}_{si}\right)\right)^2$; by the first claim, the second mean is at most $(1-\underline{\alpha})\,\ell_{ii}$, so that the gap is at least $\underline{\alpha}\,\ell_{ii} \geq \underline{\alpha}$. For the household, $\sum_{j\in\Phi_0}\omega_j\ell_{ji} = \lambda_i$, so that $\text{Var}_{\boldsymbol{\omega}}\left[\boldsymbol{\ell}_i^{col}\right] = \sum_{j\in\Phi_0}\omega_j\ell_{ji}^2 - \lambda_i^2 \geq \omega_i\ell_{ii}^2 - \lambda_i^2 \geq \omega_i - \lambda_i^2$; summing over i gives the result. ■

A.2. Local transmission at final demand

At the shared node, the household enters Propositions 1 and 2 in two ways: as a common buyer of the final goods it consumes, and as one of the buyers of a congested supplier.

PROPOSITION A.1 (Final demand as a shared node) *At first round, the household's spending on good-i responds to the price of good-j as*

$$\delta_{\to 0}[i,j] = \frac{\partial\log\left[p_i\,c_i\right]}{\partial\log p_j} = (\sigma_0 - 1)\,\omega_j$$

and the response of i's total sales in Proposition 2 gains the term $(\sigma_0-1)\,\omega_i\,\frac{\omega_j}{\lambda_i}$, that is, the sum extends to all common buyers $s \in \Phi_0 \setminus \{i,j\}$. Proposition 1 holds with the pass-through

$$\widetilde{\varphi}_s = \frac{\varphi_s}{1 + \varphi_s\left(e_{s0}\,\chi_{0s} + \sum_{m\in\Phi} e_{sm}\,\chi_{ms}\right)} \qquad \textit{with} \quad \chi_{0s} = \sigma_0\,(1-\omega_s) + \omega_s$$

Proof: the household's budget, wN, is given, so that $d\log[p_i c_i] = d\log\omega_i$; by eq. (A.1), with only p_j moving, $d\log\omega_i = (1-\sigma_0)\,(0 - \omega_j\,d\log p_j) = (\sigma_0-1)\omega_j\,d\log p_j$. Final demand accounts for a share $\frac{\omega_i}{\lambda_i}$ of i's sales, which gives the term in

$\frac{\partial \log p_i y_i}{\partial \log p_j}$; it coincides with the summand of Proposition 2 at $s = 0$, since $\lambda_0 = 1$, $\alpha_0 = 0$, $\eta_0 = 1$ and $(1-\alpha_0)\,\widehat{\alpha}_{0i} = \omega_i$. This proves the first part. Then, by eq. (A.1), with only p_s moving, $d\log c_s = d\log\omega_s - d\log p_s = \left[(1-\sigma_0)(1-\omega_s) - 1\right] d\log p_s = -\chi_{0s}\, d\log p_s$, which replaces the object $-d\log p_s$ in Step 4 of the proof of Proposition 1: market clearing now gives $d\log y_s = -e_{s0}\chi_{0s}\, d\log p_s + \sum_{m\in\Phi} e_{sm}\, d\log x_{ms}$, and the remaining steps are unchanged. ■

The household is the cleanest common buyer of the economy. Its budget is fixed, so that the vertical term of Proposition 2, $(1-\alpha_s)(1-\eta_s)$, is identically zero, and its transmission is purely horizontal: when good-j becomes cheaper, the household reallocates a given budget, and the spending on every other good moves by $(\sigma_0 - 1)\,\omega_j\, d\log p_j$, with a sign set by σ_0 alone. This is the horizontal channel of standard multi-sector models without intermediate inputs, as in the "half-spider" of Figure 2. At a common supplier, the household adds one more margin of substitution away from a congested good, χ_{0s}, which equals one in the benchmark: it changes the size of the price pass-through, not the vertical nature of the channel.

A.3. General equilibrium

The equilibrium reallocation across the whole network (Theorem 1) changes accordingly.

THEOREM A.1 (Factorization with a CES household) *To first order, Domar weights respond to sectoral supply shocks as*

$$d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'\left(\boldsymbol{\mathcal{H}} + \boldsymbol{\mathcal{H}}_0\right)\boldsymbol{\mathcal{L}}\, d\log \mathbf{z} = \left(\boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0\right) d\log \mathbf{z}$$

with $\boldsymbol{\mathcal{H}}_0 = (\sigma_0 - 1)\left[\mathrm{Diag}\left[\boldsymbol{\omega}\right] - \boldsymbol{\omega}\boldsymbol{\omega}'\right]$, *where* $\boldsymbol{\mathcal{H}}$ *and* $\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$ *are the operators of Theorem 1, and* $\boldsymbol{\mathcal{R}}_0 = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}_0\boldsymbol{\mathcal{L}}$. *Equivalently,*

$$\frac{\partial \lambda_i}{\partial \log z_j} = \sum_{s\in\Phi} (\sigma_s - 1)\,\lambda_s\,(1-\alpha_s)\,\mathrm{Cov}_{\widehat{\mathbf{a}}_s^{row}}\left[\boldsymbol{\ell}_i^{col}, \boldsymbol{\ell}_j^{col}\right] + (\sigma_0 - 1)\,\mathrm{Cov}_{\boldsymbol{\omega}}\left[\boldsymbol{\ell}_i^{col}, \boldsymbol{\ell}_j^{col}\right]$$

where the last term is the household-based covariance.

Proof: let me derive each step separately. *Step 1*. Differentiating $\boldsymbol{\lambda}' = \boldsymbol{\omega}' + \boldsymbol{\lambda}'\boldsymbol{\mathcal{A}}$, $d\boldsymbol{\lambda}' = d\boldsymbol{\omega}' + \boldsymbol{\lambda}' d\boldsymbol{\mathcal{A}} + d\boldsymbol{\lambda}'\boldsymbol{\mathcal{A}}$: Domar weights move because the household reallocates its budget, because buyers reallocate their intermediate budgets, and because the size of each buyer changes. *Step 2*. By eq. (A.1), $d\omega_i = (1-\sigma_0)\,\omega_i\left(d\log p_i - \sum_j \omega_j\, d\log p_j\right)$, that is, $d\boldsymbol{\omega} = (1-\sigma_0)\,\boldsymbol{\mathcal{C}}_0\, d\log \boldsymbol{p}$. *Step 3*. Under CES minimization, $d\alpha_{si} = (1-\alpha_s)\, d\widehat{\alpha}_{si} = (1-\alpha_s)(1-\sigma_s)\,\widehat{\alpha}_{si}\left(d\log p_i - \sum_j \widehat{\alpha}_{sj}\, d\log p_j\right)$, so that the i-th entry of $\boldsymbol{\lambda}' d\boldsymbol{\mathcal{A}}$ is $\sum_{s\in\Phi}(1-\sigma_s)\,\lambda_s\,(1-\alpha_s)\left[\boldsymbol{\mathcal{C}}_s\, d\log \boldsymbol{p}\right]_i$. Since $\lambda_0(1-\alpha_0) = 1$, Steps 2 and 3 combine into a single sum over $s \in \Phi_0$. *Step 4*. Solving Step 1 for $d\boldsymbol{\lambda}$, $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'\sum_{s\in\Phi_0}(1-\sigma_s)\,\lambda_s\,(1-\alpha_s)\,\boldsymbol{\mathcal{C}}_s\, d\log \boldsymbol{p}$. *Step 5*. Substituting $d\log \boldsymbol{p} = -\boldsymbol{\mathcal{L}}\, d\log \mathbf{z}$ gives $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'\left(\boldsymbol{\mathcal{H}} + \boldsymbol{\mathcal{H}}_0\right)\boldsymbol{\mathcal{L}}\, d\log \mathbf{z}$, where the terms $s \in \Phi$ form $\boldsymbol{\mathcal{H}}$ and the term $s = 0$ is $\boldsymbol{\mathcal{H}}_0$. The covariance form follows from $\boldsymbol{\mathcal{L}}\boldsymbol{e}_i = \boldsymbol{\ell}_i^{col}$ and $\boldsymbol{u}'\boldsymbol{\mathcal{C}}_s\boldsymbol{v} = \mathrm{Cov}_{\widehat{\mathbf{a}}_s^{row}}[\boldsymbol{u}, \boldsymbol{v}]$. ■

The theorem is the sense in which the household extends the results without changing them. The common-buyer reallocation operator, $\boldsymbol{\mathcal{R}}$, has the form derived in the text, and the household adds one more buyer to the sum that defines it: the largest buyer of the economy, of size one, whose portfolio is final demand. The household term inherits the logic of the factorization. Read from right to left, $\boldsymbol{\mathcal{L}}$ transports a shock to the price of every good, $\boldsymbol{\mathcal{H}}_0$ reallocates final expenditure according to how unevenly these prices have moved, and $\boldsymbol{\mathcal{L}}'$ transports the change in final demand for each good up to its suppliers. Two features distinguish it from the common-buyer term. First, its rivalry component is rank one: since final spending embodies sector-i by $\sum_{j\in\Phi_0}\omega_j\ell_{ji} = \lambda_i$, then

$$\boldsymbol{\mathcal{R}}_0 = (\sigma_0 - 1)\left(\boldsymbol{\mathcal{L}}'\,\mathrm{Diag}\left[\boldsymbol{\omega}\right]\boldsymbol{\mathcal{L}} - \boldsymbol{\lambda}\boldsymbol{\lambda}'\right)$$

so that the household makes every pair of sectors rivals (complements if $\sigma_0 < 1$) in proportion to the product of their Domar weights, including sectors that never sell to final demand

and compete only through the consumer goods that embody them. The common-buyer rivalry, $-\mathcal{T}'\mathcal{W}\mathcal{T}$, is instead shaped by which buyers require which goods, and it is this term that carries the horizontal geometry of the production network. Correspondingly, the unweighted counterpart of $\mathcal{H}+\mathcal{H}_0$ is $\mathcal{P}^{cb}+\boldsymbol{\omega}\boldsymbol{\omega}'$. Second, the household term is active without any production network: with $\mathcal{A}=\mathbf{0}$, $d\boldsymbol{\lambda}=\mathcal{H}_0\,d\log\boldsymbol{z}$, the reallocation of standard multi-sector models; and with Cobb-Douglas intermediates, $d\boldsymbol{\lambda}=\mathcal{R}_0\,d\log\boldsymbol{z}$, so that upstream sectors inherit, through $\mathcal{L}'$, the reallocation of the household among the consumer goods they supply.

COROLLARY A.1 (Signs and proliferation with a CES household) *Considering the final demand as an additional common downstream buyer:*

- *under Cobb-Douglas intermediates and preferences ($\sigma_s=1$, $\forall s\in\Phi_0$), $\mathcal{H}+\mathcal{H}_0=\mathbf{0}$; with only CES final demand, hence under Cobb-Douglas intermediates alone, $\mathcal{H}+\mathcal{H}_0=\mathcal{H}_0$, which is non-zero whenever $\sigma_0\neq 1$ and the household buys at least two final goods;*
- *if $\sigma_s\geq 1$ ($\sigma_s\leq 1$), $\forall s\in\Phi_0$, then $\mathcal{H}+\mathcal{H}_0$ and $\mathcal{R}+\mathcal{R}_0$ are positive (negative) semi-definite, and all off-diagonal entries of $\mathcal{H}+\mathcal{H}_0$ are non-positive (non-negative);*
- *if, in addition, $\sigma_s\neq 1$ and $\lambda_s>0$ for all $s\in\Phi_0$, then, for $i\neq j$, $\mathcal{H}_{i,j}+[\mathcal{H}_0]_{ij}\neq 0$ if and only if $\mathcal{P}^{cb}[i,j]+\omega_i\omega_j>0$, and the rivalry generated by the household, relative to its purchases, is $|\sigma_0-1|\left(1-\|\boldsymbol{\omega}\|^2\right)$.*

Proof: for the first claim, note that every term of $\sum_{s\in\Phi_0}$ carries σ_s-1; if only $\sigma_0\neq 1$, the sum reduces to $\mathcal{H}_0$, whose off-diagonal entries, $-(\sigma_0-1)\,\omega_i\omega_j$, are non-zero for any two goods with positive final demand. For the second claim, each $\mathcal{C}_s$ is a covariance matrix, hence positive semi-definite, and it enters with a weight of the sign of σ_s-1; the quadratic form of $\mathcal{R}+\mathcal{R}_0$ at $\boldsymbol{x}$ is that of $\mathcal{H}+\mathcal{H}_0$ at $\mathcal{L}\boldsymbol{x}$, and has the same sign; the off-diagonal entries of $\mathcal{C}_s$ are $-\widehat{\alpha}_{si}\widehat{\alpha}_{sj}\leq 0$. In the third claim, for $i\neq j$, then $\mathcal{H}_{i,j}+[\mathcal{H}_0]_{ij}=-\sum_{s\in\Phi_0}(\sigma_s-1)\,\lambda_s\,(1-\alpha_s)\,\widehat{\alpha}_{si}\widehat{\alpha}_{sj}$: a sum of terms of the same sign, non-zero if and only if some buyer (household included) purchases both goods, that is, if and only if $\sum_{s\in\Phi}\alpha_{si}\alpha_{sj}+\omega_i\omega_j>0$. The off-diagonal entries of $\mathcal{C}_0$ sum in absolute value to $\sum_i\omega_i\,(1-\omega_i)=1-\|\boldsymbol{\omega}\|^2$, and the household's purchases are $\lambda_0\,(1-\alpha_0)=1$. ■

The sign results of Corollary 1 extend once the household's elasticity is included in the sign conditions. Without that condition, the common-buyer statements continue to describe $\mathcal{R}$, and $\mathcal{R}_0$ is the household's contribution: with complementary intermediates and substitutable consumption goods, for instance, the network makes a sector's rivals at common buyers gain from its gain, while the household makes them lose. The last claim adds the household to Corollary 3. The household is a buyer with an exceptionally large portfolio: it supports a horizontal relation between every pair of goods with positive final demand, and its rivalry, $|\sigma_0-1|\left(1-\|\boldsymbol{\omega}\|^2\right)$, is close to its maximum when final expenditure is spread over many goods. A new consumer good becomes, at once, a rival of all the others.

A.4. Employment, aggregate output and volatility

Changes in final demand also reallocate workers upstream.

PROPOSITION A.2 (Employment reallocation with a CES household) *To first order, sectoral employment responds to supply shocks as*

$$d\log\boldsymbol{n}=\underbrace{d\log N\,\mathbf{1}}_{\textit{common}}+\underbrace{\boldsymbol{\Lambda}^{-1}\mathcal{R}\,d\log\boldsymbol{z}}_{\textit{production network}}+\underbrace{\boldsymbol{\Lambda}^{-1}\mathcal{R}_0\,d\log\boldsymbol{z}}_{\textit{household}}$$

with $\boldsymbol{\Lambda}=\mathrm{Diag}\,[\boldsymbol{\lambda}]$ and $\boldsymbol{\alpha}'\left(\mathcal{R}+\mathcal{R}_0\right)=\mathbf{0}$.

Proof: by the third property of Subsection A.1, $d\log n_i=d\log N+\frac{d\lambda_i}{\lambda_i}$, and $d\boldsymbol{\lambda}$ is given by Theorem A.1. Since $\boldsymbol{\alpha}'\boldsymbol{\lambda}=1$ at every equilibrium and $\boldsymbol{\alpha}$ is constant, $\boldsymbol{\alpha}'d\boldsymbol{\lambda}=0$ for every shock. ■

Reallocation remains zero-sum, and the household adds a component that does not require intermediate inputs (note that the production network component decomposes as the one in Proposition 3). In the half-spider of Figure 2a, where $\boldsymbol{\mathcal{A}} = \mathbf{0}$ and $\alpha_i = 1$, it holds $\boldsymbol{\lambda} = \boldsymbol{\omega}$ and the proposition reduces to $d\log n_i - d\log N = (\sigma_0 - 1)\left(d\log z_i - \sum_j \omega_j\, d\log z_j\right)$: workers move towards sectors whose productivity rises more than average if final goods are substitutes in consumption, and away from them if they are complements.

THEOREM A.2 (Aggregate output with a CES household) *Let $\boldsymbol{\varepsilon} = d\log \mathbf{z}$, with independent shocks of mean zero and variance v^2, and hold labour fixed. Then:*

- *to first order, $d\log Y = \boldsymbol{\lambda}'\boldsymbol{\varepsilon}$, so that $\mathrm{Var}\left[d\log Y\right] = v^2\|\boldsymbol{\lambda}\|^2$, as in Theorem 2;*
- *to second order, $d^2\log Y = \boldsymbol{\varepsilon}'\left(\boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0\right)\boldsymbol{\varepsilon} = \sum_{s\in\Phi}(\sigma_s - 1)\,\lambda_s\,(1-\alpha_s)\,\mathrm{Var}_{\widehat{\boldsymbol{a}}_s^{row}}\left[\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon}\right] + (\sigma_0 - 1)\,\mathrm{Var}_{\boldsymbol{\omega}}\left[\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon}\right]$;*
- *consequently, it holds $\mathbb{E}\left[\log Y\right] - \log Y^{det} = \frac{1}{2}v^2\,\mathrm{Trace}\left[\boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0\right] + o\left[v^2\right]$, where the household term has the sign of $\sigma_0 - 1$ and is bounded away from zero by*

$$\left|\,\mathrm{Trace}\left[\boldsymbol{\mathcal{R}}_0\right]\,\right| = |\sigma_0 - 1|\sum_{i\in\Phi}\mathrm{Var}_{\boldsymbol{\omega}}\left[\boldsymbol{\ell}_i^{col}\right] \geq |\sigma_0 - 1|\max\left\{1 - \|\boldsymbol{\lambda}\|^2,\ \underline{\alpha}^2\left(1 - \|\boldsymbol{\omega}\|^2\right)\right\}$$

Proof: the first claim follows from the second property of Subsection A.1. Since shocks are independent, have mean zero, and variance v^2, $\mathrm{Var}\left[d\log Y\right] = v^2\|\boldsymbol{\lambda}\|^2$. For the second claim, the same property holds at every equilibrium, so the Hessian of $\log Y$ with respect to log productivities is $\frac{\partial\boldsymbol{\lambda}}{\partial\log\mathbf{z}'} = \boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0 = \boldsymbol{\mathcal{L}}'\left(\boldsymbol{\mathcal{H}} + \boldsymbol{\mathcal{H}}_0\right)\boldsymbol{\mathcal{L}}$, by Theorem A.1. Here, $\boldsymbol{\mathcal{R}}$ captures firms' production curvature, while $\boldsymbol{\mathcal{R}}_0$ captures the additional curvature induced by the CES household. Expanding the quadratic form buyer-by-buyer gives the stated expression in terms of weighted variances, where $\widehat{\boldsymbol{a}}_s^{row}$ denotes sector-s's normalized input shares and $\boldsymbol{\omega}$ the household's expenditure shares. The corresponding elasticities of substitution are σ_s for sectoral buyers and σ_0 for the household. The household term is the curvature of the CES consumer price index. Differentiating Shephard's lemma once more gives $d^2\log P = \sum_{i\in\Phi}\omega_i d^2\log p_i + (1-\sigma_0)\,\mathrm{Var}_{\boldsymbol{\omega}}\left[d\log\boldsymbol{p}\right]$, where P is the consumer price index and p_i is the price of sector-i. Since $d\log\boldsymbol{p} = -\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon}$, this yields the household variance term in the theorem. For the third claim, the Taylor expansion in the proof of Theorem 2 applies with $\boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0$, giving $\mathbb{E}\left[\log Y\right] - \log Y^{det} = \frac{1}{2}v^2\,\mathrm{Trace}\left[\boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0\right] + o\left[v^2\right]$. Moreover, $\mathrm{Trace}\left[\boldsymbol{\mathcal{R}}_0\right] = \sum_{i\in\Phi}\left(\boldsymbol{\ell}_i^{col}\right)'\boldsymbol{\mathcal{H}}_0\boldsymbol{\ell}_i^{col} = (\sigma_0 - 1)\sum_{i\in\Phi}\mathrm{Var}_{\boldsymbol{\omega}}\left[\boldsymbol{\ell}_i^{col}\right]$, where $\boldsymbol{\ell}_i^{col}$ is the i-th column of $\boldsymbol{\mathcal{L}}$. Its sign is determined by $\sigma_0 - 1$, and the lower bound follows from Lemma A.1, summed over $i\in\Phi$ and using $\sum_{i\in\Phi}\omega_i = 1$. ■

The first-order results mirror those of the main text (Theorem 2): the household changes neither Hulten's (1978) theorem nor the variance of aggregate output, so that Corollary 2 and the diversification argument for aggregate volatility are unaffected. The household enters the curvature as one more buyer whose input prices move unevenly, with weight one. Two differences from the common-buyer bound are worth noting. First, the factor $1 - \overline{\alpha}$ disappears because the household's budget is the whole of GDP, so that the bound holds even without intermediate inputs. Second, the household term is bounded below by a statistic of Domar weights alone, $1 - \|\boldsymbol{\lambda}\|^2$, which tends to one whenever Domar weights are balanced. When $\sigma_0 - 1$ and the intermediate elasticities have opposite signs, the two components of the mean effect offset each other, and its sign is an empirical matter.

B. COMMON SUPPLIERS AND DEMAND SHOCKS

The theory of Section 3 studies supply shocks in an economy in which every sector produces at constant marginal cost. There, a shock travels downstream through prices and is reallocated horizontally at common buyers, whereas common suppliers transmit only if their marginal cost is not constant (Proposition 1) – so that their geometry plays no role in general equilibrium. This appendix asks what changes when the economy is hit by shocks to final demand, and sectors face rising marginal costs because each of them also uses a factor in fixed supply. The answer is

organized around a duality. A supply shock moves prices vertically and quantities horizontally: the Leontief inverse carries cost changes downstream, and common buyers reallocate their budgets. A demand shock moves quantities vertically and prices horizontally: the transpose of the Leontief inverse carries spending upstream, and congested suppliers raise the costs of all the sectors that require them. The horizontal operator of demand shocks is a common-supplier Gram matrix on total requirements, $\boldsymbol{\mathcal{S}}$, the dual of $\boldsymbol{\mathcal{H}}$, and the two geometries feed into each other whenever both are active. The benchmark of the main text is the special case $\boldsymbol{\mathcal{S}} = \mathbf{0}$.

B.1. Common-supplier benchmark: rising marginal costs and demand shocks

Consider a production network in which each sector-i uses, besides labour and intermediate inputs (Assumption 1), a specific primary factor F_i in fixed supply $F_i = \overline{F}_i$ rented at price r_i:

$$y_i = z_i \, F_i^{\theta_i} \, n_i^{\alpha_i} \, x_i^{1-\alpha_i-\theta_i}$$

with x_i being the Constant Elasticity of Substitution (CES) intermediate bundle of Section 3 under elasticity σ_i, where $1-\alpha_i-\theta_i = \sum_{j\in\Phi}\alpha_{ij}$, and $\theta_i \in [0, 1-\alpha_i)$ is the share of revenue paid to the specific fixed factor. Assumption 1 holds with F_i among the primary inputs: returns are constant in $\{F_i, n_i, x_i\}$, so that every row of $\boldsymbol{\mathcal{A}}$ sums to $1-\alpha_i-\theta_i$, and decreasing in $\{n_i, x_i\}$ alone. Holding production input prices fixed, sector-i's marginal cost rises with its output with elasticity $\varphi_i = \frac{\theta_i}{1-\theta_i}$, which maps into Proposition 1: θ_i measures the *congestion* of sector-i. In this set-up, intermediate shares are therefore $\widehat{\alpha}_{ij} = \frac{\alpha_{ij}}{1-\alpha_i-\theta_i}$, and $\boldsymbol{\mathcal{C}}_s$, $\boldsymbol{\mathcal{H}}$ and $\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$ are those of Theorem 1, with $1-\alpha_s-\theta_s$ in place of $1-\alpha_s$. Note that the general case in Sections 2-3 recovers under a fixed unity of primary factors with no share, $F_i = \overline{F}_i = 1$ and $\theta_i = 0$.

The household supplies $N = \sum_{i\in\Phi} n_i$ units of labour inelastically, receives the rents of the specific primary factors, and spends its income, $E = wN + \sum_{i\in\Phi} r_i\overline{F}_i$, with Cobb-Douglas shares $\boldsymbol{\omega}$. The wage is the numeraire, $w = 1$. Domar weights are then $\lambda_i = \frac{p_i y_i}{E}$, and market clearing still gives $\boldsymbol{\lambda}' = \boldsymbol{\omega}' + \boldsymbol{\lambda}'\boldsymbol{\mathcal{A}} = \boldsymbol{\omega}'\boldsymbol{\mathcal{L}}$. Labour receives a share $\Lambda_N = \boldsymbol{\alpha}'\boldsymbol{\lambda}$ of GDP and the specific factor of sector-i a share $\theta_i\lambda_i$, and, since $(\boldsymbol{\alpha}+\boldsymbol{\theta})'\boldsymbol{\lambda} = \boldsymbol{\lambda}'(\mathbf{1}-\boldsymbol{\mathcal{A}}\mathbf{1}) = \boldsymbol{\omega}'\mathbf{1} = 1$, factor shares sum to one at every equilibrium.

In this economy, a demand shock is a change in the household's expenditure shares, $d\boldsymbol{\omega}$, with $\mathbf{1}'d\boldsymbol{\omega} = 0$. Such shock to sector-$j$, $d\boldsymbol{\omega} = (\boldsymbol{e}_j - \boldsymbol{\omega})\, d\xi_j$ with $d\xi_j > 0$, shifts expenditure towards j at the expense of all commodities, in proportion to their final shares. Its effects sum a component specific to j, driven by $\boldsymbol{e}_j\, d\xi_j$, and a component common to all demand shocks, driven by $-\boldsymbol{\omega}\, d\xi_j$, not dependent on how much j participates in the production network.

Three equilibrium relations.– To first order, the economy is described by three relations, all derived in the proof of Theorem B.1. *First*, a sector's price rises with its own sales, in proportion to its congestion, and the increase is passed downstream:

$$d\log\boldsymbol{p} = \boldsymbol{\mathcal{L}}\Big[-d\log\boldsymbol{z} + \boldsymbol{\Theta}\big(\boldsymbol{\Lambda}^{-1}d\boldsymbol{\lambda} + d\log E\,\mathbf{1}\big)\Big] \tag{B.1}$$

with the $m\times m$ diagonal matrices $\boldsymbol{\Theta} = \mathrm{Diag}\,[\boldsymbol{\theta}]$ and $\boldsymbol{\Lambda} = \mathrm{Diag}\,[\boldsymbol{\lambda}]$. *Second*, nominal GDP (in wage units) moves with total rents, $d\log E = \frac{\boldsymbol{\theta}'d\boldsymbol{\lambda}}{\Lambda_N}$, and sectoral employment responds as $d\log\boldsymbol{n} = \boldsymbol{\Lambda}^{-1}d\boldsymbol{\lambda} + d\log E\;\mathbf{1}$. *Third*, final demand reaches sectors upstream, and price changes are reallocated at common buyers:

$$d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'d\boldsymbol{\omega} - \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\,d\log\boldsymbol{p} \tag{B.2}$$

Combining the first two relations gives $d\log\boldsymbol{p} = \boldsymbol{\mathcal{L}}\big(-d\log\boldsymbol{z} + \boldsymbol{\mathcal{K}}\,d\boldsymbol{\lambda}\big)$, with the $m\times m$ *congestion* matrix $\boldsymbol{\mathcal{K}} = \boldsymbol{\Theta}\boldsymbol{\Lambda}^{-1} + \frac{1}{\Lambda_N}\boldsymbol{\theta}\boldsymbol{\theta}'$, that is a matrix capturing how changes in sectoral

revenues generate marginal-cost pressures through congestion in sector-specific fixed primary factors and aggregate-income effects, which are subsequently propagated downstream through the production network by $\boldsymbol{\mathcal{L}}$.

The common-supplier operator.– Analogously to the common-buyer case, it holds

$$\boldsymbol{\mathcal{S}} = \boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{K}}\boldsymbol{\mathcal{L}}' = \sum_{s\in\Phi} \frac{\theta_s}{\lambda_s}\boldsymbol{\ell}_s^{col}\left(\boldsymbol{\ell}_s^{col}\right)' + \frac{1}{\Lambda_N}\left(\boldsymbol{\mathcal{L}}\boldsymbol{\theta}\right)\left(\boldsymbol{\mathcal{L}}\boldsymbol{\theta}\right)'$$

$$\text{so that}\quad \mathcal{S}_{i,j} = \sum_{s\in\Phi} \ell_{is}\frac{\theta_s}{\lambda_s}\ell_{js} + \frac{1}{\Lambda_N}\left[\boldsymbol{\mathcal{L}}\boldsymbol{\theta}\right]_i\left[\boldsymbol{\mathcal{L}}\boldsymbol{\theta}\right]_j$$

The first term is large when i and j require, directly or indirectly, the same congested upstream supplier-s, each weighted by its congestion relative to its Domar size: $\boldsymbol{\mathcal{S}}$ is a common-supplier Gram matrix on total requirements, whose first-tier, unweighted counterpart is $\boldsymbol{\mathcal{P}}^{cs} = \boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{A}}'$ of Section 2. It is the dual of the common-buyer operator: $\boldsymbol{\mathcal{H}}$ compares sectors as suppliers, through the columns of the Input-Output matrix normalized by buyers' intermediate budgets, while $\boldsymbol{\mathcal{R}}$ through the columns of the Leontief inverse; $\boldsymbol{\mathcal{S}}$ compares them as buyers, through its rows. For $i \neq j$, the first term is both vertical, when one sector requires the other ($s \in \{i,j\}$), and horizontal, when both require a third supplier ($s \notin \{i,j\}$). The second term is indeed aggregate: when total rents rise, nominal demand rises for every commodity relative to the wage, and every cost rises with its fixed-factor content, $\left[\boldsymbol{\mathcal{L}}\boldsymbol{\theta}\right]_i$; it is of order θ^2. Note that $\boldsymbol{\mathcal{S}}$ is symmetric, positive semi-definite and non-negative, and $\boldsymbol{\mathcal{S}} = \mathbf{0}$ if and only if $\boldsymbol{\theta} = \mathbf{0}$.

B.2. General equilibrium: a duality

The outlined common-supplier benchmark is characterised by the following dual nature.

THEOREM B.1 (Supply and demand shocks in general equilibrium) *To first order, provided that all eigenvalues of $\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}}$ exceed -1, Domar weights and prices respond to productivity shocks, $d\log\boldsymbol{z}$, and to demand shocks, $d\boldsymbol{\omega}$, as*

$$d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'\left(\boldsymbol{I} + \boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}}\right)^{-1}\left(d\boldsymbol{\omega} + \boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\,d\log\boldsymbol{z}\right)$$

$$d\log\boldsymbol{p} = -\boldsymbol{\mathcal{L}}\,d\log\boldsymbol{z} + \boldsymbol{\mathcal{S}}\left(\boldsymbol{I} + \boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}}\right)^{-1}\left(d\boldsymbol{\omega} + \boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\,d\log\boldsymbol{z}\right)$$

The eigenvalue condition holds whenever $\sigma_s \geq 1$, $\forall s \in \Phi$.

Proof: let me derive the three relations above, and then combine them. *Step 1: prices*. With a Cobb-Douglas top nest, $d\log p_i = -d\log z_i + \theta_i\,d\log r_i + \alpha_i\,d\log w + \sum_{j\in\Phi}\alpha_{ij}\,d\log p_j$, with $d\log w = 0$. The specific factor is paid $r_i\overline{F}_i = \theta_i\lambda_i E$, so that $d\log r_i = \frac{d\lambda_i}{\lambda_i} + d\log E$. Stacking and inverting the object $\boldsymbol{I} - \boldsymbol{\mathcal{A}}$ gives eq. (B.1). *Step 2: income and employment*. Employment is $n_i = \alpha_i\lambda_i E$, so that $d\log n_i = \frac{d\lambda_i}{\lambda_i} + d\log E$. Since $\sum_{i\in\Phi} n_i = \Lambda_N E = N$ is fixed, and $(\boldsymbol{\alpha} + \boldsymbol{\theta})'\,d\boldsymbol{\lambda} = 0$, then $d\log E = -\frac{\boldsymbol{\alpha}'d\boldsymbol{\lambda}}{\Lambda_N} = \frac{\boldsymbol{\theta}'d\boldsymbol{\lambda}}{\Lambda_N}$. Substituting in eq. (B.1), $\boldsymbol{\Theta}\mathbf{1}\,d\log E = \frac{1}{\Lambda_N}\boldsymbol{\theta}\boldsymbol{\theta}'d\boldsymbol{\lambda}$, so that $d\log\boldsymbol{p} = \boldsymbol{\mathcal{L}}\left(-d\log\boldsymbol{z} + \boldsymbol{\mathcal{K}}\,d\boldsymbol{\lambda}\right)$. *Step 3: Domar weights*. Differentiating market clearing, $d\boldsymbol{\lambda}' = \left(d\boldsymbol{\omega}' + \boldsymbol{\lambda}'d\boldsymbol{\mathcal{A}}\right)\boldsymbol{\mathcal{L}}$. With $\alpha_{si} = (1-\alpha_s-\theta_s)\,\widehat{\alpha}_{si}$ and $d\widehat{\alpha}_{si} = (1-\sigma_s)\,\widehat{\alpha}_{si}\left(d\log p_i - \sum_{m\in\Phi}\widehat{\alpha}_{sm}\,d\log p_m\right)$, as in the proof of Theorem 1, $\boldsymbol{\lambda}'d\boldsymbol{\mathcal{A}} = -\left(\boldsymbol{\mathcal{H}}\,d\log\boldsymbol{p}\right)'$. Transposing gives eq. (B.2). *Step 4: fixed point*. Substituting Step 2 into eq. (B.2), $\left(\boldsymbol{I} + \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{K}}\right)d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'\left(d\boldsymbol{\omega} + \boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\,d\log\boldsymbol{z}\right)$. By the push-through identity, $\left(\boldsymbol{I} + \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{Q}}\right)^{-1}\boldsymbol{\mathcal{L}}' = \boldsymbol{\mathcal{L}}'\left(\boldsymbol{I} + \boldsymbol{\mathcal{Q}}\boldsymbol{\mathcal{L}}'\right)^{-1}$ with $\boldsymbol{\mathcal{Q}} = \boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{K}}$ and $\boldsymbol{\mathcal{Q}}\boldsymbol{\mathcal{L}}' = \boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}}$, which gives the first equation of the theorem. Moreover, $\boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{Q}}$, $\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}}$ and the symmetric $\boldsymbol{\mathcal{S}}^{1/2}\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}}^{1/2}$ share their non-zero eigenvalues, which are therefore real: the inverses exist under the eigenvalue condition, which holds whenever $\boldsymbol{\mathcal{H}}$ is positive semi-definite, that is whenever $\sigma_s \geq 1$, $\forall s \in \Phi$ (Corollary 1). Finally, $d\log\boldsymbol{p} = -\boldsymbol{\mathcal{L}}\,d\log\boldsymbol{z} + \boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{K}}\,d\boldsymbol{\lambda}$ and $\boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{K}}\boldsymbol{\mathcal{L}}' = \boldsymbol{\mathcal{S}}$ give the second equation. ■

The vector $d\boldsymbol{\omega} + \boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\,d\log\boldsymbol{z}$ collects the direct spending impulses of the shocks: the change in final expenditure, and the reallocation of intermediate budgets at common buyers triggered by productivity-driven cost changes. Without congestion $\boldsymbol{\mathcal{K}}$, the Leontief inverse operator $\boldsymbol{\mathcal{L}}'$ carries these impulses upstream, and $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'d\boldsymbol{\omega} + \boldsymbol{\mathcal{R}}\,d\log\boldsymbol{z}$. With congestion $\boldsymbol{\mathcal{K}}$, every

impulse raises the sales and prices of some suppliers; $\mathcal{S}$ spreads these price increases to all the sectors that require the same suppliers, and $\mathcal{H}$ turns them into new spending impulses at common buyers. Since $\left(I+\mathcal{H}\mathcal{S}\right)^{-1}=\sum_{k=0}^{\infty}\left(-\mathcal{H}\mathcal{S}\right)^{k}$ whenever the series converges, round-k of the adjustment alternates the two geometries k times: common suppliers carry prices, and common buyers carry quantities.

COROLLARY B.1 (Duality) *Given the coexistence of idiosyncratic demand and supply shocks:*

- *under constant marginal costs ($\theta=0$), then $\mathcal{S}=0$, $d\lambda=\mathcal{L}'d\omega+\mathcal{R}\,d\log z$ and $d\log p=-\mathcal{L}\,d\log z$: demand shocks travel only upstream and leave prices unchanged, and all results of Section 3 hold;*
- *under Cobb-Douglas ($\sigma_s=1,\ \forall s\in\Phi$), then $\mathcal{H}=0$, $d\lambda=\mathcal{L}'d\omega$ and $d\log p=-\mathcal{L}\,d\log z+\mathcal{S}\,d\omega$: supply shocks leave Domar weights unchanged, while demand shocks move them only upstream, changing prices across sectors that share suppliers;*
- *the response of prices to demand shocks, $\frac{\partial\log p}{\partial\omega'}=\mathcal{S}\left(I+\mathcal{H}\mathcal{S}\right)^{-1}$, is symmetric and positive semi-definite.*

Proof: the first two claims follow from Theorem B.1. For the third, by the push-through identity $\mathcal{S}\left(I+\mathcal{H}\mathcal{S}\right)^{-1}=\left(I+\mathcal{S}\mathcal{H}\right)^{-1}\mathcal{S}$, which is its own transpose, and $\mathcal{S}\left(I+\mathcal{H}\mathcal{S}\right)^{-1}=\mathcal{S}^{1/2}\left(I+\mathcal{S}^{1/2}\mathcal{H}\mathcal{S}^{1/2}\right)^{-1}\mathcal{S}^{1/2}$, whose middle matrix is positive definite under the eigenvalue condition. ■

Supply and demand shocks therefore use the production network in opposite directions, and each horizontal channel has its own switch: substitutability at buyers for supply shocks (Corollary 1), congestion at suppliers for demand shocks (Proposition 1). Constant marginal costs are the knife-edge case in which common-supplier geometry is silent, exactly as Cobb-Douglas technologies are the knife-edge case in which common-buyer geometry is silent. Common-supplier price spillovers are also reciprocal: the price of any sector-i responds to final demand for j as the price of j responds to final demand for i, whether the two are co-customers or one supplies the other.

B.3. Common-supplier transmission between sectors

Proposition 1 characterizes transmission between co-customers at first round, holding the prices of all other commodities fixed. Since common-supplier transmission requires congestion, its general-equilibrium counterpart is characterized to first order in congestion, that is, up to terms of order θ^2, for which $\left(I+\mathcal{H}\mathcal{S}\right)^{-1}=I-\mathcal{H}\mathcal{S}+O\left[\theta^2\right]$.

COROLLARY B.2 (Common-supplier transmission in general equilibrium) *Consider a demand shock to sector-j. To first order in congestion, the components of the responses specific to j are, for every sector-i,*

$$\frac{\partial\log p_i}{\partial\xi_j}=\sum_{s\in\Phi}\ell_{is}\frac{\theta_s}{\lambda_s}\ell_{js}\qquad\text{and}\qquad\frac{\partial\lambda_i}{\partial\xi_j}=\ell_{ji}-\sum_{s\in\Phi}\mathcal{R}_{i,s}\frac{\theta_s}{\lambda_s}\ell_{js}$$

Hence:

- *the prices of j and of every other sector-i comove positively, and are related if and only if the two sectors require, directly or indirectly, a common congested supplier, which may be one of the two sectors;*
- *the employment of j and of its direct and indirect suppliers, relative to the common component, rises by $\frac{\ell_{js}}{\lambda_s}$ per unit of $d\,\xi_j$, up to a correction of order θ;*

- *let i be a co-customer of j that is not upstream of j ($\ell_{ji} = 0$). If $\sigma_s \geq 1$ ($\sigma_s \leq 1$), $\forall s \in \Phi$, and i is not a common-buyer neighbour, at any depth, of the congested suppliers of j ($\mathcal{R}^h_{i,s} = 0$ whenever $\theta_s \ell_{js} > 0$), the employment of i, relative to the common component, falls (rises). It does not move if i sells only to final demand.*

Proof: by Theorem B.1, $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}' d\boldsymbol{\omega} - \boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{K}}\boldsymbol{\mathcal{L}}' d\boldsymbol{\omega} + O\left[\theta^2\right]$, since $\boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}} = \boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{K}}\boldsymbol{\mathcal{L}}'$, and $d\log \boldsymbol{p} = \boldsymbol{\mathcal{S}}\, d\boldsymbol{\omega} + O\left[\theta^2\right]$. The component specific to j sets $d\boldsymbol{\omega} = \boldsymbol{e}_j\, d\xi_j$: then $\boldsymbol{\mathcal{L}}'\boldsymbol{e}_j = \left(\boldsymbol{\ell}^{row}_j\right)'$, and $\left[\boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{K}}\boldsymbol{\mathcal{L}}'\boldsymbol{e}_j\right]_i = \sum_{s\in\Phi} \mathcal{R}_{i,s} \frac{\theta_s}{\lambda_s} \ell_{js} + \left[\boldsymbol{\mathcal{R}}\boldsymbol{\theta}\right]_i \frac{\left[\boldsymbol{\mathcal{L}}\boldsymbol{\theta}\right]_j}{\Lambda_N}$, whose last term, as the second term of $\mathcal{S}_{i,j}$, is of order θ^2. The first claim follows from $\boldsymbol{\mathcal{L}} \geq \boldsymbol{0}$, the second from $\ell_{js} > 0$ and $d\log n_s - d\log E = \frac{d\lambda_s}{\lambda_s}$. For the third, $\ell_{ji} = 0$, and every term with $\theta_s \ell_{js} > 0$ has $\mathcal{R}_{i,s} = \mathcal{R}^v_{i,s}$, non-negative (non-positive) by Proposition 3. If i sells only to final demand, the i-th row of $\boldsymbol{\mathcal{H}}$ is zero and $\ell_{mi} = 0$ for $m \neq i$, so that $\mathcal{R}_{i,s} = \sum_{m\in\Phi} \ell_{mi} \left[\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\right]_{m,s} = \left[\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\right]_{i,s} = 0$. ■

The corollary is the general-equilibrium counterpart of Proposition 1, and shows why the latter is hard to read in employment data. In prices, common-supplier transmission is transparent: a demand shock raises the cost of every sector that shares congested suppliers with the shocked one, in proportion to their common-supplier proximity, and its sign is the sign of congestion, as at first round. In employment, the cost increase must still be turned into a change in sales, and this happens only at the co-customer's own buyers: if they treat their inputs as substitutes, they shift their budgets away from it, and its employment falls; if they treat them as complements, they spend more on it, and its employment rises. A demand shock thus reaches co-customers through common suppliers, but moves their workers only through their common buyers. The sign of employment comovement between co-customers combines congestion at the shared supplier and substitutability at the co-customer's buyers and vanishes when either is absent; meanwhile, the shocked sector and its suppliers move together.

B.4. Employment and aggregate output

Finally, I can relate the duality to changes in macroeconomic observables.

PROPOSITION B.1 (Employment reallocation under demand shocks) *To first order in shocks and in congestion, sectoral employment responds to demand shocks as*

$$d\log \boldsymbol{n} = \underbrace{d\log E\, \boldsymbol{1}}_{\text{common}} + \underbrace{\boldsymbol{\Lambda}^{-1}\boldsymbol{\mathcal{L}}' d\boldsymbol{\omega}}_{\text{vertical}} - \underbrace{\boldsymbol{\Lambda}^{-1}\boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}}\, d\boldsymbol{\omega}}_{\text{horizontal}}$$

with $d\log E = \frac{1}{\Lambda_N}\left(\boldsymbol{\mathcal{L}}\boldsymbol{\theta}\right)' d\boldsymbol{\omega}$ *and* $\boldsymbol{\alpha}'\boldsymbol{\Lambda}\, d\log \boldsymbol{n} = 0$.

Proof: by Steps 2 and 4 of the proof of Theorem B.1, $d\log \boldsymbol{n} = \boldsymbol{\Lambda}^{-1} d\boldsymbol{\lambda} + d\log E\, \boldsymbol{1}$, with $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'\left(\boldsymbol{I} - \boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}}\right) d\boldsymbol{\omega} + O\left[\theta^2\right]$ and $d\log E = \frac{\boldsymbol{\theta}' d\boldsymbol{\lambda}}{\Lambda_N} = \frac{\boldsymbol{\theta}'\boldsymbol{\mathcal{L}}' d\boldsymbol{\omega}}{\Lambda_N} + O\left[\theta^2\right]$. Finally, $\boldsymbol{\alpha}'\boldsymbol{\Lambda}\, d\log \boldsymbol{n} = \frac{1}{E}\sum_{i\in\Phi} dn_i = 0$, since N is fixed. ■

Relative to Proposition 3, the vertical component now runs upstream: a demand shock raises the employment of the shocked sector and of its suppliers together, in proportion to their exposure to its final demand relative to their size, $\frac{\ell_{js}}{\lambda_s}$. This is a second source of positive comovement between horizontally distant sectors, which requires neither congestion nor elasticities different from one. The horizontal component composes the two geometries, and requires both. The common component is no longer neutral: since $\boldsymbol{\lambda}'\boldsymbol{\theta} = 1 - \Lambda_N$, a demand shock to sector-j raises nominal GDP relative to the wage, and lowers the labour share, if and only if $\left[\boldsymbol{\mathcal{L}}\boldsymbol{\theta}\right]_j > 1 - \Lambda_N$, that is, if the supply chain of j is more intensive in specific factors than the average final good.

Aggregate output is measured by the Divisia index of real GDP, $d\log Y = \sum_{i\in\Phi} \omega_i\, d\log c_i$,

which, for given $\boldsymbol{\omega}$, is the household's utility.[I]

THEOREM B.2 (Aggregate output with rising marginal costs) *Let the reallocation operator to be* $\widetilde{\boldsymbol{\mathcal{R}}} = \boldsymbol{\mathcal{L}}'\left(\boldsymbol{I} + \boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}}\right)^{-1}\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$*, and let the eigenvalue condition of Theorem B.1 hold. Then:*

- *to first order,* $d\log Y = \boldsymbol{\lambda}' d\log \boldsymbol{z}$*, whatever the demand shock: demand shocks move the slopes of aggregate output,* $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'\left(\boldsymbol{I} + \boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}}\right)^{-1} d\,\boldsymbol{\omega}$*, but not its level;*
- *to second order,* $d^2\log Y = \boldsymbol{\varepsilon}'\widetilde{\boldsymbol{\mathcal{R}}}\,\boldsymbol{\varepsilon}$*, where* $\widetilde{\boldsymbol{\mathcal{R}}} = \left(\boldsymbol{I} + \boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{K}}\right)^{-1}\boldsymbol{\mathcal{R}}$ *is symmetric;*
- *consequently,* $\mathbb{E}\left[\log Y\right] - \log Y^{det} = \frac{1}{2}v^2\,\text{Trace}\left[\widetilde{\boldsymbol{\mathcal{R}}}\right] + o\left[v^2\right]$*, with* $\text{Trace}\left[\widetilde{\boldsymbol{\mathcal{R}}}\right] \leq \text{Trace}\left[\boldsymbol{\mathcal{R}}\right]$*, whatever the elasticities. To first order in congestion,* $\boldsymbol{\mathcal{R}} - \widetilde{\boldsymbol{\mathcal{R}}} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}}\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$*.*

Proof: for the first claim, $c_i = \frac{\omega_i E}{p_i}$ and $\sum_{i\in\Phi}\omega_i\, d\log\omega_i = \boldsymbol{1}' d\,\boldsymbol{\omega} = 0$, so that $d\log Y = d\log E - \boldsymbol{\omega}' d\log \boldsymbol{p}$. By Step 2 of the proof of Theorem B.1, $\boldsymbol{\omega}' d\log \boldsymbol{p} = -\boldsymbol{\lambda}' d\log \boldsymbol{z} + \boldsymbol{\lambda}'\boldsymbol{\mathcal{K}}\, d\boldsymbol{\lambda}$, and $\boldsymbol{\lambda}'\boldsymbol{\mathcal{K}} = \boldsymbol{\theta}' + \frac{1-\Lambda_N}{\Lambda_N}\boldsymbol{\theta}' = \frac{\boldsymbol{\theta}'}{\Lambda_N}$, so that $\boldsymbol{\omega}' d\log \boldsymbol{p} = -\boldsymbol{\lambda}' d\log \boldsymbol{z} + d\log E$ and $d\log Y = \boldsymbol{\lambda}' d\log \boldsymbol{z}$. For the second, with $d\,\boldsymbol{\omega} = \boldsymbol{0}$, $d^2\log Y = d\log \boldsymbol{z}' d\boldsymbol{\lambda}$ and $d\boldsymbol{\lambda} = \widetilde{\boldsymbol{\mathcal{R}}}\, d\log \boldsymbol{z}$ by Theorem B.1; by the push-through identity, $\widetilde{\boldsymbol{\mathcal{R}}} = \left(\boldsymbol{I} + \boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{K}}\right)^{-1}\boldsymbol{\mathcal{R}}$ and $\widetilde{\boldsymbol{\mathcal{R}}}' = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\left(\boldsymbol{I} + \boldsymbol{\mathcal{S}}\boldsymbol{\mathcal{H}}\right)^{-1}\boldsymbol{\mathcal{L}} = \widetilde{\boldsymbol{\mathcal{R}}}$. For the third, the expectation follows as in Steps *iii.1-iii.3* of the proof of Theorem 2. Moreover, again by the push-through identity, $\boldsymbol{\mathcal{R}} - \widetilde{\boldsymbol{\mathcal{R}}} = \left(\boldsymbol{I} + \boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{K}}\right)^{-1}\boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{K}}\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{K}}^{1/2}\left(\boldsymbol{I} + \boldsymbol{\mathcal{K}}^{1/2}\boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{K}}^{1/2}\right)^{-1}\boldsymbol{\mathcal{K}}^{1/2}\boldsymbol{\mathcal{R}}$. The non-zero eigenvalues of $\boldsymbol{\mathcal{K}}^{1/2}\boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{K}}^{1/2}$ are those of $\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}}$, which exceed -1: the middle matrix is positive definite, and $\boldsymbol{\mathcal{R}} - \widetilde{\boldsymbol{\mathcal{R}}}$ is positive semi-definite. To first order in congestion, $\boldsymbol{\mathcal{R}} - \widetilde{\boldsymbol{\mathcal{R}}} = \boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{K}}\boldsymbol{\mathcal{R}} + O\left[\theta^2\right] = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}}\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}} + O\left[\theta^2\right]$. ■

Three conclusions follow. First, demand shocks reallocate activity without changing aggregate output, which, as in Hulten's (1978) theorem, responds only to technology and to factor supplies; they change, however, its slopes, by moving the macroeconomic importance of the shocked sector's suppliers. Second, the curvature of aggregate output combines the two geometries: a productivity shock reallocates budgets at common buyers ($\boldsymbol{\mathcal{H}}$), the suppliers of the winners congest, their price increases spread to all the sectors that require them ($\boldsymbol{\mathcal{S}}$), and are reallocated again at common buyers ($\boldsymbol{\mathcal{H}}$). Third, congestion always lowers the second-order term, whatever the elasticities: it dampens the gains from volatility when intermediate inputs are substitutes, and amplifies the losses when they are complements. This is the network counterpart of the role that Baqaee and Farhi (2019) assign to the degree of factor reallocation: with factors in fixed supply, part of the reallocation of buyers' budgets is absorbed by congestion, whose cost falls on the sectors that share the congested suppliers. Note that $\text{Trace}\left[\widetilde{\boldsymbol{\mathcal{R}}}\right] \leq \text{Trace}\left[\boldsymbol{\mathcal{R}}\right]$ whenever the eigenvalue condition of Theorem B.1 holds – always for $\sigma_s \geq 1$, and, to first order in congestion, for any elasticities, since $\boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{S}}\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$ is positive semi-definite.

C. DATA, FIGURES AND TABLES

C.1. Data details

In order to build the main dataset with a panel of 65 private 3-digit U.S. 2017 North American Industry Classification System (NAICS) sectors used in Sections 4-5, I rely on different data

[I] This indicator provides a theoretically consistent measure of growth. It is a composite index, particularly suited to analyse variables made of multiple and differentiated elements, measuring changes in certain aggregate quantity of a given variable, and weighting all its defining components according to their relevance on that variable, that is it expresses the overall rate of output change as a weighted average of the growth rates of individual components, with time-varying weights corresponding to each component's share in total output. Conceptually, it captures the notion that aggregate GDP growth reflects both the differential growth of its components and their evolving relative importance within the economy. This formulation offers an improvement over fixed-weight indices by accommodating structural changes in the composition of economic activity over time. Refer to Oulton (2022) for further details.

sources from the Bureau of Economic Analysis (BEA), otherwise stated.[II] For years from 1998 to 2022, these information are:

(a) employment (Persons Engaged in Production by Industry, in thousands. Last revised on: September 29, 2023);

(b) value-added (Value-Added by Industry, in billions of U.S. dollars. Last revised on: December 19, 2024);

(c) aggregate Total Factor Productivity (from Fernald 2014) and Government Consumption Expenditures and Gross Investment (*e.g.*, Ramey and Zubairy 2018);[III]

(d) inter-sectoral trade and Input-Output linkages, in particular:

- sectoral share of own production, and thus the sectoral share of intermediate inputs from other sectors (Industry-by-Commodity Market Share Matrix, After Redefinitions – Summary, in producers' prices);
- total inputs required (directly and indirectly) in order to deliver one dollar of output to final users (Commodity-by-Commodity Total Requirements, After Redefinitions – Summary, in producers' prices);
- sector-specific imports and exports of goods and services, government (total, defence, non-defence) expenditure shares, as well as changes in private inventories (The Use of Commodities by Industries, Before Redefinitions – Summary, in producers' prices and in millions of U.S. dollars).

Note that "Summary" defines the level of sectoral disaggregation considered in the analysis (3-digit approximately). The BEA also provides information on Input-Output matrices to a more detailed level (6-digit), but these are on a five-yearly basis (thus it would impossible to run the analysis in Subsection 5.1), and data on other variables are not collected. Moreover, the sample begins in 1998 so to have all sectors classified given the NAICS system, and ends in 2022, the last year available when I started this project.[IV] Last revised on: March 2025.

Turn to the manipulation of Input-Output matrices. To capture the composition of commodity supply across different industries, each cell of the yearly *Industry-by-Commodity Market Share Matrix* contains the share of a specific commodity's total output produced by a given industry. Consequently, the main diagonal represents the share of a commodity produced by its primary (or corresponding) industry, while the off-diagonal cells represent secondary production (by-products or secondary activities coming from other industries). Labelling by $\boldsymbol{X}^{ms} = \left[x_{ij}^{ms}\right]$, where $x_{ij}^{ms} \geq 0$ represents the share of commodity-j produced by industry-i, I can analyse the extent of secondary production. For each commodity, the total share produced as a secondary output by alternative industries is computed as $x_i^{sh} = 1 - x_{ii}^{ms}$, where x_{ii}^{sh} is the primary industry's share. In other words, to extract the total share of commodity production that originates outside its primary sector, I compute $\boldsymbol{x}^{sh} = \mathbf{1} - \left[x_{ii}^{ms}\right]_{i\in\Phi}$, where $\mathbf{1}$ is an $m \times 1$ vector

[II] Note that I refer to 3-digit but, according to the BEA classification, four sectors are at 2-digit ("construction", "management of companies and enterprises", "educational services", and "other services, except government"), while five sectors related to finance and insurance are at 4-digit level.

[III] Here is the `website` for the TFP series. The U.S. government expenditure series is downloaded from the Federal Reserve Economic Data (FRED) database, with reference name *GCEA*; such series, net of federal defence expenditure (named *FDEFX*), represents the non-defence expenditure. Last revised on: June 15, 2026.

[IV] The 3-digit classification is the most granular if one wants to keep track of the evolution in the set of intermediate inputs, and 1998 is the second year in which the NAICS system has been adopted. Before 1997, classification follows the U.S. Standard Industrial Classification (SIC) system.

of ones. The resulting $m \times 1$ vector $\boldsymbol{x}^{sh}$ reports, for every considered year, the total share of each commodity's supply that is due to secondary production diversification rather than primary industry output. For what concerns the production network structure, I rely on the main Input-Output matrix (*Commodity-by-Commodity Total Requirements, After Redefinitions*), which I transpose to match the convention of the main text (rows as buyers, columns as suppliers) and label as $\boldsymbol{\mathcal{L}}^{BEA\ I\text{-}O} = \left[\ell_{ij}^{BEA\ I\text{-}O} \geq 0\right]$.[V] Such Leontief inverse matrix investigates its deep characteristics and, following the literature (*e.g.*, Acemoglu et al. 2012; Carvalho 2014) I disregard small transactions: this allows me to account for a narrower network amplification due to the removal of bias from meaningless inter-sectoral trade intensity. Disregarding small transactions implies that $\ell_{ij} = \frac{\ell_{ij}^{BEA\ I\text{-}O}}{\ell_i}$ is set to 0 if $\ell_{ij} \leq 0.01$, where $\ell_i = \sum_{j \in \Phi} \ell_{ij}^{BEA\ I\text{-}O}$ is sector-i's total input purchases (*i.e.*, *row sum* of matrix $\boldsymbol{\mathcal{L}}^{BEA\ I\text{-}O}$), or $\ell_i = \sum_{i \in \Phi} \ell_{ij}^{BEA\ I\text{-}O}$ is sector-i's total input sales (*i.e.*, *column sum* of matrix $\boldsymbol{\mathcal{L}}^{BEA\ I\text{-}O}$), and where the upper-script $BEA\ I\text{-}O$ identifies the coefficients directly extracted from the BEA Input-Output tables.[VI] The resulting matrix would therefore display each cell to be greater than 0.01 – *i.e.*, larger than 1%. Leontief inverse matrix, upstream and downstream sectors, and the weight associated to each of them (as in Subsections 5.1-5.2) are identified through it.

To compute network distances between sectors, I rely on the promptly-downloadable BEA Input-Output table (*i.e.*, the *Commodity-by-Commodity Total Requirements – After Redefinitions*), which reports the total output of each commodity required to satisfy one unit of final demand for any given commodity. By incorporating both direct and indirect production requirements, this table captures Input-Output linkages extending throughout the production network and along the entire supply chain. This representation is particularly suited to my purpose, as it allows me to compare sectors according to common Input-Output structures not only at the local level, through their direct relationships, but also across the full length of their respective supply chains, through the indirect linkages connecting them to further upstream and downstream stages of production. Similarities between sectors reflect commonalities in their production network exposure at different depths of the supply chain, rather than being driven solely by their immediate bilateral connections. Accordingly, the Euclidean distances (eqs. C.1-C.2) between sectors are computed from differences in their total-requirements profiles, capturing the extent to which sectors share similar buyers/suppliers across the production network. From these distances, I compute the *intensive margin* values for both network "economic" distance metrics.

(Shortest path algorithm, technical but suitable for coding) *Given the binary, symmetric $m \times m$ adjacency matrix $\boldsymbol{G}$ with zero diagonal built from $\boldsymbol{\mathcal{D}}^q$ as described in Section 4, the algorithm computes $\boldsymbol{\mathcal{D}}_{ext}^q$, whose cell $\mathcal{D}_{ext}^q[i,j]$ is the length of the shortest path between node-i and node-j:*

(a) initialize $\boldsymbol{\mathcal{D}}_{ext}^q = \boldsymbol{I}$, which marks every node as reached from itself, the path length at $k = 1$, and the reachability matrix at $\boldsymbol{G}^{(1)} = \boldsymbol{G}$;

(b) build the Boolean matrix $\boldsymbol{B}$ of newly reached pairs, with $\mathcal{P}_{ij} = 1$ if $G_{ij}^{(k)} > 0$ and $\mathcal{D}_{ext}^q[i,j] = 0$, and $\mathcal{P}_{ij} = 0$ otherwise;

(c) as long as $\boldsymbol{B}$ has a non-zero entry: set $\mathcal{D}_{ext}^q[i,j] = k$ whenever $\mathcal{P}_{ij} = 1$; update $k = k+1$

[V] In the BEA tables, row sectors identify (upstream) suppliers, while column sectors the (downstream) buyers: row sector is the origin and column sector is the destination of the circulating intermediate input.

[VI] The "sales-based" Leontief inverse (in the main text) is a normalization of the baseline Input-Output matrix by sectoral total sales, so that each entry represents the share of a sector's inputs relative to its output.

and $\boldsymbol{G}^{(k)} = \mathbb{1}\left[\boldsymbol{G}^{(k-1)}\boldsymbol{G} > 0\right]$*, element by element; rebuild* $\boldsymbol{B}$ *as in step (b);*

(d) once the loop stops, set $\mathcal{D}^{q}_{ext}[i,j] = \infty$ *for the off-diagonal pairs still at zero, which are not connected, and subtract* $\boldsymbol{I}$*, so that the diagonal is zero.*

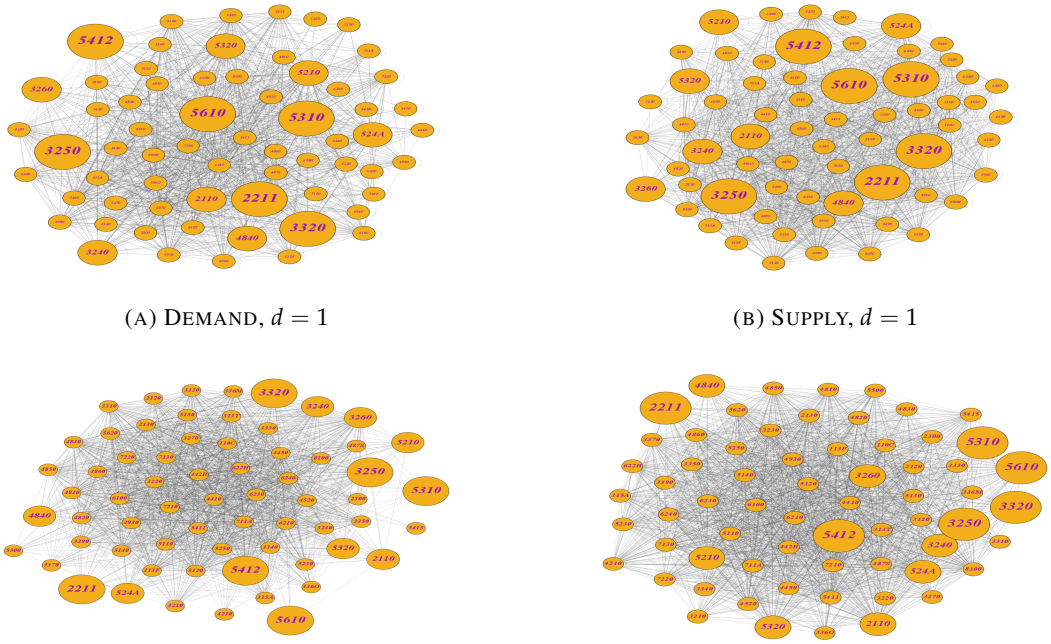
Each off-diagonal entry is then $\mathcal{D}^{q}_{ext}[i,j] \in \{1,2,\ldots,\mathcal{D}_{max}\}$*, or* ∞ *for disconnected pairs, which are excluded from the estimation.*

C.2. Characteristics of the horizontal-distance-based networks

To illustrate the constructed distance matrices Figure C.1 displays, for each extensive margin value, the horizontal linkages from both common demand and supply network ties. Importantly, each connection does not necessarily represent inter-sectoral trade between two sectors: it reflects an horizontal linkage based on common-supplier or common-buyer network distance relationships.

A few concatenated points are then worth noting. First, the distribution of horizontal interdependencies is uneven across distance levels. At very short range ($d = 1$), the common-supplier network appears slightly denser than the common-buyer one, suggesting that sectors are more frequently linked through common upstream suppliers than through shared downstream buyers. However, this asymmetry vanishes as the distance increases ($d = 2$), where both types of horizontal ties spread more diffusely across the network. Second, sectors with the highest number of Input-Output linkages tend to concentrate at the very core of the production system when considering only close connections ($d = 1$). As shown in Panels C.1a and C.1b, both the largest hubs (top 10% by connectivity) and the mid-sized hubs (top 20%) are similarly central, occupying short distances from the network's nucleus. Third, these same highly connected sectors tend to become relatively more peripheral once longer distances ($d = 2$) are considered. Panels C.1c and C.1d show that, at this range, their positions are more evenly distributed across the network, reflecting the fact that their trade structures – both upstream and downstream – are highly similar to those of other sectors. In other words, most connected sectors are less unique at longer distances, as commonalities in demand and supply linkages become more widespread.

Altogether, these observations highlight that the structure of horizontal relations depend crucially on the notion of distance. When short, horizontal linkages cluster around the key sectors whereas, at longer distances, such interdependencies are more evenly distributed. This pattern confirms that sectors typically share both upstream suppliers and downstream buyers at smaller but not at a greater distance, and the same vertical production system generates distinct horizontal geometries. This is the empirical counterpart of the last claim of Lemma 1: the U.S. Input-Output matrix is far from normal, so that the same vertical structure places sectors at different horizontal positions as buyers and as suppliers.



(A) DEMAND, $d = 1$

(B) SUPPLY, $d = 1$

(C) DEMAND, $d = 2$

(D) SUPPLY, $d = 2$

FIGURE C.1: VISUALIZATION OF PRODUCTION NETWORK DISTANCES

Note: each panel represents the distance-based inter-sectoral production network corresponding to the (*commodity-by-commodity total direct requirements*) U.S. Input-Output matrix in the year 2007. *Common-supplier* and *common-buyer* network distances are computed implementing the *shortest path algorithm* on distance matrices whose values are at the intensive margin. There are 65 3-digit U.S. 2017 NAICS private sectors, and each orange node corresponds to one of them. Two sectors connected under $d = 1$ cannot be linked under $d = 2$, and otherwise. Biggest nodes correspond to top 10% of mostly connected sectors (*i.e.*, major number of inter-sectoral connections), while intermediate ones the additional top 20%. Figure drawn with the software package *Gephi*, version 0.10, exploiting the *ForceAtlas2* layout algorithm. *Source*: BEA and own calculations.

TABLE C.1: LIST OF CONSIDERED SECTORS AND THEIR CODES

denomination	***codes***	
	NAICS 2017	*BEA*
Farms	111, 112	111CA
Forestry, Fishing and Related Activities	113, 114, 115	113FF
Oil and Gas Extraction	211	211
Mining, except Oil and Gas	212	212
Support Activities for Mining	213	213
Utilities	22	22
Construction	23	23
Wood Products	321	321
Nonmetallic Mineral Products	327	327
Primary Metals	331	331
Fabricated Metal Products	332	332
Machinery	333	333
Computer and Electronic Products	334	334
Electrical Equipment, Appliances, and Components	335	335
Motor Vehicle, Bodies and Trailers, and Parts	3361-3	3361MV
Other Transportation Equipment	3364-9	3364OT
Furniture and Related Products	337	337
Miscellaneous Manufacturing	339	339
Food and Beverage and Tobacco Products	311, 312	311FT
Textile Mills and Textile Product Mills	313, 314	313TT
Apparel and Leather and Allied Products	315, 316	315AL
Paper Products	322	322
Printing and Related Support Activities	323	323
Petroleum and Coal Products	324	324
Chemical Products	325	325
Plastics and Rubber Products	326	326
Wholesale Trade	42	42
Motor Vehicle and Parts Dealers	441	441
Food and Beverage Stores	445	445
General Merchandise Stores	452	452
Other Retail	442-4, 446-8, 45 ex. 452	4AO
Air Transportation	481	481
Rail Transportation	482	482
Water Transportation	483	483
Truck Transportation	484	484
Transit and Ground Passenger Transportation	485	485

...

Sectors are defined according to the nomenclature used by the U.S. Bureau of Economic Analysis (BEA).

TABLE C.1: LIST OF CONSIDERED SECTORS AND THEIR CODES (*continued*)

denomination	***codes***	
	NAICS 2017	*BEA*
...		
Pipeline Transportation	486	486
Other Transportation and Support Activities	487, 488, 492	487OS
Warehousing and Storage	493	493
Publishing Industries, except Internet (includes Software)	511	511
Motion Picture and Sound Recording industries	512	512
Broadcasting and Telecommunications	515, 517	513
Data Processing, Internet Publishing, and Other Information Service	518, 519	514
Federal Reserve Banks, Credit Intermediation, and Related Activities	521, 5221, 5222, 5223	521CI
Securities, Commodity Contracts, and Investments	523	523
Insurance Carriers and Related Activities	5241, 5242	524
Funds, Trusts, and Other Financial Vehicles	525	525
Housing (not considered in the analysis)	·	HS
Other Real Estate	531	ORE
Rental and Leasing Services and Lessors of Intangible Assets	532, 533	532RL
Legal Services	5411	5411
Computer Systems Design and Related Services	5415	5415
Miscellaneous Professional, Scientific, and Technical Services	541 ex. 5411, 5415	5412OP
Management of Companies and Enterprises	55	55
Administrative and Support Services	561	561
Waste Management and Remediation Services	562	562
Educational Services	61	61
Ambulatory Health Care Services	621	621
Hospitals	622	622
Nursing and Residential Care Facilities	623	623
Social Assistance	624	624
Performing Arts, Spectator Sports, Museums, and Related Activities	711, 712	711AS
Amusement, Gambling, and Recreation industries	713	713
Accommodation	721	721
Food Services and Drinking Places	722	722
Other Services, except Government	81	81

Sectors are defined according to the nomenclature used by the U.S. Bureau of Economic Analysis (BEA).

C.3. Network economic distances at the intensive margin

Section 4 of the main text characterizes distance matrices at their "extensive margin", $\boldsymbol{\mathcal{D}}^{cs}_{ext}$ and $\boldsymbol{\mathcal{D}}^{cb}_{ext}$, whose values are $\mathcal{D}^{q}_{ext}[i,j] = \{1,2,\ldots,\mathcal{D}_{max}\}$ for $q = \{cs, cb\}$, exploiting the *shortest path algorithm* on the "intensive margin" distances from the Leontief inverse production network, $\boldsymbol{\mathcal{L}}$, that accounts for direct and indirect sectoral connections. Label with $\ell^{BEA\ I\text{-}O}_{ij}$ each *Total Requirements* coefficient displayed in the BEA Input-Output table. To characterize the "intensive margin" matrices $\boldsymbol{\mathcal{D}}^{cs}$ and $\boldsymbol{\mathcal{D}}^{cb}$ I exploit the system of *Cartesian coordinates* typical of any production network (*e.g.*, Conley and Dupor 2003).

Given $\ell^{cs}_{is} = \frac{\ell^{BEA\ I\text{-}O}_{is}}{\sum_{i\in\Phi}\ell^{BEA\ I\text{-}O}_{is}}$, the generic element of the *common-supplier* distance matrix is computed as

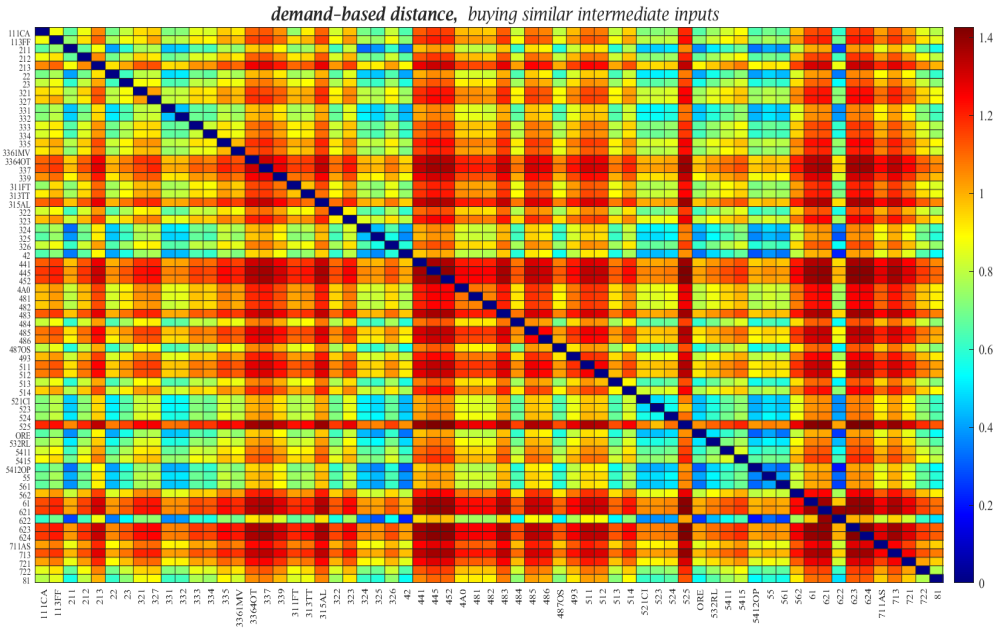
$$\mathcal{D}^{cs}[i,j] = \left\{\sum_{s\in\Phi}\left[\ell^{cs}_{is} - \ell^{cs}_{js}\right]^2\right\}^{\frac{1}{2}} \in [0,\sqrt{2}] \tag{C.1}$$

which defines the length of the line segment (*i.e.*, the Euclidean distance) connecting sectors i and j when both are buying from the same other sector-s, $\forall s \in \Phi$, so that $\boldsymbol{\mathcal{D}}^{cs} = [\mathcal{D}^{cs}[i,j]]$ with generic element $\mathcal{D}^{cs}[i,j] > 0$. Demand distance compares the input profiles of two buyers, each normalized by the supplier's total intermediate sales, so that ℓ^{cs}_{is} is the share of sector-i in sector-s's intermediate supply. Hence, it is the difference between columns constructed by dividing each cell by its row sum. Why? A row sum represents the total sales of the sector in question, so dividing each entry by its associated row sum yields the share of that (row) sector's output purchased by each of the column sectors. This normalization allows for a meaningful comparison across dyads of sectors: by examining these shares, one can assess the relative distance between two (column) sectors in terms of their purchases from common upstream suppliers, thereby pointing to the strength of the horizontal effect where the upstream pass-through occurs (share of total sales of the common sector), Proposition 1.

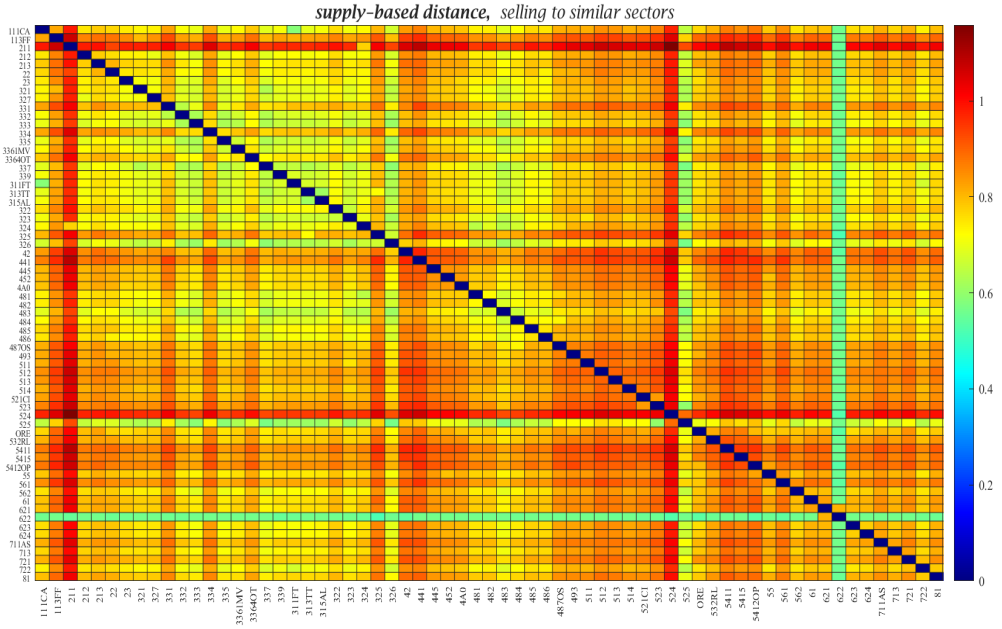
In an analogous way, given $\ell^{cb}_{si} = \frac{\ell^{BEA\ I\text{-}O}_{si}}{\sum_{i\in\Phi}\ell^{BEA\ I\text{-}O}_{si}}$, the generic element of the *common-buyer* distance matrix is shaped by

$$\mathcal{D}^{cb}[i,j] = \left\{\sum_{s\in\Phi}\left[\ell^{cb}_{si} - \ell^{cb}_{sj}\right]^2\right\}^{\frac{1}{2}} \in [0,\sqrt{2}] \tag{C.2}$$

tracing a Euclidean distance between sectors $\{i,j\}$ when both are selling to the same sector-s, $\forall s \in \Phi$, so that $\boldsymbol{\mathcal{D}}^{cb} = [\mathcal{D}^{cb}[i,j]]$ with $\mathcal{D}^{cb}[i,j] > 0$. Supply distance is the difference between rows constructed by dividing each cell by its column sum. Why? A column sum represents the total purchases of the sector in question, so dividing each entry by its associated column sum yields the share of that (column) sector's input purchased from each of the row sectors. This normalization allows for a meaningful comparison across dyads of sectors: by examining these shares, one can assess the relative distance between two (row) sectors in terms of their traded quantities to common downstream buyers, thereby pointing to the strength of the horizontal effect where the downstream pass-through occurs (share of total purchases of the common sector), Proposition 2.



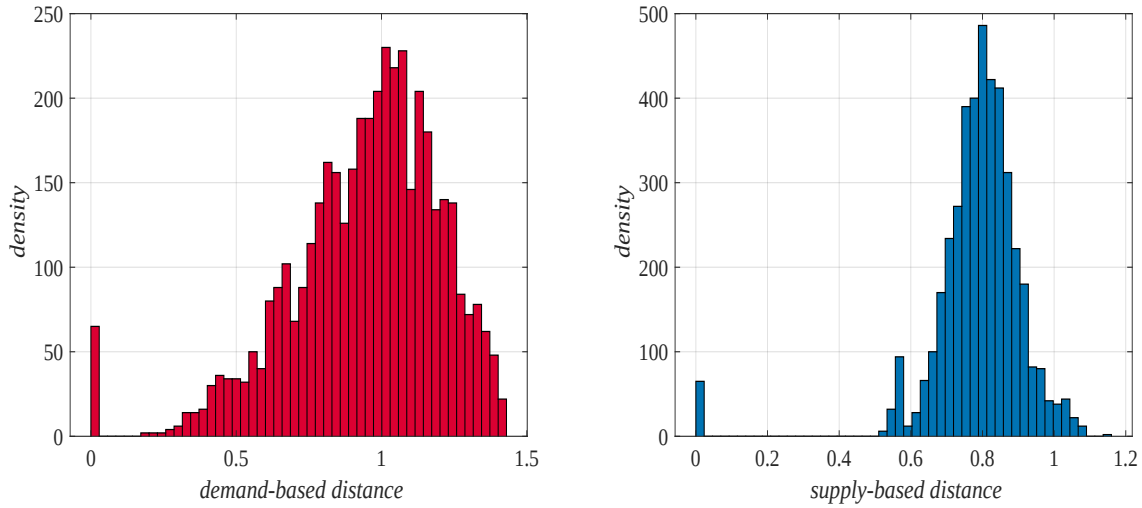
(A) COMMON-SUPPLIER



(B) COMMON-BUYER

FIGURE C.2: INTENSIVE MARGIN MATRICES FOR NETWORK ECONOMIC DISTANCE

Note: heatmaps of sectoral distances characterizing the U.S. production network in year 2007 for 3-digit U.S. 2017 NAICS sectors. Entry (i, j) represents the intensive marginal value, $\mathcal{D}^q[i, j] \geq 0$ for $q = \{cs, cb\}$, of the horizontal relationship between sector-i and sector-j under *common-supplier* network distance (*i.e.*, demand linkages across sectors given their common upstream suppliers, Panel C.2a) and *common-buyer* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers, Panel C.2b). Diagonal sparsity (*i.e.*, lots of 0s), is a direct consequence of the fact that the difference of inter-sectoral trade intensities between a sector and itself is zero. Distance matrices are at the *intensive margin*, as in eqs. (C.1)-(C.2), computed from the *directed* sectoral production network. *Source*: BEA and own calculations.



(A) COMMON-SUPPLIER (B) COMMON-BUYER

FIGURE C.3: DISTRIBUTION OF NETWORK ECONOMIC DISTANCES

Note: these figures depict the distributions for *common-supplier* network distance (*i.e.*, demand linkages across sectors given their common upstream suppliers, Panel C.3a) and *common-buyer* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers, Panel C.3b). Distance values on the x-axis are at the *intensive margin*, as in eqs. (C.1)-(C.2), computed from the *directed* sectoral production network *Source*: BEA and own calculations.

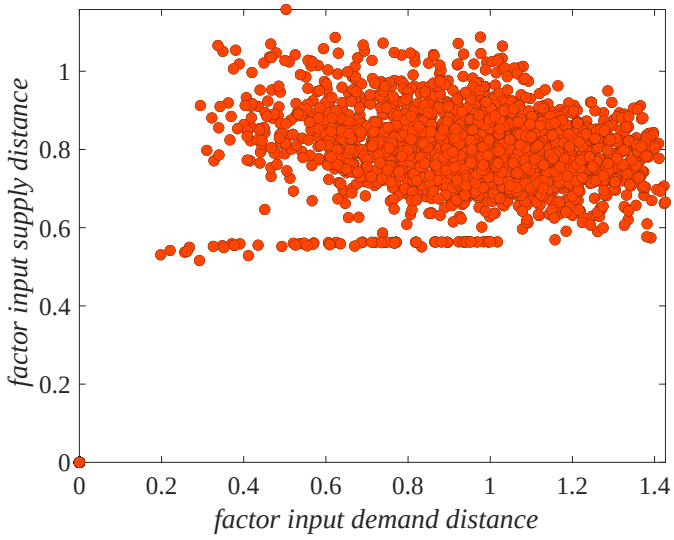
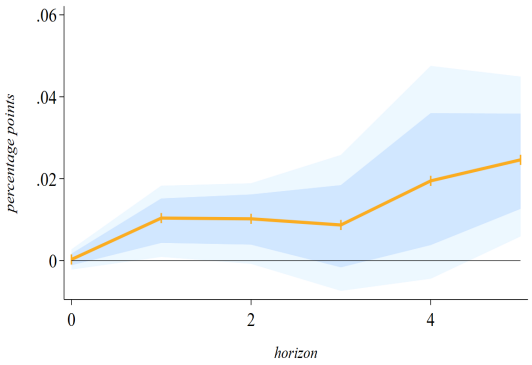


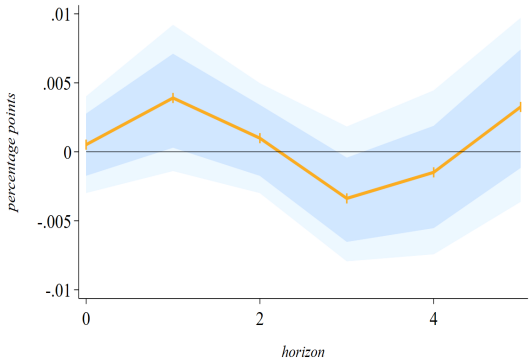
FIGURE C.4: CORRELATION OF NETWORK ECONOMIC DISTANCES

Note: correlating the measures for *common-supplier* network distance (*i.e.*, demand linkages across sectors given their common upstream suppliers, x-axis) and *common-buyer* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers, y-axis). Distances are at the *intensive margin*, as in eqs. (C.1)-(C.2), computed from the *directed* sectoral production network *Source*: BEA and own calculations.

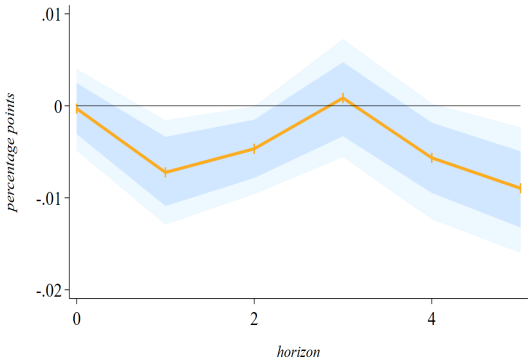
D. COMPLEMENTARY RESULTS



(A) SECTOR-SPECIFIC



(B) $d = 1$



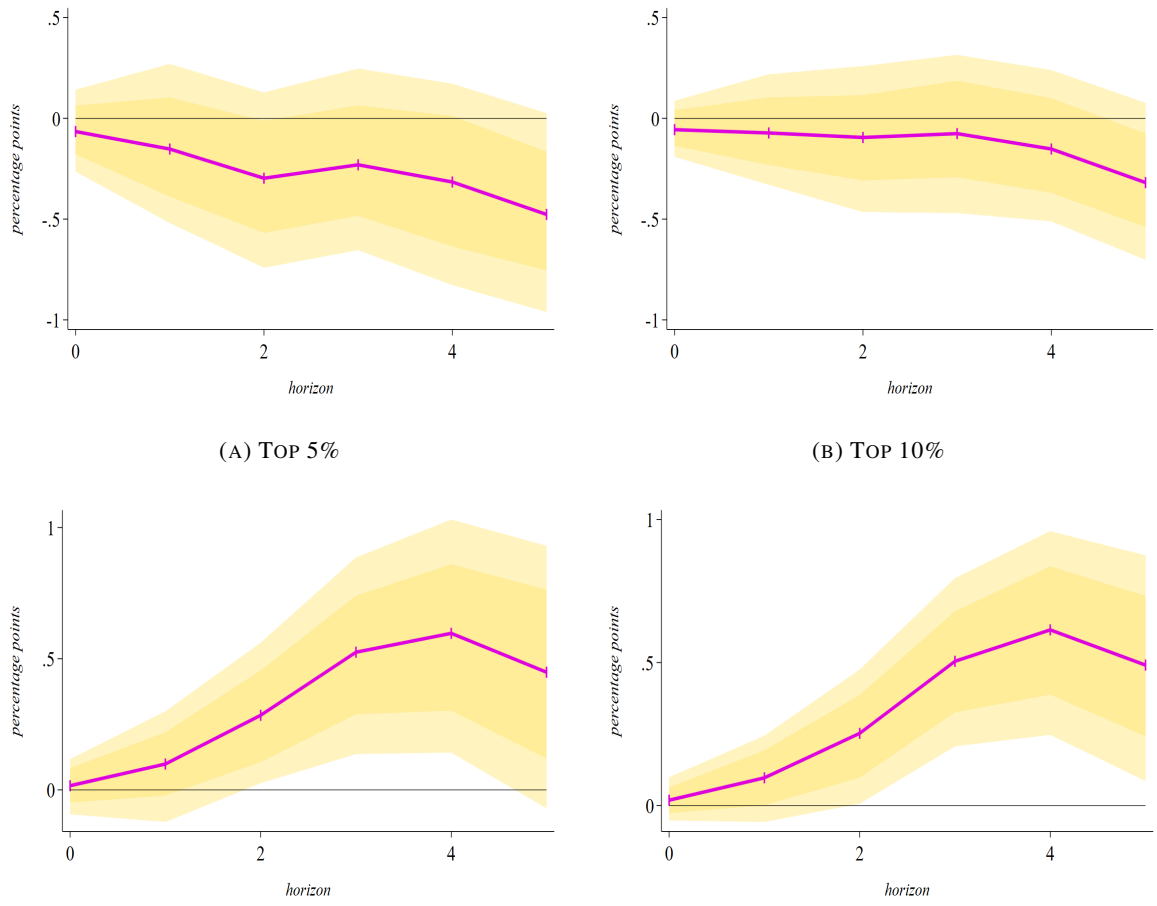
(C) $d = 2$

FIGURE D.1: SECTORAL EMPLOYMENT RESPONSE TO INTERMEDIATE INPUTS, COMMON-SUPPLIER

Note: given the *common-supplier* network distance (*i.e.*, demand linkages across sectors given their common upstream suppliers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposures) characterizing the North American production network in year 2007, it is shown the response of sectoral employment to a 1% increase in the sector-specific share of intermediate inputs (the share of the sector's corresponding commodity produced by alternative sectors) for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Solid-orange line corresponds to the average response of employment across sectors, while shadow-blue and shadow-light blue areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eqs. (5)-(6). *Source*: BEA and own calculations.

D.1. Plotting separated results for highly interlinked sectors

Unpacking all the combined plots, below I report the estimated Local Projection (LP) dynamics for top 5%, top 10%, top 20%, and top 30% of more connected sectors in the production network (*i.e.*, major number of Input-Output linkages) separately.[VII] The purpose is to show the significance of the responses appearing in Figures 7c-7d and 8c-8d, Subsection 5.2.



(A) TOP 5%

(B) TOP 10%

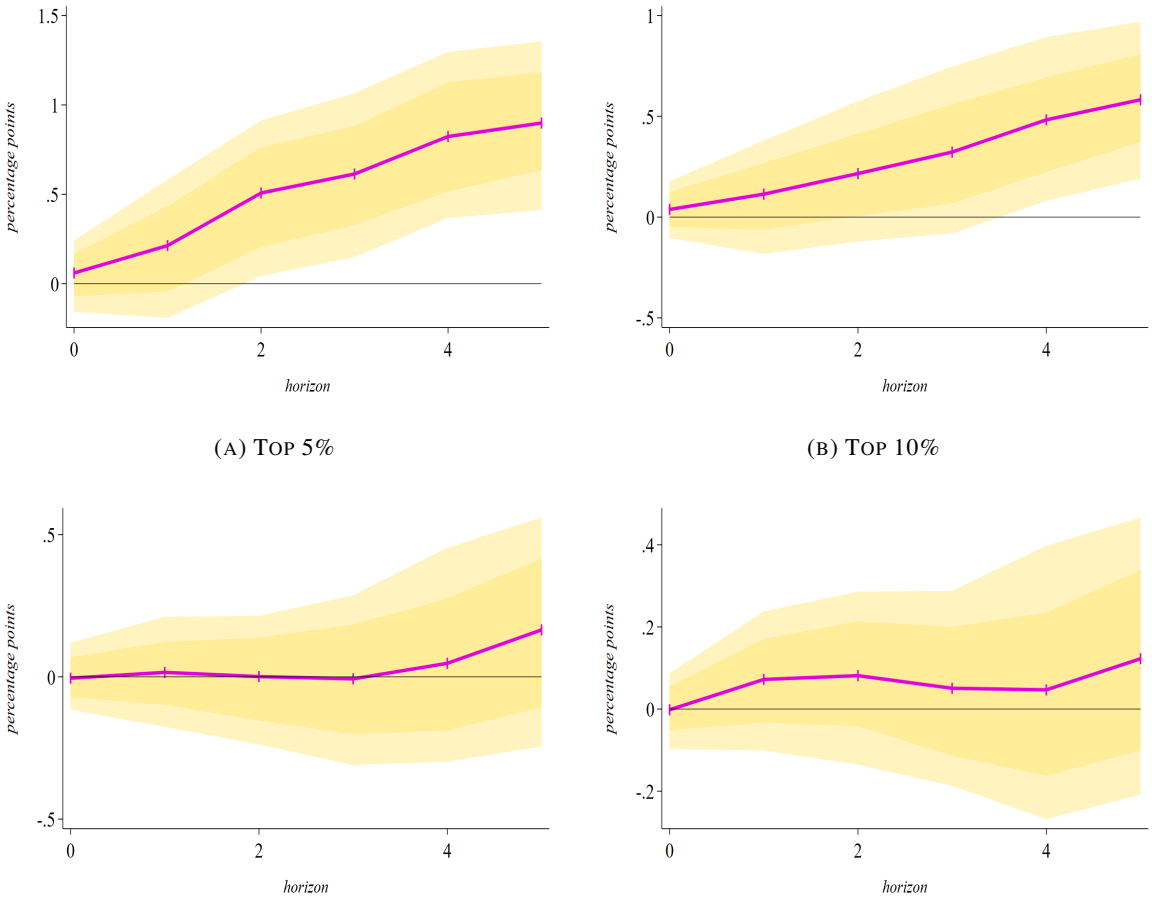
(C) TOP 20%

(D) TOP 30%

FIGURE D.2: COMMON-SUPPLIER ($d = 1$) EFFECTS FOR INTERLINKED SECTORS

Note: given the *common-supplier* network distance (*i.e.*, demand linkages across sectors given their common upstream suppliers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposures) characterizing the North American production network in year 2007, it is shown the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 7c of Figure 7, each panel of this figure shows the employment response of top 5% (Panel D.2a), 10% (Panel D.2b), 20% (Panel D.2c), and 30% (Panel D.2d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). *Source*: BEA and own calculations.

[VII] Besides my network-based effects outlined in the main text (Proposition 3), further theories beyond employment comovement are in Rogerson (1987) and Boldrin et al. (2001).



(A) TOP 5%

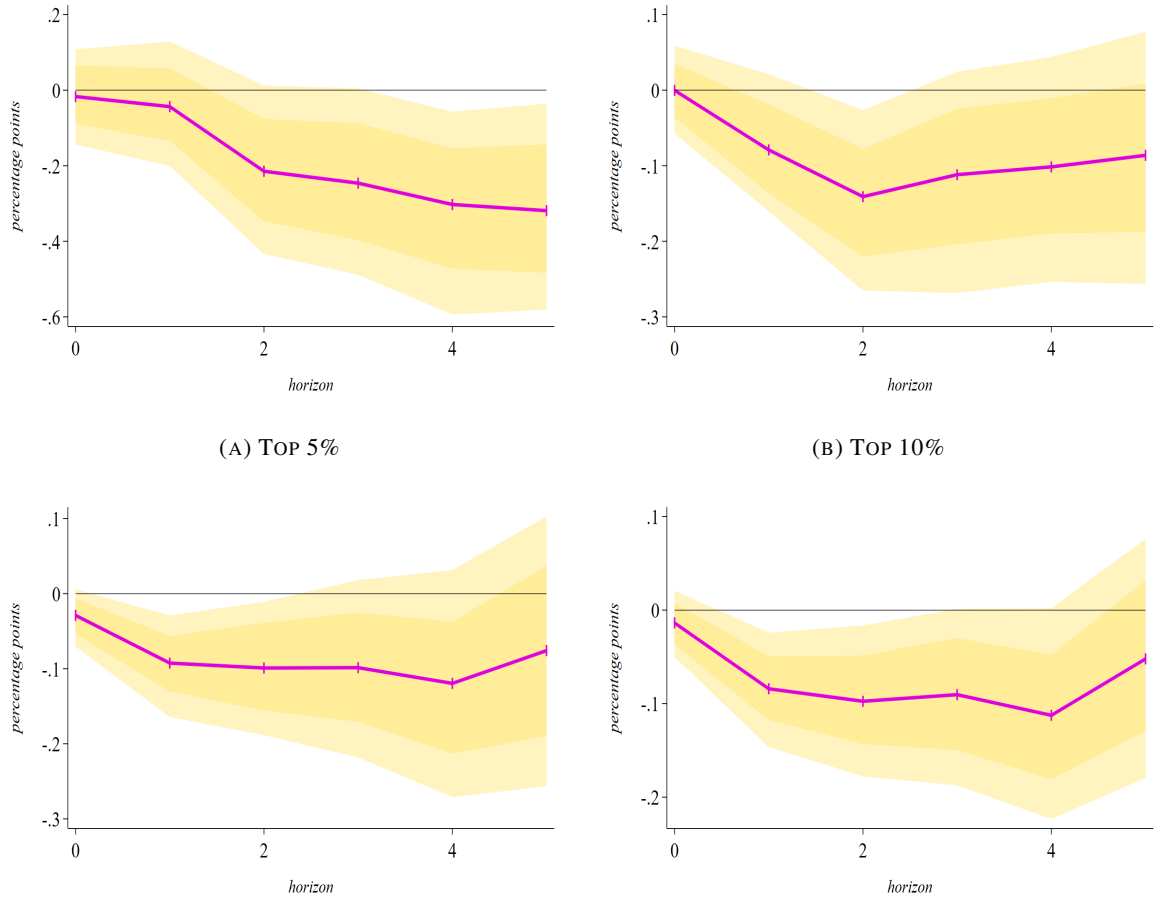
(B) TOP 10%

(C) TOP 20%

(D) TOP 30%

FIGURE D.3: COMMON-SUPPLIER ($d = 2$) EFFECTS FOR INTERLINKED SECTORS

Note: given the *common-supplier* network distance (*i.e.*, demand linkages across sectors given their common upstream suppliers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposures) characterizing the North American production network in year 2007, it is shown the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 7d of Figure 7, each panel of this figure shows the employment response of top 20% (Panel D.3c) and 30% (Panel D.3d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) to employment changes in their further (distance equal to 2) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). *Source*: BEA and own calculations.



(A) TOP 5%

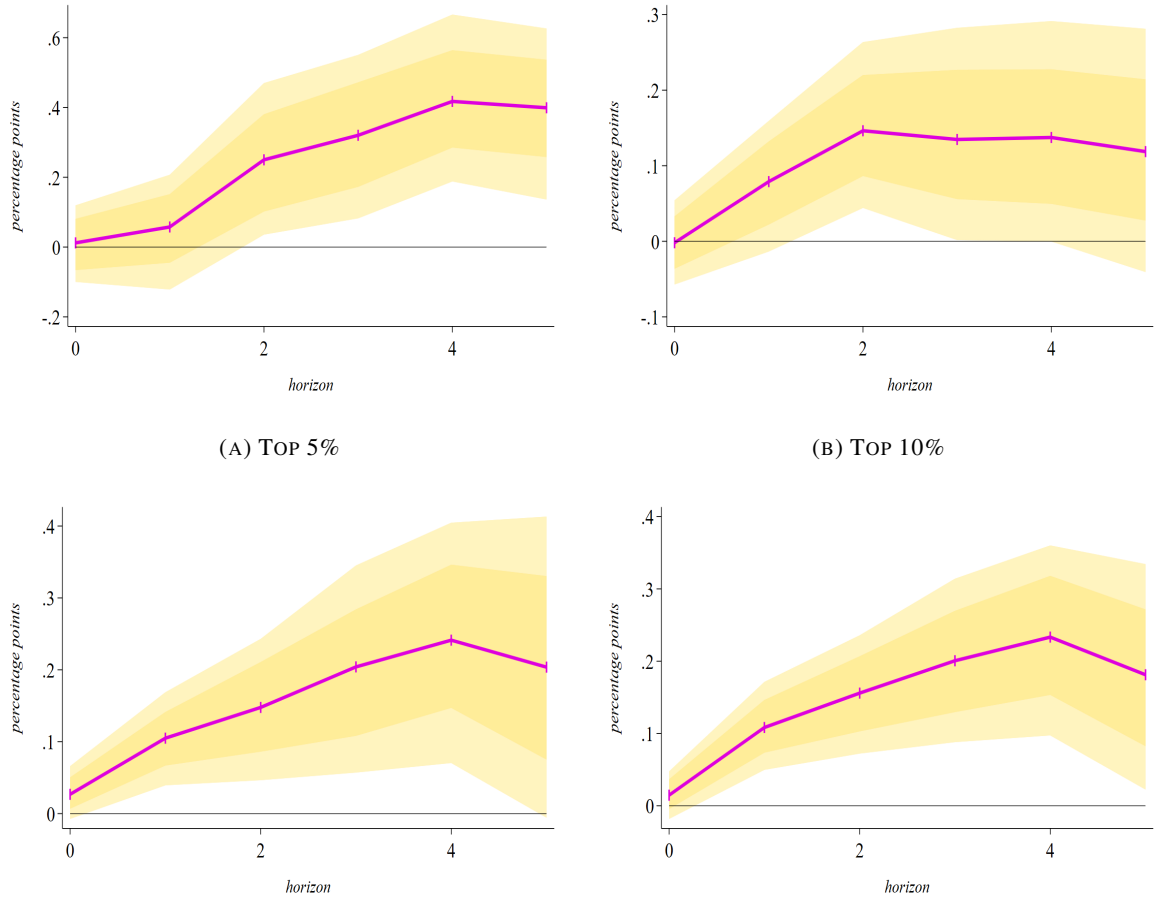
(B) TOP 10%

(C) TOP 20%

(D) TOP 30%

FIGURE D.4: COMMON-BUYER ($d = 1$) EFFECTS FOR INTERLINKED SECTORS

Note: given the *common-buyer* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposures) characterizing the North American production network in year 2007, it is shown the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 8c of Figure 8, each panel of this figure shows the employment response of top 5% (Panel D.4a), 10% (Panel D.4b), 20% (Panel D.4c), and 30% (Panel D.4d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). *Source*: BEA and own calculations.



(A) TOP 5%

(B) TOP 10%

(C) TOP 20%

(D) TOP 30%

FIGURE D.5: COMMON-BUYER ($d = 2$) EFFECTS FOR INTERLINKED SECTORS

Note: given the *common-buyer* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposures) characterizing the North American production network in year 2007, it is shown the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 8d of Figure 8, each panel of this figure shows the employment response of top 5% (Panel D.5a), 10% (Panel D.5b), 20% (Panel D.5c), and 30% (Panel D.5d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their further (distance equal to 2) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). *Source*: BEA and own calculations.

D.2. Robustness with centrality measures

To identify the centrality degree of each sector, I follow the approach outlined in Carvalho (2014) to measure the sector-level Bonacich-Katz centrality.[VIII] Using the Leontief-inverse matrix $\boldsymbol{\mathcal{L}}$ for the U.S. production network as of 2007, the Bonacich-Katz centrality $m \times 1$ vector $\boldsymbol{BK}$ is defined as:

$$\boldsymbol{BK} = \sum_{k \geq 0} \zeta^k \boldsymbol{\mathcal{L}}^k \mathbf{1} = \left(\boldsymbol{I} - \zeta \, \boldsymbol{\mathcal{L}} \right)^{-1} \mathbf{1} \tag{D.1}$$

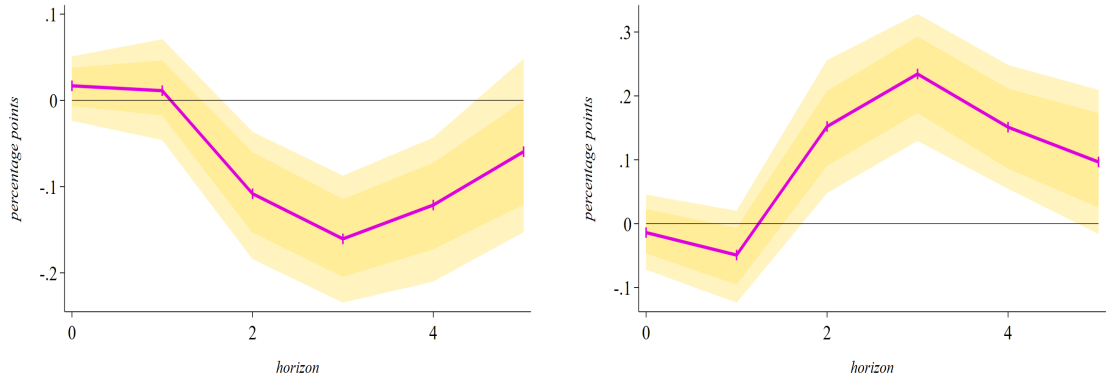
where index-k refers to the length of the paths in the production network – that is, the number of steps (or network "hops") connecting one sector to another – and, given a finite set of sectors $i, j \in \Phi = \{1, \ldots, m\}$:

- $\mathbf{1}$ is an $m \times 1$ vector of ones, reflecting the base influence;
- $\zeta \in \left(0, 1 - \rho\left[\boldsymbol{\mathcal{A}}\right]\right)$ is a dampening scalar parameter, where $\rho\left[\boldsymbol{\mathcal{A}}\right]$ is the spectral radius of $\boldsymbol{\mathcal{A}}$, so that $\frac{1}{1-\rho[\boldsymbol{\mathcal{A}}]}$ is the largest eigenvalue of $\boldsymbol{\mathcal{L}}$;
- $\boldsymbol{I}$ is the $m \times m$ identity matrix.

The value of ζ determines the weight assigned to indirect linkages: smaller values place more emphasis on direct connections, while larger values account for deeper propagation within the network. Following Carvalho (2014), a value of $\zeta = 0.5$ is typically used, which lies safely below the inverse of the largest eigenvalue of most empirically observed Input-Output matrices, ensuring convergence of the series. This measure reflects the idea that a sector is central not only if it is directly connected to many others, but also if it is connected to sectors that are themselves highly central.

This formulation ensures that the centrality score of each sector captures recursive amplification along both its direct connections and its indirect influence through other sectors due to the use of matrix $\boldsymbol{\mathcal{L}} = (\boldsymbol{I} - \boldsymbol{\mathcal{A}})^{-1} = \left[\ell_{ij} \geq 0\right]$, where $\boldsymbol{\mathcal{A}} = \left[\alpha_{ij} \geq 0\right]$ is the directed production network matrix defining the intensity of good produced by sector-j in the total intermediate inputs used by sector-i, with $\alpha_{ij} = 0$ indicating that sector-i does not make use of the good produced by sector-j in producing its own intermediate good.

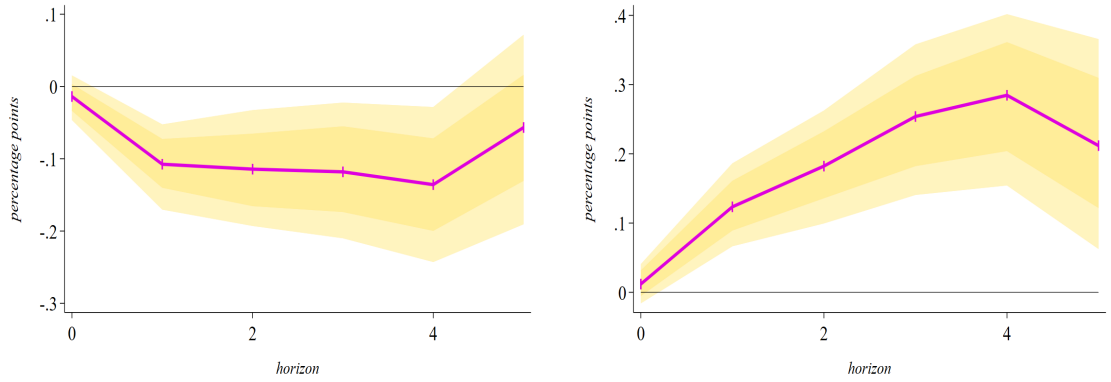
[VIII] Outlined by Bonacich (1987), it is a measure of a sector's overall importance within an Input-Output system, capturing not only its direct connections to other sectors but also the centrality of those it is connected to. In essence, a sector is considered central if it is linked to other central sectors, creating a recursive structure of influence. This measure goes beyond simple counts of linkages by incorporating indirect connections – weighted by a dampening factor – to reflect the diminishing influence of more distant sectors in the network. In the context of production network, a sector with high Bonacich-Katz centrality is one that plays a key role in the propagation of shocks, as its position allows it to influence (or be influenced by) large portions of the economy through both direct and indirect channels. Compared to other centrality measures, Bonacich-Katz is particularly well-suited for economic applications, as it accounts for the intensity of inter-sectoral relationships and the structure of the network in a realistic and nuanced way.



(A) $d = 1$ (B) $d = 2$

FIGURE D.6: COMMON-SUPPLIER LINKAGES AND COMOVEMENT IN PERIPHERAL SECTORS

Note: given the *common-supplier* network distance (*i.e.*, demand linkages across sectors given their common upstream suppliers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposures) characterizing the North American production network in year 2007, it is shown the response of sectoral employment of most central sectors (with *centrality* defined according to sector's relevance in final consumption) to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). *Source*: BEA and own calculations.



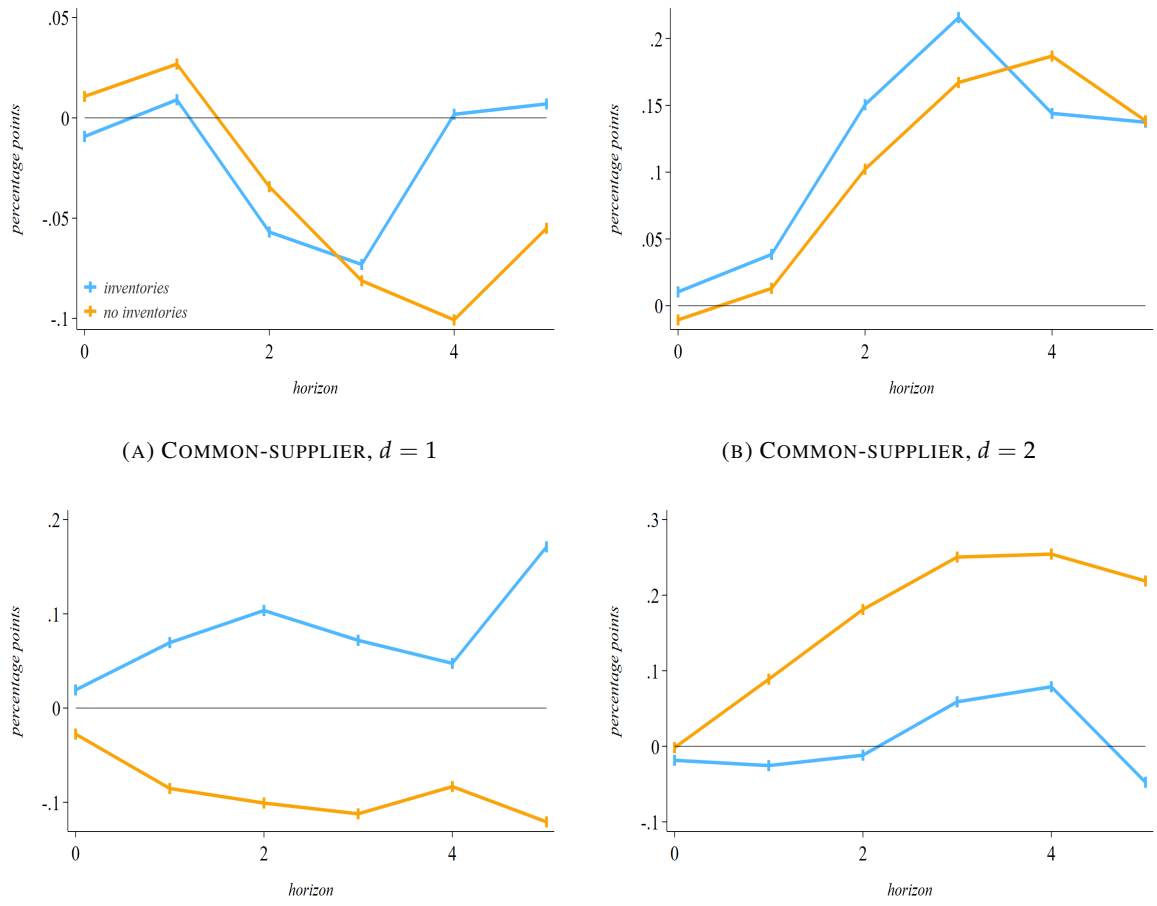
(A) $d = 1$ (B) $d = 2$

FIGURE D.7: COMMON-BUYER LINKAGES AND COMOVEMENT IN PERIPHERAL SECTORS

Note: given the *common-buyer* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposures) characterizing the North American production network in year 2007, it is shown the response of sectoral employment of most central sectors (with *centrality* defined according to sector's relevance in final consumption) to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). *Source*: BEA and own calculations.

D.3. Robustness with private inventories

Responses for comovement in sectoral employment levels when controlling for changes in sectoral inventories, distinguishing further between sectors with inventories and those without. The results are shown in Figure D.8. Note that the plots refer dynamic changes in inventories and the Input-Output matrix held at the baseline year 2007.



(A) COMMON-SUPPLIER, $d = 1$

(B) COMMON-SUPPLIER, $d = 2$

(C) COMMON-BUYER, $d = 1$

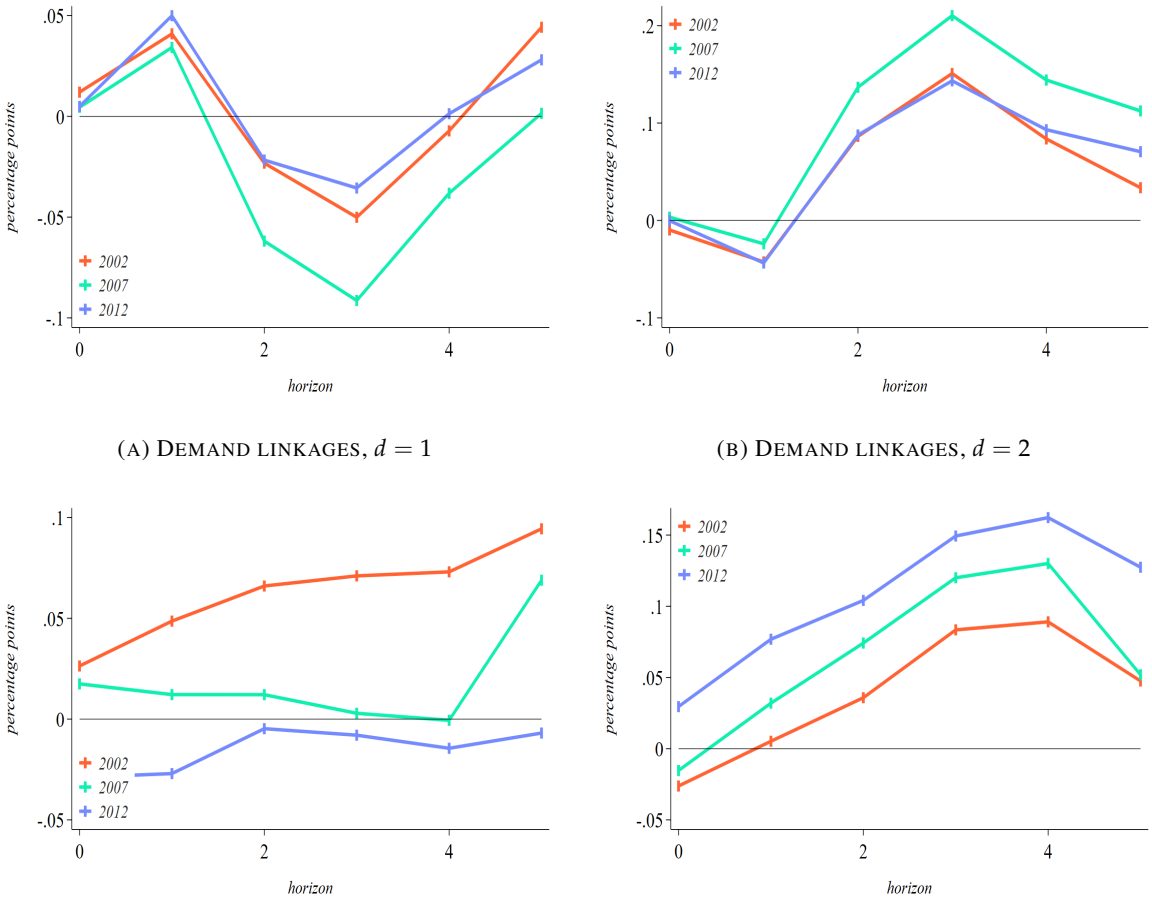
(D) COMMON-BUYER, $d = 2$

FIGURE D.8: COMOVEMENT UNDER CHANGES IN SECTORAL INVENTORIES

Note: given both the *common-supplier* (*i.e.*, demand linkages across sectors given their common upstream suppliers, at the top) and the *common-buyer* (*i.e.*, supply linkages across sectors given their common downstream buyers, at the bottom) network distances, and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposures) characterizing the North American production network in year 2007, it is shown the response of sectoral employment of sectors to a 1% increase in the employment level of other sectors when controlling for the changes in private inventories, for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Solid lines corresponds to the average response of employment across sectors, given 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (8). Responses are referred to employment changes in closer (distance equal to 1) and further (distance equal to 2) sectors. Since not all sectors have inventories, the panels are disentangled by subgroups that make use or not of private inventories. *Source*: BEA and own calculations.

D.4. Robustness with production network at different base years

Responses for comovement in sectoral employment levels in the main analysis of Section 5 are centred on the implementation of a sectoral production network represented by an Input-Output matrix in the baseline year 2007. Differently, Figures D.9, D.10 and D.11 perform again all the main analysis by referring, instead, to a production network structure five years before (2002) and five years after (2012) the baseline year.



(A) DEMAND LINKAGES, $d = 1$

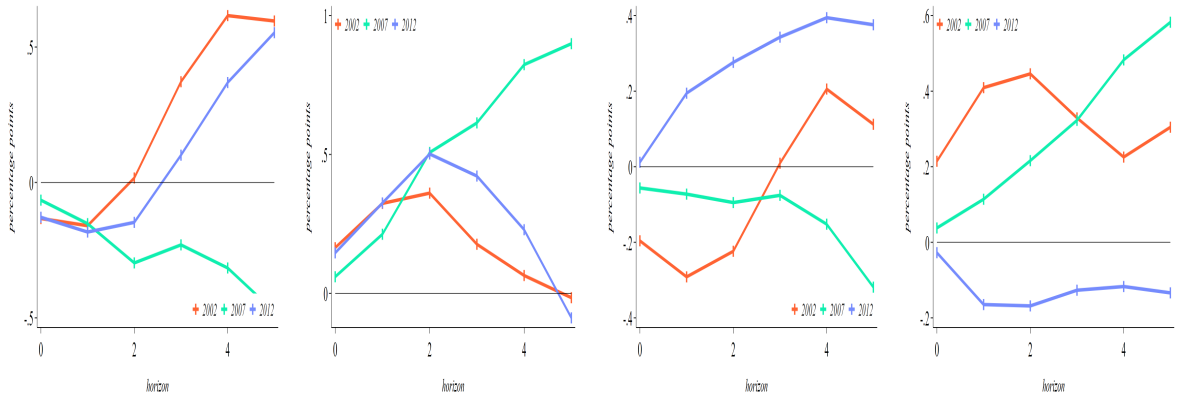
(B) DEMAND LINKAGES, $d = 2$

(C) SUPPLY LINKAGES, $d = 1$

(D) SUPPLY LINKAGES, $d = 2$

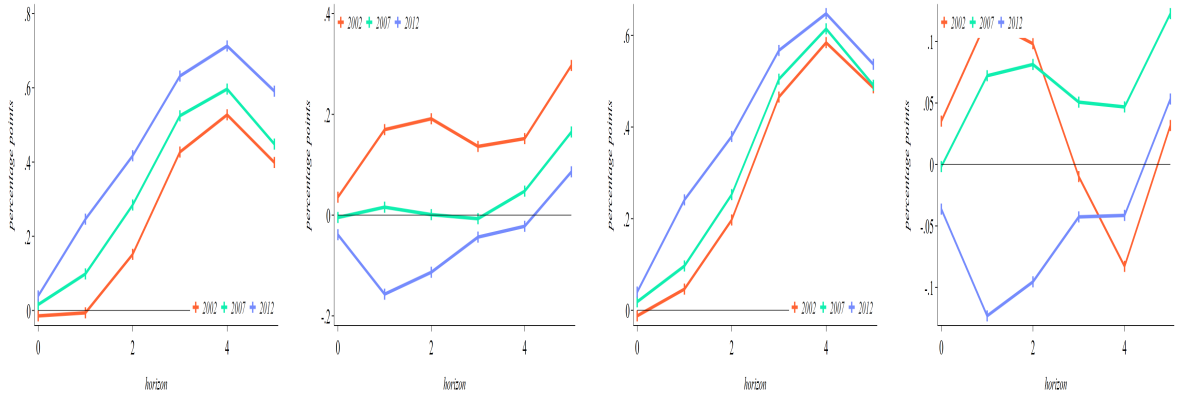
FIGURE D.9: COMOVEMENT OVER DIFFERENT BASELINE YEARS

Note: given both the *common-supplier* (*i.e.*, demand linkages across sectors given their common upstream suppliers) and the *common-buyer* (*i.e.*, supply linkages across sectors given their common downstream buyers) network distances, and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposures) characterizing the North American production network over different years (2002, 2007, and 2012), it is shown the response of sectoral employment of sectors to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Expanding to different production network structures the results in the main text: Panels D.9a-D.9b expand their relatives of Figure 7, while Panels D.9c-D.9d do so for their relatives in Figure 8. Solid lines corresponds to the average response of employment across sectors, while 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (8), roughly correspond to the associated figures in the main text. Responses are referred to employment changes in closer (distance equal to 1) and further (distance equal to 2) sectors. *Source*: BEA and own calculations.



$d = 1$ $d = 2$ $d = 1$ $d = 2$

(A) TOP 5% (B) TOP 10%

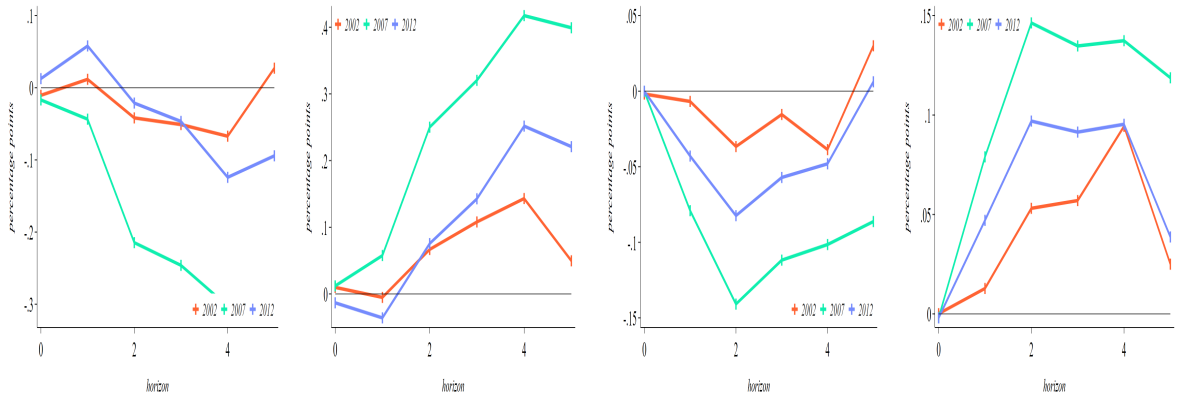


$d = 1$ $d = 2$ $d = 1$ $d = 2$

(C) TOP 20% (D) TOP 30%

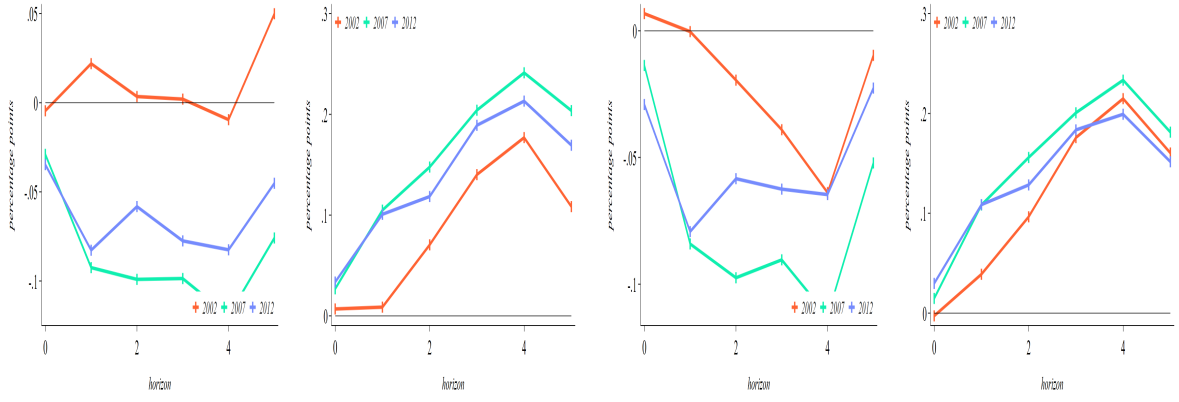
FIGURE D.10: COMMON-SUPPLIER LINKAGES AND INTERLINKED SECTORS OVER THE YEARS

Note: given both the *common-supplier* network distance (*i.e.*, demand linkages across sectors given their common upstream suppliers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposures) characterizing the North American production network over different years (2002, 2007, and 2012), it is shown the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Expanding to different production network structures the responses of Panels 7c-7d of Figure 7, each panel of this figure shows the employment response of top 5% (Panel D.10a), 10% (Panel D.10b), 20% (Panel D.10c), and 30% (Panel D.10d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) or further (distance equal to 2) sectors. Solid lines corresponds to the average response of employment across sectors, while 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (8), roughly correspond to the associated figure. *Source*: BEA and own calculations.



$d = 1$ $d = 2$ $d = 1$ $d = 2$

(A) TOP 5% (B) TOP 10%



$d = 1$ $d = 2$ $d = 1$ $d = 2$

(C) TOP 20% (D) TOP 30%

FIGURE D.11: COMMON-BUYER LINKAGES AND INTERLINKED SECTORS OVER THE YEARS

Note: given both the *common-buyer* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposures) characterizing the North American production network over different years (2002, 2007, and 2012), it is shown the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Expanding to different production network structures the responses of Panels 8c-8d of Figure 8, each panel of this figure shows the employment response of top 5% (Panel D.11a), 10% (Panel D.11b), 20% (Panel D.11c), and 30% (Panel D.11d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) or further (distance equal to 2) sectors. Solid lines corresponds to the average response of employment across sectors, while 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (8), roughly correspond to the associated figure. *Source*: BEA and own calculations.

E. Illustrative episodes of horizontal transmission

In conclusion, a set of recent episodes that exemplify the formalized mechanisms of horizontal shock transmission are presented. Firms/sectors not directly linked through bilateral transactions nonetheless exhibit simultaneous outcomes because they share a common downstream buyer (common-buyer distance) or a common upstream supplier (common-supplier distance). These episodes illustrate how network-based horizontal economic distance, rather than observed vertical exposure, can generate comovement.

Technological shock in Samsung and common-buyer spillovers to competing memory suppliers (2026).– On February 2, 2026, Samsung Electronics announced significant technological progress in the development and production of next-generation High Bandwidth Memory 4 (HBM4) chips. HBM is a vertically stacked dynamic memory technology optimized for high-speed data processing and is a critical input for Artificial Intelligence (AI) accelerators produced by firms such as NVIDIA. The announcement constituted a positive, firm-specific productivity shock for Samsung in the high-performance memory segment. Because Samsung and other memory producers – most notably Micron and SK Hynix – compete to supply advanced memory chips to the same downstream buyers (including NVIDIA), market expectations regarding future procurement patterns were immediately revised. Financial press coverage reported that investors anticipated a potential reallocation of orders toward Samsung, reflecting its improved technological position. Importantly, Samsung and its rival suppliers do not maintain vertical trade relationships with one another. Nevertheless, the positive shock in Samsung generated adverse valuation effects for competing firms, reflecting expected substitution by the common downstream buyer.[IX]

Common-buyer Apple's reallocation of AirPods assembly orders (2022).– In 2022, Apple reduced and partially suspended assembly orders for AirPods 2 at Goertek, a major Chinese contract manufacturer. The shock was idiosyncratic to Goertek, stemming from operational issues that impaired its ability to meet Apple's standards and delivery schedules. Rather than reducing overall AirPods production proportionally, Apple responded by shifting demand from the negatively shocked supplier to its competitor, reallocating a substantial share of the affected orders to Luxshare Precision Industry, another contract manufacturer producing similar components for Apple. As a result, Luxshare experienced an increase in expected output and revenues, while Goertek faced declining orders and downward revisions in market expectations. Goertek and Luxshare are not vertically linked to each other; however, they are horizontally proximate because they depend on the same downstream buyer (Apple). The negative shock in one supplier therefore translated into a positive demand shift toward its competitor: Luxshare experienced an increase in expected orders and revenues despite having no direct vertical linkage with Goertek.[X]

Common-buyer Apple's reallocation for OLED panels (2026).– In January 2026, Apple shifted a significant portion of its iPhone OLED (Organic Light-Emitting Diode) panel orders from BOE Technology Group in China to the South Korean company Samsung Display after persistent production and yield issues affected BOE's ability to meet quality and volume requirements. OLED panels are critical display components used in iPhones, and Apple relies on

[IX] Reported by financial outlets such as `Barron's`, `investing.com`, and `MarketWatch` in February 2026.

[X] Covered by `Yicai Global`, `Euronew` and `MacRumors` in November 2022.

multiple suppliers to ensure supply stability and bargaining power. The operational difficulties at BOE constituted a negative firm-specific shock. In response, Apple reallocated millions of display panel orders to Samsung Display, which had available capacity and superior production reliability. The reallocation of downstream demand increased expected revenues for Samsung Display while reducing BOE's projected sales and market share. As in the AirPods episode, the two suppliers are not connected through vertical production relationships; rather, they are horizontally linked because they sell to the same downstream buyer. The transmission of the shock occurred through Apple's procurement decision, not through direct trade between BOE and Samsung Display.[XI]

AI-driven surge in memory chips demand (2025).– In 2025, the rapid expansion of Artificial Intelligence (AI) applications led to an unprecedented increase in global demand for advanced memory chips, particularly DRAM (Dynamic Random Access Memory) and HBM (High Bandwidth Memory). DRAM constitutes the primary volatile memory used in computing systems, while HBM is a high-performance, vertically stacked memory technology optimized for AI accelerators and data-intensive workloads. The surge in AI-related computing, especially for large language models and generative AI systems, generated a sharp increase in procurement by major cloud service providers and AI hardware producers. Because global production capacity for advanced memory is concentrated among a small number of firms (*e.g.*, Samsung, SK Hynix, Micron), the downstream demand shock tightened upstream supply conditions and increased prices. Downstream sectors relying on these same memory inputs – including consumer electronics manufacturers, enterprise hardware producers, and specialized computing firms – faced rising input costs and constrained availability, even though they were not directly linked to each other through trade.[XII]

Cloud-based services expansion (2023-2024).– Between 2023 and 2024, large-scale investment in cloud computing infrastructure – driven by AI adoption and data center expansion – significantly increased downstream demand for advanced semiconductors, including graphics processing units (GPUs) and specialized AI accelerators. This expansion placed substantial pressure on upstream semiconductor manufacturing capacity. Because leading-edge chip fabrication is highly concentrated – particularly in firms such as Taiwan Semiconductor Manufacturing Company (TSMC) – the increase in cloud-sector demand reduced effective upstream supply available to other industries relying on the same fabrication processes. Automotive manufacturers, consumer electronics firms, and industrial equipment producers faced delays and supply constraints despite having no direct trade relationship with cloud service providers. The resulting synchronized production adjustments across otherwise unrelated sectors reflected shared exposure to upstream fabrication capacity rather than vertical trade interdependence.[XIII]

Reallocation of TSMC supply toward AI customers (2023-2026).– In 2023, the rapid expansion of Artificial Intelligence (AI) applications triggered a sharp increase in downstream demand for high-performance computing chips, particularly graphics processing units (GPUs). These chips rely on advanced fabrication nodes supplied by TSMC, the world's leading contract chip manufacturer, operating as a pure-play foundry, producing chips designed by numerous fa-

[XI] Reported by `MacRumors`, `TechNode`, and `DisplayDaily` in January 2026.

[XII] Reported in 2025 by `investing.com`, `Business Standard`, and `Reuters`.

[XIII] Documented by the `CNBC`, `Reuters`, and `Financial Times` during 2023-2024.

bless semiconductor firms across industries, and whose cutting-edge production capacity is both technologically complex and capacity-constrained. Given the long planning horizon of semiconductor manufacturing – with wafer capacity typically allocated several quarters, and often years, in advance – the AI-driven surge in orders during 2023 induced TSMC to reprioritize production toward high-margin AI customers: in 2024, TSMC reallocated portions of its advanced fabrication capacity toward AI chip designers such as NVIDIA, responding to exceptionally strong demand for AI accelerators. Industry reports indicate that such reallocations affected fabrication schedules extending into 2025–2026, as a growing share of advanced-node capacity was reserved for AI-related chips. Given finite leading-edge fabrication capacity, TSMC's prioritization of AI-related chip production effectively reduced available output for other clients, including smartphone and consumer electronics chip designers. As a consequence, other downstream clients competing for the same foundry capacity faced relatively tighter access or lower priority in allocation. These experienced longer lead times and constrained access to advanced nodes, despite not interacting directly with the AI hardware sector. The shock therefore originated in heightened demand from one downstream segment (AI hardware) but propagated through TSMC as a shared upstream production node, generating horizontal effects on multiple unrelated industries.[XIV]

Across all episodes, the defining feature is that idiosyncratic variations are mediated by the horizontal geometry of the production network rather than by bilateral trade, where shared downstream buyers (*common-buyer* distance) or shared upstream suppliers (*common-supplier* distance) determine the strength and the direction of micro-level shock transmission. These cases provide qualitative empirical grounding for the paper's central theoretical insight: comovement can arise from horizontal network proximity even in the absence of vertical Input-Output linkages. Note that all the common-buyer episodes hinge on a downstream reallocation if demand (Section 3), while all the common-supplier episodes hinge on a congestion effect, central in Subsection 3.4 and Supplemental Appendix B. The three common-buyer episodes are instances of Proposition 2 and Corollary 1: a firm-specific shock to one supplier makes the common buyer reallocate a given budget, and its rivals move in the opposite direction; with close substitutes, as for suppliers of the same component, the shocked firm gains when its productivity rises and loses when it falls. The three common-supplier episodes are instances of Proposition 1 and Corollary B.2: a demand surge at suppliers with constrained capacity raises the costs of all their customers, which comove in costs although they do not trade with each other. In these episodes, part of the adjustment takes the form of delays and allocation of capacity rather than prices, a non-price form of congestion that the model captures only through rising marginal costs.

[XIV] Reported in January-February 2026 by `MacRumors`, `Business Insider`, and `AppleInsider`.

REFERENCES

Acemoglu, D., V. M. Carvalho, A. Ozdaglar, and A. Tahbaz-Salehi (2012), "The network origins of aggregate fluctuations", *Econometrica*, 80(5):1977–2016

Baqaee, D. R. and E. Farhi (2019), "The macroeconomic impact of microeconomic shocks: beyond Hulten's Theorem", *Econometrica*, 87(4):1155–1203

Boldrin, M., L. J. Christiano, and J. D. M. Fisher (2001), "Habit persistence, asset returns, and the business cycle", *American Economic Review*, 91(1):149–166

Bonacich, P. (1987), "Power and centrality: a family of measures", *American Journal of Sociology*, 92(5):1170–1182

Carvalho, V. M. (2014), "From micro to macro via production networks", *Journal of Economic Perspectives*, 28(4):23–48

Conley, T. G. and B. Dupor (2003), "A spatial analysis of sectoral complementarity", *Journal of Political Economy*, 11(2):311–352

Fernald, J. G. (2014), "A quarterly, utilization-adjusted series on Total Factor Productivity", *Federal Reserve Bank of San Francisco Working Paper 2012-19*

Hulten, C. R. (1978), "Growth accounting with intermediate inputs", *Review of Economic Studies*, 45(3):511–518

Oulton, N. (2022), "The Divisia approach to measuring output and productivity: with an application to the BEA-BLS integrated industry-level production account, 1987-2020", *Discussion Paper 2022-17, Centre For Macroeconomics*

Ramey, V. A. and S. Zubairy (2018), "Government spending multipliers in good times and in bad: evidence from US historical data", *Journal of Political Economy*, 126(2):850–901

Rogerson, R. (1987), "An equilibrium model of sectoral reallocations", *Journal of Politcal Economy*, 95(4):824–834